

Critical-Point Defects in Surreal Arithmetic

Large-cardinal embeddings, exact support thresholds,
descent to the target model, omnific independence,
surcomplex fields, omnific integers, and first failures

A merged research report from four manuscripts

Manuscripts 06, 07, 09 (batch 30) and 10 (batch 32), prepared for Vladimir Reshetnikov (07, 09, 10 with ChatGPT), 23 September 2026; 07 is the base text

Main result. An elementary embedding $j : V \rightarrow M$ with critical point κ induces two ordered-field embeddings $J, H_j : \mathbf{No} \rightarrow \mathbf{No}$. They agree on every real number, ordinal, and Conway monomial, but

$$J(x) = H_j(x) \iff |\text{supp}(x)| < \kappa.$$

Their images have an exactly determined intersection. A single coefficient of their discrepancy recovers the normal measure derived from j . [merge] For every individual normal form of length λ , the value $H_j(x)$ lies in the target model exactly when $j \restriction \lambda \in M$; for a normal-measure ultrapower this happens exactly when x has at most κ terms [06]. Without any large cardinal, almost-disjoint families give algebraically independent omnific integers over the whole short-support field [09]. Countably strong linear maps first fail at measurable cardinals [10].

Scope and status. All central assertions below have written proofs under explicit hypotheses. The general Hahn and large-cardinal ingredients are classical; the exact defect and intersection package is offered as a candidate original contribution. Publication priority has not been certified, and the article is neither refereed nor Lean-verified. It does not claim a solution to a named published conjecture or a new consistency proof for a large-cardinal axiom.

Repository context: VladimirReshetnikov/Surreal, snapshot 0865f043aec1.

Standalone source, compiled PDF, proof-status notes, and finite algebraic regression checks accompany this article. [merge] All three manuscripts were written against the same snapshot; the merged report was placed in commit 21375f8. Sources, notation and merge decisions: Section 1.5 and Appendix D; non-claims of every source: Appendix C.

Abstract

Large-cardinal embeddings preserve every finitary arithmetic identity, but need not preserve the external strong sums that define surreal normal forms. We make this distinction exact. Let $j : V \rightarrow M$ be an amenable elementary embedding into a transitive inner class, with critical point κ . Its restriction to surreal sign sequences gives an ordered-field embedding J . The same action on Conway monomials has a unique strongly $\mathbb{R}$ -linear extension H_j . If x has normal form indexed by θ , then $J(x) - H_j(x)$ is precisely the subseries of the transformed normal form indexed by $j(\theta) \setminus j \text{ `` } \theta$. Consequently the equalizer of J and H_j is exactly the real-closed field of surreal numbers with fewer than κ outer normal-form terms. Moreover, their two image fields intersect exactly in the common image of that equalizer, and are its incomparable logarithmic and strong-sum closures. For a positive strongly summable family, preservation by J holds exactly when the index set has cardinality less than κ .

The defect has a first term at the first missing index κ . For $h_A = \sum_{\alpha \in A} \omega^{-\alpha}$, $A \subseteq \kappa$, its coefficient at $\omega^{-\kappa}$ is the indicator of membership in the normal measure derived from j . This gives a coefficient realization of classical ultrafilter evaluation and a precise measurable-cardinal criterion for the associated Hadamard mask algebra. The field results transfer to surcomplex numbers and to omnific integers; a purely infinite omnific defect encodes the same measure while its constant term vanishes. We also obtain a logarithmic defect homomorphism, an exact exponential compatibility locus, and a supercompact seed-coefficient construction. The article distinguishes these contributions from standard Hahn reindexing, classical derived measures, and the repository's earlier work on birthday expansions and forcing. Twelve further research questions and a formalization plan conclude the account.

[merge] This report merges three manuscripts on the same pair of maps. The text above is the abstract of the base manuscript 07. Manuscript 06 adds descent: the companion value $H_j(x)$ belongs to the target model exactly when the seed $j \text{ `` } \lambda$ does, where λ is the normal-form length of x ; in a set ultrapower this is closure of M under λ -sequences, and for a normal-measure ultrapower it gives an exact three-way classification at support sizes $< \kappa$, κ , $> \kappa$, without GCH. It also adds bounded and omnific measure codes, and support-cover and exact-descent forms of strong compactness and supercompactness. Manuscript 09 adds a large-cardinal-free theory of the short-support field (real closed at every uncountable cardinal, closed and nowhere dense, with an omnific integer part), algebraically independent omnific masks over that whole field from almost-disjoint families (2^κ of them in one set-sized Hahn field when $2^{<\kappa} = \kappa$), a compatible-field-embedding criterion for measurability, seed coefficients over arbitrary index sets, and an omnific prefix-tree form of weak compactness. The strong-compactness, supercompactness and weak-compactness statements are credited translations of classical ultrafilter and tree characterizations. As a consequence, if there is a measurable cardinal then there is a field embedding of **No** that preserves and reflects the omnific integers and their constant terms but is not strongly additive. There are 34 research questions in all, and a list of every source's non-claims.

[merge] Manuscript 10 (batch 32, Sections 22–26) proves a converse for countably strong maps. If a coefficient-linear map between full real or complex Hahn fields preserves every countable strong sum, then it preserves the summability of every set-indexed family, and its first failure to commute with a sum, if any, occurs at a measurable cardinal; the proof localizes to one output coefficient on one support and uses the classical finite complete-ultrafilter factorization of product functionals. Every such map is the sum of a canonical strong part and a countably strong remainder vanishing on all monomials; for J these are H_j and D_j . Conversely a measure on κ gives order-preserving additive automorphisms of **No** that stabilize **Oz**, fix every monomial and any prescribed set of parameters, and first fail at κ ; they are not multiplicative. The least size of a full Hahn field with such a failure is 2^κ , attained by 09's field K_κ . Hence the absence of measurable cardinals is equivalent to strongness of every countably strong linear map, and, with J , of every countably strong coefficient-fixing **Oz**-preserving field embedding. The merge deduces that 09's criterion for measurability needs no Hadamard product. Manuscript 10 adds twelve questions, for 46 in all.

Keywords. Surreal number; Conway normal form; strong summation; measurable cardinal; elementary embedding; omnific integer; surcomplex number; Hahn field; ultrafilter; supercompactness.

Contents

1	The questions answered and the novelty boundary	1
1.1	The distinction that drives the construction	1
1.2	Assumption and headline statements	1
1.3	The headline statements of the other two sources [merge]	2
1.4	What is not being asserted	3
1.5	Sources, conventions and notation [merge]	4
2	Foundations, normal forms, and the meaning of a sum	6
2.1	Surreal conventions	6
2.2	Elementary embeddings and external sets	7
2.3	The absoluteness needed to compare the two fields	8
3	The field of short normal forms [09, 06]	11
4	The strong companion	13
4.1	General Hahn reindexing	13
5	The exact normal-form defect	15
5.1	The transformed index set	15
5.2	Why small support is not small birthday	17
6	The equalizer field and the exact image intersection	17
6.1	A canonical isomorphism by erasing nonimage exponents	18
7	Explicit witnesses and incomparability	19
8	The exact strong-summation threshold	20
8.1	An omnific monomial witness [09]	21
8.2	An intrinsic closure theorem for the image	22
9	Descent of the companion to the target model [06]	23
9.1	Landing is equivalent to the seed	23
9.2	Why the seed measures sequence closure in an ultrapower	24
9.3	Normal measures: the exact three-way classification	25
10	Recovering a normal measure from one coefficient	26
10.1	Boolean masks at ordinal exponents	26
10.2	Real-valued masks and the Hadamard product	29
10.3	A compatible-field-embedding criterion [09]	31
11	Omnific integers and a purely infinite measure code	33
11.1	The ring and its short-support part	33
11.2	A defect with no finite residue	34
12	Surcomplex numbers and the Gaussian omnific lattice	35
13	Algebraic independence over the short-support field [09]	37
13.1	A coordinate separation lemma	37
13.2	The independence theorem	38
13.3	Maximal cardinality inside a fixed Hahn field	39
14	Defect algebra and a logarithmic homomorphism	40
14.1	What all old-image coefficients fail to see	40
14.2	Passing from a multiplicative discrepancy to an additive one	41

15 An exact exponential compatibility locus	42
16 Supercompactness as a seed-coefficient system	45
16.1 Coding an arbitrary index set [09]	45
16.2 The supercompact seed [07]	46
16.3 Covering seeds and exact seeds [09]	47
17 Strong compactness as a support cover; supercompactness as exact descent [06]	48
17.1 The set-theoretic principles and their witnesses	48
17.2 Binary normal forms and coefficientwise covers	48
17.3 What the distinction says, and what it does not say	50
18 Weak compactness as an omnific prefix principle [09]	50
18.1 Length-preserving binary codes	50
18.2 The branch is not an order-topological limit	51
19 What large cardinals change, and what they do not	51
19.1 No change to finite algebraic truth	51
19.2 Why there is no Kunen contradiction	52
19.3 A hierarchy of information	52
19.4 Choice and the scope of set formation	53
19.5 What is visible at each level of structure [09]	53
19.6 What has actually been established [06]	54
19.7 Hypotheses which cannot be silently removed [06]	54
20 Position relative to the literature and the supplied repository	55
20.1 Classical ingredients and proposed contributions	55
20.2 The repository snapshot actually inspected	56
20.3 The collection at the placement commit [merge]	57
20.4 Relation to current automorphism work	58
21 Further research questions and concrete next steps	59
21.1 Questions of manuscript 06 [06]	61
21.2 Questions of manuscript 09 [09]	63
22 Measurable first failures: source, conventions and headline [10]	65
22.1 The problem and the headline statement	66
22.2 Why the result is more than an example	66
22.3 Prior work and proposed contributions	67
22.4 Conventions, notation and the pin [merge]	67
23 Countable detection and complete-ultrafilter functionals [10]	69
23.1 Fields, supports, and class size	69
23.2 Hahn summability	69
23.3 The omnific subring	70
23.4 Products on a fixed support are only coefficient products	70
23.5 Countability detects admissibility, not the sum	70
23.6 The product-functional theorem	71
24 The least failure is measurable, and the canonical strong part [10]	73
24.1 A coefficient-by-coefficient description	74
24.2 A canonical strong part and a monomial-invisible remainder	74
24.3 Multiplicative maps and derivations	76
25 Measurable transvections, set-sized workspaces and the surcomplex transfer [10]	77

25.1	A bounded support interval	77
25.2	The symmetry and all its preservation properties	77
25.3	Arbitrary set parameters can be fixed	79
25.4	Set-sized workspaces and a sharp cardinal bound	79
25.5	Surcomplex and Gaussian-omnific consequences	81
26	Field embeddings, target tests, and what manuscript 10 answers [10]	81
26.1	The elementary-embedding input, recalled	81
26.2	An explicit unrepresentable target test	83
26.3	Consequences for the questions of Section 21 [merge]	84
26.4	Proof audit and limits of the conclusions [10]	85
26.5	Questions of manuscript 10 [10]	87
26.6	What manuscript 10 does not claim [10]	89
27	Conclusion	90
A	Proof dependencies and a formalization plan	91
A.1	Dependency order	91
A.2	A proposed Lean interface, not a formalization claim	91
A.3	06’s dependency map and formalization route [06]	92
A.4	09’s dependency ledger and formalization boundary [09]	93
B	Verification, limitations, and an audit checklist	93
B.1	Finite regression checks	93
B.2	Points requiring particular attention in an independent review	94
B.3	06’s audit checklist [06]	94
B.4	09’s specific error checks [09]	95
B.5	Claim summary	95
C	Non-claims, by source [merge]	96
D	Provenance and merge decisions [merge]	100
	References	102

1 The questions answered and the novelty boundary

1.1 The distinction that drives the construction

For a set-theoretic elementary embedding, the image $j(I)$ of an infinite set is generally larger than its pointwise image $j \restriction I$. For a surreal normal form, this is not merely a difference between two encodings of the same sum. The additional indices can carry additional nonzero monomials. The resulting discrepancy is visible in arithmetic, even when every original summand is individually fixed.

This article answers the following concrete question under a measurable-cardinal hypothesis: *how much of a surreal number is determined by the action of an embedding on all real coefficients and all monomials?* The answer is exactly its short-support part. There are two explicitly associated embeddings that agree on all those generators, and even on every ordinal, but disagree on every surreal whose outer normal form has at least κ terms. The distinction is a support-cardinality threshold, not a birthday cutoff.

Conway’s normal-form theorem and the generalized-series presentation of **No** are the classical background [1, 2, 3, 4]. Strongly additive maps and monomial reindexing also have a substantial literature. In particular, Kaplan–Krapp–Serra distinguish strong from non-strong automorphisms and study their relation to extra surreal structure [5]. The contribution proposed here is not the bare possibility of non-strong behavior. It is the exact critical-point formula, equalizer, intersection, and coefficient reconstruction for the specific large-cardinal construction.

1.2 Assumption and headline statements

Throughout the main construction, assume that

$$j : V \longrightarrow M \subseteq V \tag{1.1}$$

is an elementary class embedding, M is transitive and contains every ordinal, and $\kappa = \text{crit}(j)$ exists. Class parameters are allowed in the Separation and Replacement instances used below. A sufficient, familiar instance is a normal ultrapower embedding from a measurable cardinal. One can carry out that instance in ZFC using the measure as a set parameter; the use of classes is notation for definable constructions. The general class-embedding version can be read in GBC with the indicated embedding as an amenable class. These hypotheses are discussed in Section 2.

Let

$$\mathcal{S}_{<\kappa} = \{x \in \mathbf{No} : |\text{supp}(x)| < \kappa\}, \tag{1.2}$$

where $\text{supp}(x)$ means the set of *outer exponents* in its Conway normal form. Zero has empty support. Define $J(x) = j(x)$ on surreal sign sequences, and define

$$H_j \left(\sum_{\alpha < \theta} r_\alpha \omega^{a_\alpha} \right) = \sum_{\alpha < \theta} r_\alpha \omega^{J(a_\alpha)}. \tag{1.3}$$

The central conclusions, proved in Sections 4–8, are

$$\text{Eq}(J, H_j) = \mathcal{S}_{<\kappa}, \tag{1.4}$$

$$J(\mathbf{No}) \cap H_j(\mathbf{No}) = J(\mathcal{S}_{<\kappa}) = H_j(\mathcal{S}_{<\kappa}), \tag{1.5}$$

$$J \left(\sum_{i \in I} x_i \right) = \sum_{i \in I} J(x_i) \iff |I| < \kappa \quad (x_i > 0 \text{ and strongly summable}). \tag{1.6}$$

The two image fields in (1.5) are incomparable, not merely different. Their common image is real closed. Coefficient projection canonically identifies the two images as immediate valued extensions of that common field (Theorem 6.4). An intrinsic version shows that every positive strong sum of at least κ image elements escapes the elementary image (Theorem 8.6).

For $A \subseteq \kappa$, put $h_A = \sum_{\alpha \in A} \omega^{-\alpha}$. One then obtains

$$[\omega^{-\kappa}] (J(h_A) - H_j(h_A)) = \begin{cases} 1, & \kappa \in j(A), \\ 0, & \kappa \notin j(A). \end{cases} \quad (1.7)$$

Thus the usual derived measure is recovered from a distinguished coefficient of the arithmetic defect. The derived measure itself is classical [6]; the coefficient realization is the new construction being proposed.

1.3 The headline statements of the other two sources [merge]

Manuscripts 06 and 09 prove the equalizer and the defect formula independently, under essentially the same hypotheses, and state them in their own headline theorems. We print both, with the numbering of this report in the proof pointers. The support cardinal $|\text{supp}(x)|$, the number of nonzero terms in the Conway normal form of x , is 06's $\text{sc}(x)$; the short-support field $\mathcal{S}_{<\kappa}$ is 06's $K_{<\kappa}$ and 09's F_κ (Section 1.5).

Theorem 1.1 (Main comparison and descent [06]). *Under the standing embedding hypotheses (2.3), the following assertions hold.*

- (i) *J and H_j are ordered-field embeddings fixing $\mathbb{R}$ and agreeing on all monomials. The map H_j preserves every set-indexed strong sum; J preserves every such sum indexed by a set of cardinality $< \kappa$, but not every one indexed by κ .*
- (ii) *For every surreal x , $J(x) = H_j(x) \iff |\text{supp}(x)| < \kappa$.*
- (iii) *For every normal form of ordinal length λ , $H_j(x) \in \mathbf{No}^M \iff j \text{ “ } \lambda \in M$. When j is a set-ultrapower embedding, these conditions are equivalent to ${}^\lambda M \cap V \subseteq M$.*
- (iv) *If j is the ultrapower by a normal measure on κ , then $H_j(x) \in \mathbf{No}^M \iff |\text{supp}(x)| \leq \kappa$. Consequently equality, unequal descent, and failure of descent occur exactly at support sizes $< \kappa$, κ , and $> \kappa$.*

The proof is distributed among Theorem 4.3, Theorem 5.1, Theorem 8.1, Theorem 9.1, Lemma 9.3 and Theorem 9.7. No cardinal-arithmetic hypothesis such as GCH is needed for the last clause [06].

Two further manifestations of the same phenomenon are in 06's introduction. For $x = p(x) + r(x) + \varepsilon(x)$, with $p(x)$ purely infinite, $r(x) \in \mathbb{R}$ and $\varepsilon(x)$ infinitesimal,

$$\frac{H_j(\exp x)}{\exp(H_j x)} = \exp(J(p(x)) - H_j(p(x))),$$

so the Hahn lift commutes with exponentiation at x exactly when the positive-exponent part has fewer than κ terms (Theorem 15.3, proved by 06 and 07). Exact descent of $H_j(h_\lambda)$ characterizes a supercompactness witness, whereas an internally short binary-support *cover* characterizes a strong compactness witness; multiplication by a monomial gives the same tests inside the omnific integers (Section 17) [06].

Theorem 1.2 (Main package [09]). *Assume that κ is measurable and let $j : V \rightarrow M$ be the transitive ultrapower embedding by a normal measure on κ . There are ordered field embeddings $J, H_j : \mathbf{No} \rightarrow \mathbf{No}$ with the following properties.*

- (i) Both fix $\mathbb{R}$ and agree on every Conway monomial. The map J commutes with Conway's omega-map, and H_j preserves every set-indexed strongly summable family.
- (ii) Their exact equalizer is $\text{Eq}(J, H_j) = \mathcal{S}_{<\kappa}$. For a normal form of length θ , the difference is the subseries indexed by $j(\theta) \setminus j \prec \theta$ in the normal form of $J(x)$.
- (iii) The map J preserves every strongly summable family of cardinality less than κ , but fails to preserve a specified family of size κ consisting of purely infinite omnific monomials.
- (iv) The equalizers on omnific integers and on the specified complexification are $\mathbf{Oz} \cap \mathcal{S}_{<\kappa}$ and $\mathcal{S}_{<\kappa}(i)$, respectively.
- (v) There are 2^κ explicitly described omnific integers, algebraically independent over $\mathcal{S}_{<\kappa}$, on each of which J and H_j disagree.
- (vi) The normal measure can be recovered from one designated normal-form coefficient of $J(Y_A)$, where $A \subseteq \kappa$ and Y_A is an explicit omnific code of A .

Proof pointers: (i) Proposition 2.10 and Theorem 4.3; (ii) Theorem 5.1; (iii) Theorems 8.1 and 8.4; (iv) Theorems 11.1 and 12.1; (v) Theorem 13.5 and Corollary 13.6; (vi) Theorem 10.3. According to 09, part (ii), its full difference formula, and the coefficient-independent algebraic construction underlying part (v) are its main proposed contributions. The hypotheses and the exact formulas matter. Merely constructing a Hahn lift, observing that elementary embeddings fail to preserve some long sequences, or translating an ultrafilter into a numerical code would not by itself constitute the same result [09]. Part (ii) is proved independently by all three sources and printed once, as Theorem 5.1.

1.4 What is not being asserted

We do not establish that arbitrary non-strong surreal embeddings require a measurable cardinal. We do not prove a new consistency result about measurability, supercompactness, or elementary self-embeddings of the set universe. We do not identify the two embeddings as automorphisms. Nor do we claim that their equality on monomials implies equality on a completed field: the whole point is that the relevant external sums need not be preserved by J . [merge] *Re-scoped in batch 32*. The first sentence stays true of arbitrary embeddings, but manuscript 10 establishes the converse for embeddings, indeed for all $\mathbb{R}$ -linear maps, that preserve every countable strong sum: their first failure is a measurable cardinal (Theorem 24.1, Theorem 26.2). An embedding that fails a countable sum is outside that theorem (Remark 26.3).

The proofs are mathematical arguments, not output from a theorem prover. The accompanying finite checks test finite identities used in the arguments; they cannot test an elementary embedding of the universe or a nonprincipal complete ultrafilter. A targeted literature and repository comparison appears in Section 20; it is not an exhaustive priority search.

[09] The results do not establish the existence or consistency of any large cardinal. They do not produce a nontrivial elementary embedding $(V, \in) \rightarrow (V, \in)$, and do not settle the outstanding structure questions for all surreal exponential automorphisms. The research questions at the end include genuine gaps in the present theory; they are not presented as an exhaustive list of previously published open problems.

[06] Manuscript 06 is an AI-assisted research draft with written proofs, not a refereed or machine-checked paper. Its proposed contributions are the exact comparison, defect and descent statements, not the underlying Hahn-lifting or ultrapower-seed machinery. They were not located in the sources it inspected; this is not a priority certification or a claim to resolve a named published conjecture. All its large-cardinal implications retain their hypotheses; no consistency strength beyond the corresponding classical ultrafilter principle is claimed. Appendix C lists every non-claim of every source.

1.5 Sources, conventions and notation [merge]

Three manuscripts. The report is built from manuscripts 06, 07 and 09 of batch 30. All three were written on 23 September 2026 against the repository snapshot 0865f04, and all three study the same pair of maps (J, H_j) induced by an elementary embedding with critical point κ . They prove the same spine independently: the companion lift, the first missing index, the exact defect formula, the short-support equalizer, the small-family summation threshold and the measure coefficient. [merge] A fourth manuscript, 10 (manuscript 05 of batch 32, pinned to 7af8a30), was added later as Sections 22–26; its conventions, renamed symbols and pin are in Section 22.4, and its labels carry the sub-prefix `lce:mf`: (“measurable failures”).

Source	Manuscript	Contributes
07	<i>Critical-Point Defects in Surreal Arithmetic: Large-cardinal embeddings, exact support thresholds, surcomplex fields, and omnific integers</i>	The base text: every section not marked otherwise, with its exact image intersection and projection isomorphism, positive-family threshold and escape theorem, mask algebra and mask criterion, omnific and surcomplex transfers, logarithmic defect, exponential locus and two closures, supercompact seed, twelve questions, formalization plan and finite regression.
06	<i>Critical Supports and Large Cardinals in Surreal Arithmetic: Ultrapower transport, Hahn lifts, and omnific descent</i>	A proof of the critical-point facts (Lemma 2.3); descent of $H_j(x)$ to M , the landing spectrum, the seed–closure lemma, the cover obstruction and the normal-measure trichotomy (Section 9); bounded and omnific measure codes with a standard-part reading; formal evaluation of power series and the exponential scale corollary; support covers and exact descent for compactness principles (Section 17); the fixed class of H_j ; nine questions and an audit checklist.
09	<i>Large Cardinals and the Support Geometry of Surreal Embeddings: Exact equalizers, additional normal-form terms, and omnific algebraic independence</i>	A second proof of the absoluteness lemma with its limit step exposed; the short-support field for every uncountable κ (Section 3); omnific monomial witnesses; the omega-map incompatibility; order continuity; almost-disjoint omnific independence and the 2^κ family (Section 13); the compatible-field-embedding criterion; seed coefficients over arbitrary index sets; weak compactness as an omnific prefix principle (Section 18); the visibility hierarchy; thirteen questions.
10	<i>Measurable Cardinals and the First Failure of Surreal Summation: Exact failure spectra, canonical strong parts, and omnific-preserving additive symmetries</i> (batch 32)	The first failure of a countably strong linear map is measurable; complete-ultrafilter localization; the canonical strong part; measurable transvections; the workspace criterion and the 2^κ bound; the exact boundary for countably strong embeddings; the target test ω^κ ; twelve questions (Sections 22–26).

How the merge is marked. A statement proved by several sources is printed once, with its sources in brackets, for example [06, 07, 09]. Material from 06 or 09 carries its source tag, and its labels carry the sub-prefixes `lce:cs`: (06, “critical supports”) and `lce:sg`: (09, “support geometry”); labels of the base text carry `lce`: alone. Where two sources give genuinely different proofs, the second is printed as a marked second route. Text written for the merge, rather than taken from a source, is marked [merge]. Base-text passages without a tag are 07’s. Every proof imported from 06 or 09 was re-read against the hypotheses of this report; the places where a source’s hypotheses differ are noted at the statement.

Hypotheses of the three sources. 07 assumes an elementary class embedding $j : V \rightarrow M \subseteq V$ into a transitive class containing every ordinal, with a critical point, and class-parameter Separation and Replacement (an amenable class in GBC, or a normal ultrapower in ZFC). 06 assumes the same, stated as (2.3) with $M \models \text{ZFC}$ added. 09 states its main package for the ultrapower by a normal measure, and says that its Sections on the equalizer, the threshold, the omnific and surcomplex consequences and the measure apply to any such class embedding with uncountable critical point. On the shared theorems 06 and 07 thus have the weakest stated hypotheses, and 09's arguments need no more; 07 is the base because it is the broader of 06 and 07 (placement commit 21375f8). 09's independence theorems assume no embedding at all, and 06's descent results add the hypotheses stated with them (a set-ultrapower presentation, or an index set of size κ).

Notation. The base text's symbols are kept. Symbols of 06 and 09 were renamed where they collide with the base text or with each other; no normalization changed. The table lists every renaming.

Source symbol	Here	Reason and meaning
09's T, D	H_j, D_j	the strong companion and the defect $J - H_j$, as in 06 and 07
06's $K_{<\kappa}$, 09's F_κ	$\mathcal{S}_{<\kappa}$	surreals with fewer than κ normal-form terms; 06's $K_{\leq\kappa}$ is $\mathcal{S}_{\leq\kappa}$
09's I_κ	$\mathcal{F}_{<\kappa} = \mathbf{Oz} \cap \mathcal{S}_{<\kappa}$	short-support omnific integers
06's $\text{sc}(x)$	$ \text{supp}(x) $	the support cardinal
06's $\ell_{\text{nf}}(x)$, 09's $\text{len}_{\text{NF}}(x)$	$\text{len}(x)$	normal-form length; 06 writes λ and 09 writes μ for it, kept as λ in Section 9
06's u_λ , 09's Q_θ	h_λ, h_θ	$\sum_{\alpha < \lambda} \omega^{-\alpha}$, 07's (7.1)
06's c_A	$\tilde{h}_A = h_\kappa + h_A$	the bounded code $\sum_{\alpha < \kappa} (1 + \mathbf{1}_A(\alpha)) \omega^{-\alpha}$; the letter c is kept for coefficients
06's $z_A = \omega^\kappa c_A$	$\tilde{z}_A = z_\kappa + z_A$	07's z_A is $\omega^\kappa h_A$
06's $v_\lambda = \omega^\lambda u_\lambda$	$\zeta_\lambda = \omega^\lambda h_\lambda$	v is 07's valuation; $\zeta_\kappa = z_\kappa$
06's $x^\uparrow$	$p(x)$	the purely infinite part, as in 07
06's LM, LC	LT	leading monomial and coefficient enter only through the leading term
06's binary series Q, Q^+	$\widehat{B}, \widehat{B}^+$	07's Question 21.1 uses P, Q for embeddings
06's W_s	U_s	$\{X : s \in j(X)\}$, as in 07 and 09; 06's letter W for a fine ultrafilter is kept locally
09's e_α and q_α	$q_\alpha = \omega^{-\alpha}$	09 had two names for one number; e is 07's enumeration of an index set
09's $u_\alpha = \omega^{e_\alpha}$	$\tau_\alpha = \omega^{q_\alpha}$	u is 06's detector and 07's infinitesimal
09's $Z_\theta, Z_A, W_f, Z_{A,h}$	$Y_\theta, Y_A, Y_f, Y_{A,e}$	omnific masks $\sum \tau_\alpha$; z_A is 07's shifted mask
09's $c_\gamma(x)$	$[\omega^{q_\gamma}]x$	the coefficient of τ_γ ; 07's coefficient notation
09's bijection $h : \theta \rightarrow I$	e	07's name; h is 07's mask
09's $E = \bigoplus \mathbb{Q}e_\alpha, H$	G_κ, Ξ	E is 09's embedding in Theorem 10.10; H collides with H_j
09's polynomial P ; exponents h_0, h ; coefficient a_0	$\Phi; d_0, d'; \varrho$	P_θ is 07's missing-index set; a is an exponent
09's index j in $\bigcup_{j \neq i} A_j$	k	j is the embedding
09's $\chi(A)$	$\chi_E(A)$	07's χ acts on the mask algebra
09's strings $s \in 2^\alpha$, tree S , injections q_α , lengths D_α	$\sigma, \mathcal{R}, \iota_\alpha, \Delta_\alpha$	s is a seed, $\mathcal{S}_{<\kappa}$ the short field, q_α an exponent, D_j the defect
09's $P_\kappa \lambda, \text{Ord}$	$P_\kappa(\lambda), \mathbf{On}$	base-text spelling

Some tempting readings are false and are excluded throughout. The equalizer $\mathcal{S}_{<\kappa}$ is not a fixed field: $J(\kappa) = H_j(\kappa) = j(\kappa) \neq \kappa$ (Warning 5.4). Short support is not a birthday bound: every monomial lies in $\mathcal{S}_{<\kappa}$ (Section 3). The Hadamard product $\odot$ is not surreal multiplication. ω^a is always Conway's omega-map, never $\exp(a \log \omega)$. Two different letters denote measure codes: h_A has coefficients 0, 1 and $\tilde{h}_A$ has coefficients 1, 2; the omnific masks z_A (exponents $\kappa - \alpha$) and Y_A (exponents $\omega^{-\alpha}$) are different numbers. The class $\mathcal{L}(j)$ of Section 9 is a landing spectrum, not the logarithmic defect L_j .

2 Foundations, normal forms, and the meaning of a sum

2.1 Surreal conventions

We use the sign-sequence realization of **No**: a surreal is a function from an ordinal to $\{-, +\}$. Its birthday $b(x)$ is the domain of that function. The numerical order is the first-disagreement order, with $- < \text{undefined} < +$. Simplicity is proper prefix inclusion. Ordinals are identified with all-plus sequences. The distinction between an ordinal and its all-plus sign code will normally be suppressed, but all applications of j commute with that identification.

The Conway monomial map is denoted by

$$\Omega(a) = \omega^a.$$

Here ω^a is the Conway omega-map, not an abbreviation for $\exp(a \log \omega)$ with an arbitrarily chosen exponential structure. It satisfies $\omega^{a+b} = \omega^a \omega^b$, and is strictly increasing. Every nonzero surreal has a unique normal form

$$x = \sum_{\alpha < \theta} r_\alpha \omega^{a_\alpha}, \quad r_\alpha \in \mathbb{R} \setminus \{0\}, \quad a_\alpha > a_\beta \quad (\alpha < \beta < \theta), \quad (2.1)$$

where θ is an ordinal. All supports in this article are sets, even though the full coefficient and exponent universes are classes. Write $\text{len}(x) = \theta$ for the length of (2.1) and $\text{supp}(x) = \{a_\alpha : \alpha < \theta\}$. For $x \neq 0$, define

$$\text{LT}(x) = r_0 \omega^{a_0}, \quad v(x) = -a_0.$$

This is the additive natural valuation with values in $(\mathbf{No}, +)$; $v(0) = \infty$. The infinitesimal ideal of the finite-surreal valuation ring is

$$\mathfrak{m} = \{0\} \cup \{x \in \mathbf{No} : v(x) > 0\}.$$

These conventions use decreasing omega-exponents; sources that use t^g with increasing support have the opposite sign convention for exponents.

Definition 2.1 (Strong summability). A set-indexed family $(x_i)_{i \in I}$ is *strongly summable* if $\bigcup_{i \in I} \text{supp}(x_i)$ is reverse well ordered and each exponent occurs in the support of only finitely many members. Its strong sum is the unique normal form obtained by adding those finitely many coefficients at each exponent and discarding zero coefficients. A map is *strongly $\mathbb{R}$ -linear* if it is $\mathbb{R}$ -linear and preserves all such external set-indexed sums, including their summability.

The field laws for generalized series, in particular the finiteness of contributions in a product, are part of the Hahn–Neumann support theorem [2, 3, 5]. We use them for sets of exponents. There is no summation over a proper-class support.

Remark 2.2 (Not order-topological convergence). A displayed strong sum is not defined as the supremum of its partial sums, and is not asserted to be an order-topological limit in all of **No**. For example, the sum h_κ constructed below exists by Definition 2.1. This makes no assertion that its partial sums converge against every positive surreal neighborhood. Confusing these two notions would invalidate the threshold argument.

2.2 Elementary embeddings and external sets

The standard large-cardinal facts used here are that the critical point of (1.1) is an uncountable inaccessible cardinal, that j fixes V_κ , and that a measurable cardinal supplies such an embedding by a normal ultrapower [6]. In particular,

$$j(\mathbb{R}) = \mathbb{R}, \quad j(r) = r \ (r \in \mathbb{R}), \quad \mathbb{R}^M = \mathbb{R}^V. \quad (2.2)$$

We take these large-cardinal facts as background, not as results of this article. When a normal measure U is chosen, the ultrapower is first formed on set functions with domain κ and then collapsed to its transitive class model. Countable completeness gives well-foundedness. All individual images and all set-indexed images used below are sets.

We distinguish $j(I)$, an element of M , from the external pointwise image $j \text{ `` } I = \{j(i) : i \in I\}$, formed in V . The latter need not belong to M . Every cardinality without a superscript is computed in V . In particular, $|\text{supp}(x)| < \kappa$ always concerns the original external support. No cardinal arithmetic in M is silently substituted for it.

06's class conventions and standing hypotheses [06]. The concrete large-cardinal statements concern a countably complete ultrafilter on a set and its well-founded ultrapower. They can be read in ZFC using definable classes. For statements about an arbitrary class embedding, work in GBC with the embedding as a class parameter, or assume the corresponding class-parameter Separation and Replacement instances. In particular, restrictions of j to sets and their ranges are sets. No object consisting of all class embeddings is formed. 06's standing hypotheses are

$$j : V \longrightarrow M \subseteq V, \quad M \text{ transitive}, \quad M \models \text{ZFC}, \quad \mathbf{On}^M = \mathbf{On}^V, \quad \text{crit}(j) = \kappa, \quad (2.3)$$

with j elementary for set theory; the canonical sign-sequence presentation identifies $\mathbf{No}^M$ with a subclass of $\mathbf{No}^V$, and when j comes from an ultrapower its values are understood after the Mostowski collapse. For a function $a : \lambda \rightarrow V$,

$$j(a)(j(\alpha)) = j(a(\alpha)) \quad (\alpha < \lambda), \quad (2.4)$$

but $j(a)$ has values at every index below $j(\lambda)$, not just at indices in $j \text{ `` } \lambda$. Hypotheses (2.3) are those of (1.1), with the model-theoretic reading made explicit.

Where 07 takes the critical-point facts as background, 06 proves them; the proof is short and is printed here, since Section 9 uses the power-set clause.

Lemma 2.3 (Critical point facts [06]). *Under (2.3), the cardinal κ is uncountable, regular, and strong limit. The embedding fixes every real number, and M and V have the same reals. Moreover $\mathcal{P}(\kappa)^M = \mathcal{P}(\kappa)^V$.*

Proof. Define the derived ultrafilter $U_j = \{A \subseteq \kappa : \kappa \in j(A)\}$. It is a set under the class conventions. Preservation of complements makes it an ultrafilter. It is nonprincipal since $j(\alpha) = \alpha$ for $\alpha < \kappa$. For a family of fewer than κ members, j fixes the indexing ordinal, so κ belongs to the image of their intersection. Thus U_j is κ -complete. No set of size $< \kappa$ belongs to it, by intersecting the complements of its singletons. The critical point cannot be finite or ω , since these ordinals are fixed by their set-theoretic definitions.

A cofinal sequence in κ of length $< \kappa$ would partition κ into fewer than κ bounded, hence null, sets, contradicting completeness. For the strong-limit property, suppose $\mu < \kappa$ and choose distinct subsets $A_\xi \subseteq \mu$ for $\xi < \kappa$. For each coordinate $\eta < \mu$, choose the U_j -large side of the partition of κ according to whether $\eta \in A_\xi$. Intersecting these μ large sets yields a large set on which all A_ξ agree, contradicting their distinctness. Consequently $2^\mu < \kappa$. The same standard argument shows that κ is a cardinal; alternatively, a bijection from a smaller ordinal onto κ contradicts the definition of the critical point after applying j .

By induction on rank below κ , using inaccessibility and the fact that sets of size $< \kappa$ have pointwise images, j fixes V_κ pointwise. In particular it fixes all real numbers. Every ambient real therefore belongs to M , and transitivity gives the other inclusion. Finally, for $A \subseteq \kappa$, the identity $A = j(A) \cap \kappa$ puts A in M . Since every member of the internal power set is an actual subset, the two power sets agree. $\square$

Remark 2.4 (Why the target has all real coefficients [09]). A second route to the reals clause: a real can be coded by a subset of ω , and j fixes that subset coordinate by coordinate since $j(\omega) = \omega$. Thus every real of V belongs to M and $\mathbb{R}^M = \mathbb{R}^V$, so Lemma 2.6 applies to the target. It does not assert that M contains every set of reals or every set of old surreals.

Lemma 2.5 (The first missing index [06, 07, 09]). *For any set I ,*

$$j(I) = j \text{ `` } I \iff |I| < \kappa.$$

For any ordinal θ , define $P_\theta = j(\theta) \setminus j \text{ `` } \theta$. Then

$$P_\theta = \emptyset \quad (\theta < \kappa), \quad \min P_\theta = \kappa \quad (\theta \geq \kappa).$$

Proof. Choose a bijection $e : \mu \rightarrow I$, where μ is the cardinality of I . If $\mu < \kappa$, then $j(\mu) = \mu$, and every member of $j(I)$ has the form $j(e)(\alpha) = j(e(\alpha))$ for some $\alpha < \mu$. Conversely, suppose $\mu \geq \kappa$. The ordinal κ belongs to $j(\mu)$, but is not $j(\alpha)$ for any ordinal α : below κ all ordinals are fixed, while for $\alpha \geq \kappa$ we have $j(\alpha) \geq j(\kappa) > \kappa$. Thus $j(e)(\kappa)$ is in $j(I)$ but not in $j \text{ `` } I$, by injectivity of $j(e)$.

For ordinal $\theta < \kappa$, every index is fixed. If $\theta \geq \kappa$, then $j(\theta) > \kappa$, all smaller indices are in $j \text{ `` } \theta$, and κ is not. This proves both assertions. $\square$

06 (its Lemma “The first omitted index”) and 09 state and prove the ordinal clause with the same argument; the first clause, for an arbitrary set I , is 07’s.

2.3 The absoluteness needed to compare the two fields

There are two universe changes in play. Applying j moves an input into M . Then the identity inclusion regards the resulting old surreal sign sequence as a surreal of V . We need the standard constructions to agree under this second passage.

Lemma 2.6 (Absolute surreal constructions [06, 07, 09]). *Let $M \subseteq V$ be transitive models or inner classes of ZFC with the same ordinals and the same reals. On inputs that belong to M , the identity inclusion $\mathbf{No}^M \subseteq \mathbf{No}^V$ preserves order, the field operations, the Conway omega-map, and canonical normal forms. An M -set strongly summable family in M is externally strongly summable in V , with the same sum.*

Proof. Order and prefix comparison are comparisons of ordinal-length sign functions, so they are absolute. The simplest separator of a given set cut is constructed by following the forced signs until the first permissible termination; this construction depends only on its input sign sequences. Thus the cut construction is absolute.

Well-founded recursion on canonical sign options gives the same negation, addition, multiplication, and omega-map in both universes. Inverses then agree by uniqueness in the ordered field.

For normal forms we use the constructive sign-sequence version of the Conway–Gonshor normal-form theorem [2, Chapter 5], not just the existence of an unspecified isomorphism with a Hahn field. The conversion from a reverse well-ordered list of real-coefficient monomials to its surreal sign sequence is given by the normal-form sign expansion: ordinal sign concatenations with the prescribed deletions determined by the input exponent signs and preceding terms. The ordinal operations, restrictions, and sign comparisons in that construction are absolute on a given set of inputs. Hence an M -normal form represents the same sign sequence in V . Uniqueness of the normal form in V shows that it acquires no extra terms on passing to V .

Finally, a reverse well ordering of an M -set has an ordinal order-type witness in M , which remains a witness in V ; finiteness of an old fiber is absolute. The old family is therefore still strongly summable. Its coefficientwise finite sums agree, as do the normal forms just considered. $\square$

All three sources prove this lemma. 06’s proof follows the same constructive route as the one above, adds that the sign-recursion algorithm and its ordinal stopping stage are absolute because the rank bound on the option sign lengths bounds the construction in both universes, and includes Gonshor’s exponential in the statement: it is treated by its canonical option recursion with finite Taylor polynomials, the real constants and natural indices agreeing (this is Lemma 15.1). 06 states the lemma under (2.3), and 09 for a transitive inner class of ZFC containing all ordinals and all reals of V . 09’s proof is genuinely different at the limit step and does not invoke the sign-expansion algorithm; it is printed as a second route.

Second route [09]. Sign comparison and sign restriction are absolute. Every sign prefix of an old sign sequence is old: it is obtained by restricting that sequence to an ordinal in M .

For the simplest-number operation, let a nonempty convex class be given by an absolute condition on old parameters and suppose that M has a member of that class. Its externally simplest member is an initial sign segment of every member, in particular of the internally simplest member. It therefore belongs to M . Membership is absolute, so internal minimality forces the two simplest members to coincide. Applied recursively to the canonical option sets, this proves absoluteness of addition, negation, multiplication and the omega-map. Inversion then follows from uniqueness of a multiplicative inverse. The relevant recursions use sign prefixes, finite arithmetic, and sets of options; their domains for any specified input are sets. For the omega-map one can use its defining cut with positive integers and their reciprocals, so there is no quantification over newly available coefficients.

For normal forms, let $(r_\alpha, a_\alpha)_{\alpha < \theta} \in M$ be a normal-form sequence, and argue by induction on θ . The zero and successor steps use finite field operations. Suppose that θ is a nonzero limit and all proper partial sums p_η have already been identified in the two universes. Consider the convex class defined by

$$|z - p_{\alpha+1}| < \frac{\omega^{a_\alpha}}{n} \quad (\alpha < \theta, 1 \leq n < \omega). \quad (2.5)$$

Both universes have their own proposed full sum in this class. The condition is absolute on old z because its quantifiers range over the old ordinal θ and the ordinary natural numbers, and every boundary in (2.5) is already absolute.

In either universe, a member of this class has the prescribed sequence as an initial segment of its normal form. To see this, compare its normal form with the prescribed sequence at the first exponent or coefficient at which they differ; the corresponding inequality in (2.5) fails. A discrepancy after all prescribed terms is allowed, but then the prescribed full sum is a normal-form truncation of that member. The classical truncation-simplicity theorem says that a proper normal-form truncation is simpler than the original surreal; see [14, Theorem 19.2(4)]. It follows that the full sum is the simplest member of (2.5) in each universe.

The simplest-member argument of the first paragraph now identifies the two full sums. This completes the induction. This reasoning is also consistent with the usual recursive construction of normal-form sums, whose limit stage chooses a simplest approximant [14, Theorem 10.2].

For $x \in M$, take its normal-form sequence computed in M ; it sums externally to the same x , and uniqueness of normal forms gives the claim about extraction of coefficients and exponents. Finally, for a strongly summable family in M , its union of supports has in M an ordinal enumeration witnessing reverse well-ordering. This witness is still valid in V . The finitely many occurrences at each exponent and their coefficient sums are absolute. The resulting normal-form sequence belongs to M , so its sum is absolute by the preceding argument. $\square$

09 stresses the point of this route: it is not enough to say that normal forms are definable and therefore absolute, since general definable objects need not be absolute [09]. 06 adds that the lemma is a foundational interface, not a claim that the full Hahn fields of M and V coincide.

Remark 2.7 (Old monomials, new sums [06]). An external sequence of old monomials may fail to be an element of M , and its sum may be a new surreal. That failure is precisely what the descent theorem of Section 9 detects.

Remark 2.8 (Where a classical theorem is used). The normal-form sign-expansion theorem is a classical input to Lemma 2.6; it is not reproved from the axioms of surreal games here. This is the one place where an arbitrary Hahn-field identification would be insufficient: the identification must be the canonical Conway normal form. A formal verification should first prove this absoluteness lemma from that canonical construction, rather than assuming that every isomorphism chosen in M is absolute.

Remark 2.9 (An instance for the definable-surreals report [merge]). Lemma 2.6 supplies the absoluteness that the definable-surreals report leaves unproved in its question on intermediate support fields (dsn:q:support in docs/foundations-and-computation/definable-surreals-and-omnific-integers). There, a proper subfield $K \subsetneq \mathbf{No}$ with $\omega \in K$, $\omega^{-(\alpha+1)} \in K$ for every ordinal α , and $\text{supp}(a) \subseteq K$ for $0 \neq a \in K$ (dsn:eq:Khyp) carries an **On**-indexed algebraically independent omnific family (dsn:thm:amplify), and the report records that $\mathbf{No}^M$ for a proper same-ordinal inner model M “would satisfy” these hypotheses if the omega map and normal forms were absolute, while the universes report proves only the arithmetic and cuts (univ:prop:absolute). Let M be a transitive inner class of ZFC with $\mathbf{On}^M = \mathbf{On}^V$ and $\mathbb{R}^M = \mathbb{R}^V$, and $M \neq V$. Then $K = \mathbf{No}^M$ satisfies dsn:eq:Khyp: it is a subfield containing the ordinal ω by the arithmetic part of the lemma; each $\omega^{-(\alpha+1)}$ computed in M is the ambient one by the omega-map part; and the normal form of $a \in \mathbf{No}^M$ computed in M is its ambient normal form, so its exponents are elements of $\mathbf{No}^M$. It is proper: a sign sequence of length α codes a subset of α , so $\mathbf{No}^M = \mathbf{No}^V$ would put every set of ordinals of V into M , and a transitive inner model of ZFC containing every set of ordinals is all of V . Hence the support-subfield amplification applies to $\mathbf{No}^M$. The targets M of the embeddings (1.1) are instances (Lemma 2.3); for the ultrapower by a normal measure U , $M \neq V$ since $U \notin M$. Same-ordinal inner models without all reals, such as L when $\mathbb{R} \not\subseteq L$, and HOD when it lacks a real, are not covered: all three sources state the lemma for models with the same reals (06 through (2.3) and Lemma 2.3), and this report does not claim more.

Proposition 2.10 (The elementary-induced surreal embedding [06, 07, 09]). *The map $J : \mathbf{No} \rightarrow \mathbf{No}$, $J(x) = j(x)$ on sign codes, is an ordered-field embedding that fixes $\mathbb{R}$, and*

$$J(\omega^a) = \omega^{J(a)}, \quad \mathbf{b}(J(x)) = j(\mathbf{b}(x)).$$

It preserves and reflects prefix simplicity. It is elementary in the ordered-field language.

Proof. Elementary transfer into M preserves the defining surreal constructions; Lemma 2.6 then identifies their values with the values in V . The all-plus coding of ordinals is definable, so it commutes with j . The

domain of $j(x)$ is $j(\text{dom } x)$, proving the birthday formula. Prefix simplicity is a definable statement about sign functions and is absolute after transfer. Fixing the reals follows from (2.2). An embedding between real-closed ordered fields is elementary by quantifier elimination for RCF. Applied formula by formula, the same argument works for these class fields: it makes no demand for a class-sized satisfaction set. $\square$

06 (its Theorem “The two lift constructions”) and 09 (its Proposition “The set-induced action”) prove the same statement, and add two clauses: the range of J is contained in $\mathbf{No}^M$, and J maps $\mathbf{Oz}$ into $\mathbf{Oz}$ (Theorem 11.1). 09 also records here the external normal form (5.1) of $J(x)$, whose domain is $j(\theta)$, not $j \text{ “ } \theta$, and notes that injectivity follows either from elementarity of equality or from strict order preservation [06, 09].

Remark 2.11 (The first moved birthday [09]). Every surreal of birthday less than κ is fixed by J : its sign sequence has a fixed domain of length less than κ and fixed values. The ordinal κ itself is moved to $j(\kappa)$. Thus κ is also the least birthday of a surreal moved by J . This is a *fixed-point* statement about J , distinct from the *equalizer* statement about J and H_j below.

3 The field of short normal forms [09, 06]

This section needs no elementary embedding: κ is an arbitrary uncountable cardinal, and $\mathcal{S}_{<\kappa}$ is defined by (1.2). 07 proves that $\mathcal{S}_{<\kappa}$ is real closed through the two embeddings (Theorem 6.1); 09 proves it directly for every uncountable κ , regular or singular. Because κ is a cardinal,

$$\text{len}(x) < \kappa \iff |\text{supp}(x)| < \kappa, \quad (3.1)$$

since an ordinal of cardinality below κ is itself below κ . We use the “large monomial” convention: for a set-sized divisible ordered subgroup $G \subseteq (\mathbf{No}, +)$, the normal forms supported in G form the Hahn field $\mathbb{R}((\omega^G))$ with reverse well-ordered supports; this is the increasing-support convention with $t^g = \omega^{-g}$. The standard Hahn-field theorem says that $\mathbb{R}((\omega^G))$ is real closed when G is divisible [1, 2, 14].

Theorem 3.1 (Short-support real closed field [09]). *For every uncountable cardinal κ , regular or singular, $\mathcal{S}_{<\kappa}$ is a real closed ordered subfield of $\mathbf{No}$. It contains $\mathbb{R}$, every Conway monomial and every ordinal. It is closed under normal-form truncation, but is not a birthday cutoff $\{x : \text{b}(x) < \lambda\}$ for any ordinal λ .*

Proof. Let S be the union of the supports of finitely many elements of $\mathcal{S}_{<\kappa}$. Then $|S| < \kappa$. The rational vector space $G = \text{span}_{\mathbb{Q}}(S) \subseteq \mathbf{No}$ is a set-sized divisible ordered group, with $|G| \leq \max(|S|, \aleph_0) < \kappa$. Every Hahn series supported in G consequently belongs to $\mathcal{S}_{<\kappa}$. The selected elements lie in the real closed field $\mathbb{R}((\omega^G))$.

The fields so obtained form a directed family: replace two exponent groups by their rational span to include both. Their union is exactly $\mathcal{S}_{<\kappa}$. This proves closure under the field operations. Positive square roots and roots of odd-degree polynomials can be found in one such real closed subfield containing the finitely many coefficients. Thus the union is real closed. No union of $\text{cf}(\kappa)$ unrelated supports was taken, so no regularity assumption has been used.

Real numbers and monomials have support of size at most one. The Cantor normal form of an ordinal is finite and agrees with its surreal normal form, so ordinals also belong to $\mathcal{S}_{<\kappa}$. Truncation only deletes support. Finally, $\mathcal{S}_{<\kappa}$ contains ordinals of arbitrarily large birthday but omits a normal form of length κ , for instance h_κ of (7.1) (09 uses the omnific number Y_κ of Theorem 8.4). No ordinal birthday cutoff contains ordinals of arbitrarily large birthday, while the full class $\mathbf{No}$ also contains the omitted number. $\square$

Corollary 3.2 (Truncation-closed short fields [06]). *Let κ be the critical point of (2.3). The class $\mathcal{S}_{<\kappa}$ is exactly the equalizer of J and H_j , and it is a truncation-closed subfield containing every monomial. The class $\mathcal{S}_{\leq\kappa} = \{x \in \mathbf{No} : |\text{supp}(x)| \leq \kappa\}$ is also a truncation-closed subfield.*

Proof. The equalizer assertion is Theorem 5.1, and an equalizer of field embeddings is a subfield. Direct support counting also proves both field statements. Unions and pairwise sums of fewer than κ support exponents still have size $< \kappa$. For an inverse write $x = r\omega^a(1+\varepsilon)$ and use $(1+\varepsilon)^{-1} = \sum_{n \geq 0} (-\varepsilon)^n$. Its support lies in the additive monoid generated by $\text{supp}(\varepsilon)$, of cardinality at most $\max(\aleph_0, |\text{supp}(\varepsilon)|)$. Regularity and uncountability give the strict bound. The same estimate gives the nonstrict bound κ for $\mathcal{S}_{\leq\kappa}$. Truncation only removes terms. Every monomial has one term. $\square$

[merge] The counting proof uses only that κ is uncountable: the product of two cardinals below an infinite κ is below κ , and the monoid bound needs no regularity. So both field statements hold for every uncountable cardinal, in agreement with Theorem 3.1; 06 states them at the critical point, where they are needed.

For example, ω^a belongs to the equalizer no matter how large the birthday or support of a is. The map need not fix this monomial: it sends it to $\omega^{J(a)}$. Equality of the two lifts is different from being a fixed point of either lift [06].

Remark 3.3 (The countable boundary [09]). The finite-support collection $\mathcal{S}_{<\omega}$ is not a field. For example, $1 - \omega^{-1}$ has finite support, but its inverse is the strongly summable geometric series $\sum_{n < \omega} \omega^{-n}$. This explains the uncountability hypothesis in Theorem 3.1.

Corollary 3.4 (Individual transcendence [09]). *If $x \in \mathbf{No} \setminus \mathcal{S}_{<\kappa}$, then x is transcendental over $\mathcal{S}_{<\kappa}$.*

Proof. A real closed field has no proper ordered algebraic extension. A surreal algebraic over $\mathcal{S}_{<\kappa}$ would generate such an extension and must already belong to $\mathcal{S}_{<\kappa}$. $\square$

The corollary is much weaker than algebraic independence of many specified long-support numbers. Two transcendental numbers can satisfy a polynomial relation. Section 13 proves the stronger assertion by an explicit coefficient-separation argument [09].

Proposition 3.5 (Order-closed but not an order-dense approximation [09]). *For uncountable κ , the class $\mathcal{S}_{<\kappa}$ is closed and nowhere dense in the order topology of $\mathbf{No}$. It is not definable, even with finitely many surreal parameters, in the pure ordered-field language.*

Proof. Take $x \notin \mathcal{S}_{<\kappa}$ and its first κ normal-form terms $r_\alpha \omega^{a_\alpha}$. Choose an exponent $b < a_\alpha$ for all $\alpha < \kappa$, using a cut below this set of exponents. If $|y - x| < \omega^b$, then every exponent in the support of $y - x$ is at most b , and hence is smaller than every a_α . Thus all the first κ terms of x remain unchanged in y , and $y \notin \mathcal{S}_{<\kappa}$. The complement is open.

To see that $\mathcal{S}_{<\kappa}$ has no interior, fix a nonzero $z \notin \mathcal{S}_{<\kappa}$. If an interval about $y \in \mathcal{S}_{<\kappa}$ of radius $\varepsilon > 0$ were contained in $\mathcal{S}_{<\kappa}$, choose a positive ordinal $\beta > |z|/\varepsilon$. Then $\beta \in \mathcal{S}_{<\kappa}$ and $0 < |z/\beta| < \varepsilon$, but $y + z/\beta \notin \mathcal{S}_{<\kappa}$: otherwise field closure would imply $z \in \mathcal{S}_{<\kappa}$. This is a contradiction. Closedness and empty interior give nowhere density.

Quantifier elimination for real closed fields makes every definable one-variable set a finite union of intervals and points [20]. The class $\mathcal{S}_{<\kappa}$ is infinite and has no interior, so it cannot be such a set. $\square$

The proposition explains why agreement on monomials plus order continuity does not force the two embeddings to agree everywhere (Proposition 8.5). Strong summability is a different extension principle from approximation in the order topology [09].

[merge] *Related (batch 32).* Part II of the first- κ report (docs/surcomplex/first-kappa-coefficients, prefix `fkc:`) studies the same class, written $K_{\kappa, \mathbb{R}}(\mathbf{No})$ there, for every uncountable κ and without large cardinals. It proves the closedness and nowhere density above for valuation neighbourhoods, in $\mathbf{No}$ and inside $\mathcal{S}_{<\lambda}$ for every $\lambda > \kappa$ (`fkc:sb:prop:closed`); that every set-presented gap of $\mathcal{S}_{<\kappa}$ has cofinality pair $(\text{cf } \kappa, \text{cf } \kappa)$ (`fkc:sb:thm:gapcharacter`); that $\mathcal{S}_{<\kappa}$ is ν -saturated exactly for infinite $\nu \leq \text{cf } \kappa$ (`fkc:sb:cor:saturation`); and that every strongly summable family of fewer than $\text{cf } \kappa$ members of $\mathcal{S}_{<\kappa}$ has its sum in $\mathcal{S}_{<\kappa}$, while some family of $\text{cf } \kappa$ members does not (`fkc:sb:thm:sums`; compare Theorem 8.4, which concerns J). None of these is used here.

Proposition 3.6 (Short-support integer part [09]). *For uncountable κ , the ring $\mathcal{I}_{<\kappa} = \mathbf{Oz} \cap \mathcal{S}_{<\kappa}$ is an integer part of $\mathcal{S}_{<\kappa}$: it is discretely ordered and, for every $x \in \mathcal{S}_{<\kappa}$, there is $z \in \mathcal{I}_{<\kappa}$ with $z \leq x < z + 1$.*

Proof. Write $x = p + r + \varepsilon$, where p is its positive-exponent part, $r \in \mathbb{R}$ is its constant coefficient, and ε is its negative-exponent part. The last term is zero or infinitesimal. Put $n = \lfloor r \rfloor$, except that if $r \in \mathbb{Z}$ and $\varepsilon < 0$, put $n = r - 1$. Then $z = p + n$ has the required inequalities. Its support is a subset of $\text{supp}(x)$ together with possibly zero, so $z \in \mathcal{I}_{<\kappa}$. The normal-form description (11.1) of $\mathbf{Oz}$ shows closure under the ring operations. A nonzero omnific integer is either an ordinary nonzero integer or has infinite magnitude, so 1 is the least positive one. $\square$

Warning 3.7 (Three unrelated bounds [09]). The conditions $\text{b}(x) < \kappa$, $|\text{supp}(x)| < \kappa$, and “the support of x is bounded in an exponent group” are not interchangeable. The short-support field has no birthday bound and includes arbitrarily large exponents. In the main independence example (Theorem 13.3), all selected exponent supports lie in the order interval $(0, 1]$, while their cardinality is κ .

4 The strong companion

4.1 General Hahn reindexing

We record the standard mechanism in the form needed here. It does not use large cardinals.

Lemma 4.1 (Reindexing exponents). *Let $f : (\mathbf{No}, +, <) \rightarrow (\mathbf{No}, +, <)$ be an additive order embedding. Then*

$$T_f \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{f(a)}$$

defines a strongly $\mathbb{R}$ -linear ordered-field embedding of $\mathbf{No}$ into itself. Its image consists exactly of the surreal numbers whose exponent supports are contained in $f(\mathbf{No})$. It preserves and reflects the condition of being an omnific integer.

Proof. An order embedding carries each reverse well-ordered set to a reverse well-ordered set of the same order type. Thus the displayed normal form exists and is unique. Coefficientwise addition and scalar multiplication commute with reindexing. For multiplication, $f(a + b) = f(a) + f(b)$, and injectivity shows that exactly the same finitely many contributing pairs occur at corresponding exponents. Hence $T_f(xy) = T_f(x)T_f(y)$. Also $T_f(1) = 1$, so inverses are preserved by the field equation $T_f(x)T_f(x^{-1}) = 1$ when $x \neq 0$. A nonzero leading coefficient is unchanged; signs, strict order, and injectivity follow.

For a strongly summable family, the union of its supports and every finite coefficient fiber are simply reindexed by f . This proves strong $\mathbb{R}$ -linearity for arbitrary external set-indexed families.

One inclusion in the image description is immediate. For the other, take a set support $S \subseteq f(\mathbf{No})$. Injectivity and class-parameter Replacement form the set of its unique inverse images. That set is reverse well ordered, and reindexing its series gives the desired preimage. No assertion is made that the inverse-image class of an arbitrary class is a set.

An omnific integer has only nonnegative exponents and has integer coefficient at exponent zero. An additive order embedding fixes zero, preserves and reflects nonnegativity, and does not change coefficients. This proves the last assertion. $\square$

The preceding proof is the usual Hahn reindexing argument; compare [3, 5] and the exponent-embedding results described in the repository [8]. Its role is to produce a canonical comparison map, not to supply the critical-point obstruction.

Definition 4.2 (Strong companion and defect). For the embedding in (1.1), let $H_j = T_j$ and set

$$D_j(x) = J(x) - H_j(x), \quad \Gamma_j = J(\mathbf{No}).$$

The subgroup Γ_j is being used as an *exponent group* when it occurs inside a support condition. Write

$$\mathcal{H}(\Gamma_j) = \{y \in \mathbf{No} : \text{supp}(y) \subseteq \Gamma_j\}.$$

Theorem 4.3 (Existence and uniqueness of the companion [06, 07, 09]). *The map H_j is a strongly $\mathbb{R}$ -linear ordered-field embedding and*

$$H_j(\mathbf{No}) = \mathcal{H}(\Gamma_j). \tag{4.1}$$

It is the unique strongly $\mathbb{R}$ -linear map that agrees with J on every Conway monomial. The maps J and H_j agree on every real number and on every ordinal, as well as on every monomial.

Proof. Apply Lemma 4.1 to the additive order embedding J . For any strongly $\mathbb{R}$ -linear map with the stated monomial values, applying strong linearity to the normal form (2.1) forces (1.3), proving uniqueness.

Agreement on monomials follows from Proposition 2.10. Real coefficients are fixed. Every ordinal has a finite Cantor normal form $\omega^{\beta_0}n_0 + \cdots + \omega^{\beta_{m-1}}n_{m-1}$, which is also its Conway normal form. Finite additivity therefore gives agreement on every ordinal, irrespective of its size. $\square$

06 (Theorem “The two lift constructions”) and 09 (Proposition “Canonical strong lift”) prove the same theorem with the same reindexing argument; 09 notes that Replacement for j makes the image of a support a set, and that the proof is the usual Hahn lifting argument, included to identify exactly which hypotheses are used. Its novelty is not claimed: the proposed contribution is its comparison with J [06, 09]. Both also note that H_j maps $\mathbf{Oz}$ into $\mathbf{Oz}$ (Lemma 4.1).

Remark 4.4 (A crucial noncommutation). The formula for the companion is

$$H_j(\omega^a) = \omega^{J(a)},$$

not $\omega^{H_j(a)}$ in general. In the former expression, J acts on an *exponent*; in $H_j(x)$, it reindexes the outer exponents of x . Failing to distinguish these two levels would incorrectly make the two embeddings identical.

5 The exact normal-form defect

5.1 The transformed index set

Fix a nonzero x with normal form (2.1). Regard the coefficient list r and the exponent list a as set functions with domain θ . Put

$$r^* = j(r), \quad a^* = j(a), \quad P_\theta = j(\theta) \setminus j \text{ `` } \theta.$$

The transformed lists have domain $j(\theta)$. By elementary transfer and Lemma 2.6, the full *external* normal form of $J(x)$ is

$$J(x) = \sum_{\beta < j(\theta)} r_\beta^* \omega^{a_\beta^*}. \quad (5.1)$$

For every $\alpha < \theta$,

$$r_{j(\alpha)}^* = r_\alpha, \quad a_{j(\alpha)}^* = J(a_\alpha). \quad (5.2)$$

The extra indices in P_θ will be called *missing-image indices*. They are ordinary set indices in V , not informal nonstandard positions.

Theorem 5.1 (Exact defect and first term [06, 07, 09]). *For x as above,*

$$D_j(x) = \sum_{\beta \in P_\theta} r_\beta^* \omega^{a_\beta^*}. \quad (5.3)$$

Every exponent occurring on the right lies outside $\Gamma_j = J(\mathbf{No})$. Consequently

$$D_j(x) = 0 \iff \theta < \kappa \iff |\text{supp}(x)| < \kappa. \quad (5.4)$$

If $\theta \geq \kappa$, then

$$\text{LT}(D_j(x)) = r_\kappa^* \omega^{a_\kappa^*}, \quad v(D_j(x)) = -a_\kappa^*. \quad (5.5)$$

The same statements include $x = 0$ by giving it empty support.

Proof. The normal form (5.1) is a strongly summable family. Split it into the subfamilies on $j \text{ `` } \theta$ and P_θ . Subfamilies of a strongly summable family remain strongly summable. By (5.2), the first subfamily sums to $H_j(x)$. Subtracting it leaves exactly (5.3).

Suppose, toward a contradiction, that $a_\beta^* = J(b)$ for some $\beta \in P_\theta$ and $b \in \mathbf{No}$. Then in M , $j(b) \in \text{ran}(j(a))$. Elementarity gives $b \in \text{ran}(a)$ in V , so $b = a_\alpha$ for some $\alpha < \theta$. But then $a_\beta^* = J(a_\alpha) = a_{j(\alpha)}^*$. Since a^* is strictly decreasing, it is injective, and $\beta = j(\alpha)$, a contradiction. Thus all the missing-image exponents lie outside Γ_j .

All r_β^* are nonzero, and all a_β^* are distinct. There is therefore no cancellation between the terms of (5.3). The defect vanishes precisely when P_θ is empty, which by Lemma 2.5 is precisely when $\theta < \kappa$. Since κ is an initial ordinal, the latter condition is equivalent to $|\theta| < \kappa$. The bijection $\alpha \mapsto a_\alpha$ converts it into the last condition of (5.4).

For $\theta \geq \kappa$, the least missing-image index is κ . The exponent list is strictly decreasing, so its term at that index is the leading term of the defect, proving (5.5). $\square$

This is 06's “missing-index formula” and “exact equalizer” and 09's “exact normal-form discrepancy”, proved by all three sources by splitting the transformed normal form into its pointwise-image subseries and the complementary subseries. 09 adds that the complement of a subset of a well-ordered ordinal is still well ordered, so both subseries are legitimate Hahn sums read in the inherited increasing order of the indices. The exclusion of image exponents is 07's. All three sources stress that every coefficient of the transformed normal form is nonzero; otherwise a missing index could disappear and the equalizer conclusion would fail.

Corollary 5.2 (Same leading scale, different tails [06, 07, 09]). *For every $x \neq 0$,*

$$\text{LT}(J(x)) = \text{LT}(H_j(x)) = r_0 \omega^{J(a_0)}, \quad v(J(x)) = v(H_j(x)) = J(v(x)).$$

If $D_j(x) \neq 0$, its valuation is strictly larger than this common value.

Proof. Index zero is fixed and is never a missing-image index. The first missing index, when present, is later in a strictly decreasing exponent list. $\square$

06 states the same as $D_j(x) \prec H_j(x)$ and $J(x)/H_j(x) \in 1 + o(1)$, where $y \prec z$ means $|y| < c|z|$ for every positive real c ; 09 as (5.6) below. 09 adds a sharper form.

Corollary 5.3 (Agreement to all earlier normal-form positions [09]). *If $\text{len}(x) \geq \kappa$, then $J(x)$ and $H_j(x)$ have the same first κ terms, but are not equal. The discrepancy is strictly dominated by every monomial occurring at one of those earlier positions:*

$$|D_j(x)| \prec \omega^{J(a_\alpha)} \quad (\alpha < \kappa).$$

For all $x \neq 0$, the two images have the same leading term and

$$\frac{J(x)}{H_j(x)} = 1 + \varepsilon_x, \quad \varepsilon_x = 0 \text{ or } |\varepsilon_x| < 1/n \text{ for every } n \geq 1. \quad (5.6)$$

Proof. Every index below κ is fixed and lies in the pointwise-image subseries. The first missing exponent lies strictly below all their exponents. Division by the common leading monomial establishes the stated Archimedean comparisons. If x has fewer than κ terms the images are equal; otherwise the same argument at index zero proves (5.6). $\square$

This separates leading-scale information from exact Hahn information. Even the relative error is always infinitesimal, while the absolute discrepancy can itself be an infinite omnific integer (Theorem 8.4) [09]. The word “small” needs care here: the defect is infinitesimal *relative to the original value*; it need not itself be an infinitesimal. Multiplying a bounded witness by a large monomial can make its defect purely infinite without changing its leading coefficient [06].

Warning 5.4 (Equalizer is not fixed field [09]). An element of $\mathcal{S}_{<\kappa}$ need not be fixed by either map. For example, κ has finite Cantor normal form, so it belongs to $\mathcal{S}_{<\kappa}$ and $H_j(\kappa) = J(\kappa) = j(\kappa) \neq \kappa$. The equalizer consists of inputs on which the two procedures agree, not inputs on which they are the identity.

Proposition 5.5 (The fixed class of the Hahn lift [06]). *For a real or complex normal form $x = \sum_{\alpha < \lambda} r_\alpha \omega^{a_\alpha}$,*

$$H_j(x) = x \iff J(a_\alpha) = a_\alpha \text{ for every } \alpha < \lambda.$$

Thus the fixed class of the Hahn lift consists exactly of the set-supported series over the fixed exponent group $\text{Fix}(J)$.

Proof. The two normal forms have the same length and the same coefficient array. Equality of their values forces equality of their unique ordered exponent arrays, giving necessity. Sufficiency is termwise. The fixed exponent group is an ordered subgroup because J preserves addition and negation. $\square$

The measure codes h_A and $\tilde{h}_A$ of Section 10 exemplify this observation: their exponents are fixed, even though transporting the entire code can detect the measure. There is no conflict with the equalizer theorem. “Fixed by H_j ” and “sent to the same value by J and H_j ” are different conditions [06]. (06 places this proposition in its surcomplex section; the complex case uses Section 12.)

Corollary 5.6 (Exact omega-map compatibility locus [06, 07, 09]). *For every $x \in \mathbf{No}$,*

$$H_j(\omega^x) = \omega^{H_j(x)} \iff x \in \mathcal{S}_{<\kappa}.$$

Thus J preserves the omega-map globally but is not strongly additive, whereas H_j is strongly additive but does not preserve the omega-map globally.

Proof. By Proposition 2.10 and the companion formula, $H_j(\omega^x) = \omega^{J(x)}$. The omega-map is injective, so the desired equality is equivalent to $J(x) = H_j(x)$. Apply Theorem 5.1. The existence of a long-support witness is established explicitly in Section 7. $\square$

06 (its Corollary on compatibility with the monomial map) and 09 (the first part of its Theorem “Exact omega-equivariance locus of the lift”) prove the same criterion. 09 adds an incompatibility for the prescribed monomial action.

Theorem 5.7 (Omega-map versus strong additivity [09]). *There is no ordered field embedding $E : \mathbf{No} \rightarrow \mathbf{No}$ agreeing with J on all monomials that both commutes with the omega-map everywhere and preserves every strongly summable family of size κ .*

Proof. If E had the two stated properties, then for every x ,

$$\omega^{E(x)} = E(\omega^x) = J(\omega^x) = \omega^{J(x)},$$

so $E = J$ by injectivity of the omega-map. The monomial family $(\tau_\alpha)_{\alpha < \kappa}$ of Theorem 8.4 (or the family $(\omega^{-\alpha})_{\alpha < \kappa}$ of Theorem 7.1) contradicts the asserted summation property. $\square$

This is an incompatibility for a *specified* monomial action. It is not a theorem that all omega-preserving strongly additive embeddings are impossible; the identity is an immediate counterexample to that broader statement [09].

5.2 Why small support is not small birthday

Every monomial belongs to $\mathcal{S}_{<\kappa}$, however large its exponent or birthday. For example, $\omega^b \in \mathcal{S}_{<\kappa}$ for every surreal b . This excludes any interpretation of (5.4) as agreement only on a set-sized initial segment of $\mathbf{No}$. Conversely, the simplest witness below has κ outer terms and falls outside the equalizer even though all its individual monomials are fixed by J . The cardinal bound applies to the number of terms, not to the size of a single exponent.

The distinction from the classical bounded birthday fields $\mathbf{No}(\lambda)$ is substantial. Van den Dries–Ehrlich analyze those fields and their restricted analytic and exponential closures [3, Proposition 4.7 and Corollary 5.5]. We are not renaming those set-sized fields; $\mathcal{S}_{<\kappa}$ is a proper class because it contains every monomial. Section 3 develops $\mathcal{S}_{<\kappa}$ without the embedding, and Warning 3.7 separates it from birthday bounds and from order-bounded supports [09].

6 The equalizer field and the exact image intersection

Theorem 6.1 (Equalizer and common image). *Let $C_j = J(\mathcal{S}_{<\kappa}) = H_j(\mathcal{S}_{<\kappa})$. Then*

$$\text{Eq}(J, H_j) = \mathcal{S}_{<\kappa}, \tag{6.1}$$

$$J(\mathbf{No}) \cap H_j(\mathbf{No}) = C_j, \tag{6.2}$$

$$C_j = \{y \in \mathbf{No} : |\text{supp}(y)| < \kappa \text{ and } \text{supp}(y) \subseteq \Gamma_j\}. \tag{6.3}$$

Both $\mathcal{S}_{<\kappa}$ and C_j are real-closed subfields of $\mathbf{No}$.

Proof. The equalizer assertion is Theorem 5.1. Suppose $y = J(x) \in H_j(\mathbf{No})$. The image description (4.1) says that all exponents in the normal form of y lie in Γ_j . If x had at least κ normal-form terms, Theorem 5.1 would exhibit a nonzero term of $J(x)$ with exponent outside Γ_j . Hence $x \in \mathcal{S}_{<\kappa}$, and $y = J(x) = H_j(x) \in C_j$. The opposite inclusion is immediate. Reindexing preserves the cardinality of support, and the inverse-support construction in Lemma 4.1 proves (6.3).

The equalizer of two field embeddings is a subfield, so $\mathcal{S}_{<\kappa}$ is a subfield. To prove real closedness without relying on a separate bounded-support theorem, let $0 \neq p \in \mathcal{S}_{<\kappa}[X]$. If $r_1 < \dots < r_m$ are all its distinct real roots in $\mathbf{No}$, then $J(r_1) < \dots < J(r_m)$ and $H_j(r_1) < \dots < H_j(r_m)$ are both the full ordered list of real roots of the same transformed polynomial. Indeed, the images of $\mathbf{No}$ under these embeddings are real closed, and an ordered extension of a real-closed field cannot add a new real algebraic element. Each ordered position therefore agrees: $J(r_i) = H_j(r_i)$. Every such root lies in $\mathcal{S}_{<\kappa}$.

In particular, each positive member of $\mathcal{S}_{<\kappa}$ has its positive square root in $\mathcal{S}_{<\kappa}$, and each odd-degree polynomial over $\mathcal{S}_{<\kappa}$ has a root there. This characterizes real closedness. The isomorphic common image C_j is real closed as well. $\square$

The real closedness of $\mathcal{S}_{<\kappa}$ has a second, embedding-free proof, valid at every uncountable cardinal: Theorem 3.1 [09].

Remark 6.2 (Domain equalizer versus image intersection). The equalizer is $\mathcal{S}_{<\kappa}$, whereas the intersection is its *image* C_j . The two are not being identified. An arbitrary short-support series can have exponents outside $J(\mathbf{No})$, and then it is not in C_j . In (6.3), both the support bound and the exponent-image condition are essential.

Corollary 6.3 (Transcendental separation). *Every element of $J(\mathbf{No}) \setminus C_j$ is transcendental over $H_j(\mathbf{No})$, and every element of $H_j(\mathbf{No}) \setminus C_j$ is transcendental over $J(\mathbf{No})$. In particular, each is transcendental over C_j .*

Proof. Both image fields are real closed. An element of one outside their intersection is not in the other. A real-closed field has no proper algebraic ordered extension, so that element cannot be algebraic over the other field. $\square$

The same argument over the domain field is 09's Corollary 3.4: every surreal outside $\mathcal{S}_{<\kappa}$ is transcendental over $\mathcal{S}_{<\kappa}$.

6.1 A canonical isomorphism by erasing nonimage exponents

Define a coefficient projection on all surreals by

$$\Pi_j(y) = \sum_{a \in \text{supp}(y) \cap \Gamma_j} ([\omega^a] y) \omega^a. \quad (6.4)$$

This retains a subseries; it is not generally an initial normal-form truncation. Its support is a set, formed by Separation with the embedding parameter.

Theorem 6.4 (Isomorphic immediate image fields). *The projection Π_j is strongly $\mathbb{R}$ -linear and idempotent, with image $H_j(\mathbf{No})$, and*

$$\Pi_j \circ J = H_j. \quad (6.5)$$

Its restriction is an ordered-field isomorphism

$$\Pi_j \upharpoonright J(\mathbf{No}) : J(\mathbf{No}) \xrightarrow{\cong} H_j(\mathbf{No}) \quad (6.6)$$

that fixes C_j pointwise and fixes every Conway monomial in either image. Both image fields are immediate valued extensions of C_j : they have the same value group Γ_j and residue field $\mathbb{R}$. The projection is not a field homomorphism on all of $\mathbf{No}$.

Proof. Taking a subseries is compatible with coefficientwise finite sums, and commutes with an arbitrary strong sum: one either retains or deletes all contributions at each exponent. This proves strong $\mathbb{R}$ -linearity. The image and idempotence follow from (4.1). The defect theorem states that $J(x)$ consists of $H_j(x)$ supported on Γ_j , plus a series supported outside Γ_j . Therefore $\Pi_j(J(x)) = H_j(x)$.

The maps J and H_j are ordered-field isomorphisms onto their respective images. Hence (6.5) identifies the restriction in (6.6) with $H_j \circ J^{-1}$, which is an ordered-field isomorphism. It fixes every element of the intersection, since those supports are contained in Γ_j . The monomials in either image are exactly $\{\omega^{J(a)} : a \in \mathbf{No}\}$, and they already lie in C_j . For the assertion about the monomials of $J(\mathbf{No})$, a monomial image must have normal-form length one and leading coefficient one, so its preimage also does.

Every nonzero member of either image has valuation in Γ_j , and the monomials just listed realize every value in Γ_j . They belong to C_j , so its value group is the same. All three fields contain the constant copy of $\mathbb{R}$ and have residue field $\mathbb{R}$. Equality of these value groups and residue fields is precisely immediacy; the value groups here are allowed to be classes.

Finally choose $a \notin \Gamma_j$, for instance $a = \kappa$. Then also $-a \notin \Gamma_j$, and

$$\Pi_j(\omega^a) = \Pi_j(\omega^{-a}) = 0, \quad \Pi_j(\omega^a \omega^{-a}) = \Pi_j(1) = 1.$$

Thus global multiplicativity fails. The fact that multiplicativity holds on $J(\mathbf{No})$ is a genuine restriction on the tails of that image field. $\square$

This gives transcendence, not algebraic independence between the two image fields. Exact intersection alone does not establish linear disjointness; that stronger question is left open in Section 21.

7 Explicit witnesses and incomparability

For every ordinal η , define the genuine strong sum

$$h_\eta = \sum_{\alpha < \eta} \omega^{-\alpha}. \quad (7.1)$$

The set $\{-\alpha : \alpha < \eta\}$ is strictly decreasing in its displayed order, and each exponent occurs once, so this is a legitimate normal form. We always use surreal negation and subtraction in exponents; they are not ordinal subtraction.

Theorem 7.1 (The basic witness). *For h_κ one has*

$$H_j(h_\kappa) = h_\kappa, \quad (7.2)$$

$$J(h_\kappa) = h_{j(\kappa)}, \quad (7.3)$$

$$D_j(h_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \omega^{-\beta} = \omega^{-\kappa}(1 + \varepsilon), \quad 0 < \varepsilon \in \mathfrak{m}. \quad (7.4)$$

Moreover,

$$h_\kappa \in H_j(\mathbf{No}) \setminus J(\mathbf{No}), \quad h_{j(\kappa)} \in J(\mathbf{No}) \setminus H_j(\mathbf{No}).$$

Thus $J(\mathbf{No})$ and $H_j(\mathbf{No})$ are incomparable real-closed subfields of $\mathbf{No}$, although the embeddings agree on all reals, ordinals, and monomials.

Proof. Each $\alpha < \kappa$ is fixed, so $J(-\alpha) = -\alpha$, and (7.2) follows. The definition of the whole list in (7.1) transfers to the list of length $j(\kappa)$, giving (7.3). Their difference is (7.4). Factoring out its first monomial leaves 1 plus a nonempty series with strictly negative exponents and positive coefficients. That remainder is a positive infinitesimal. It is nonempty because $j(\kappa)$ is a limit cardinal strictly above κ .

If $h_\kappa = J(x)$, then the length of its normal form is $j(\text{len}(x)) = \kappa$, contradicting that κ is not in the ordinal range of j . Hence h_κ is not in $J(\mathbf{No})$. On the other hand, the normal form of $h_{j(\kappa)}$ contains the exponent $-\kappa$. This exponent is not $J(b)$ for any surreal b : if it were, $J(-b) = \kappa$, and elementarity would make $-b$ an ordinal, again contradicting the missing-index fact. The image description for H_j therefore excludes $h_{j(\kappa)}$ from its image. $\square$

Example 7.2 (A short-support element outside the common image). By Proposition 2.10, $J(\kappa) = j(\kappa) > \kappa$. Consequently $\omega^{-\kappa}$ is not in C_j : its exponent is outside Γ_j . This is a one-term surreal, so it nevertheless belongs to $\mathcal{S}_{<\kappa}$. It concretely illustrates Remark 6.2. The two assertions “ x has short support” and “ x lies in the common image” really are different assertions.

Remark 7.3 (Not a geometric series [06]). The expression h_λ is a transfinite Hahn sum. It must not be simplified as though every ordinal exponent were a natural multiple of 1. In particular the ordinary geometric-series identity for $\sum_{n<\omega} \omega^{-n}$ does not identify h_κ with that countable sum. Surreal addition of ordinal exponents is not ordinary ordinal addition.

Proposition 7.4 (Both maps are proper [06]). *Both J and H_j are proper embeddings of $\mathbf{No}^V$ into itself.*

Proof. The normal-form length of $J(x)$ is $j(\text{len}(x))$. No ordinal has image κ , by Lemma 2.5; hence h_κ is not in the range of J . Every exponent in the support of an H_j -image belongs to $J(\mathbf{No})$. The monomial ω^{h_κ} therefore cannot be in the range of H_j . More explicitly, injectivity on exponent supports would force a preimage to have one term, with exponent a satisfying $J(a) = h_\kappa$, which is impossible. $\square$

The first half is contained in Theorem 7.1, whose second half uses the witness $h_{j(\kappa)}$ instead of ω^{h_κ} . Thus J , although preserving the monomial map and exponentiation, is not an automorphism. These results do not answer the automorphism classification in Question 5.6 of [5]. Nor does an elementary embedding of sets into a proper inner model contradict a theorem excluding a nontrivial elementary embedding of the universe into itself [06].

8 The exact strong-summation threshold

The previous results compare the images of a *single normal form*. There is a separate, equally sharp statement about an arbitrary family whose individual members may themselves have very long supports.

Theorem 8.1 (Positive-family threshold [06, 07, 09]). *Let $F : I \rightarrow \mathbf{No}$ be an external set-indexed strongly summable family. Then $(J(F(i)))_{i \in I}$ is strongly summable. If $|I| < \kappa$, then*

$$J\left(\sum_{i \in I} F(i)\right) = \sum_{i \in I} J(F(i)). \quad (8.1)$$

If in addition $F(i) > 0$ for every $i \in I$, then (8.1) holds if and only if $|I| < \kappa$. For $|I| \geq \kappa$ in this positive case,

$$J\left(\sum_{i \in I} F(i)\right) - \sum_{i \in I} J(F(i)) > 0. \quad (8.2)$$

Proof. The family $j(F)$ is strongly summable over $j(I)$ in M , and hence externally in V by Lemma 2.6. Its sum is $J(\sum_{i \in I} F(i))$. Restrict it to the old-index part $j \restriction I$: its values there are $j(F)(j(i)) = J(F(i))$. This proves external summability of the termwise image family. Splitting the full family yields

$$J\left(\sum_{i \in I} F(i)\right) - \sum_{i \in I} J(F(i)) = \sum_{u \in j(I) \setminus j \restriction I} j(F)(u). \quad (8.3)$$

If $|I| < \kappa$, the right side is empty by Lemma 2.5.

Suppose now that every $F(i) > 0$. Transfer gives $j(F)(u) > 0$ for all $u \in j(I)$. A nonempty strongly summable family of positive surreals has positive sum: take the largest exponent in the union of its supports. Every term contributing at that exponent has it as a leading exponent, so its coefficient is positive; only finitely many contribute, and their sum is positive. When $|I| \geq \kappa$, the missing-index part is nonempty by Lemma 2.5. Applying this observation to (8.3) proves (8.2). $\square$

Remark 8.2 (Why the positivity hypothesis is necessary). Let $I = \kappa \times \{0, 1\}$ and put $F(\alpha, 0) = \omega^{-\alpha}$, $F(\alpha, 1) = -\omega^{-\alpha}$. This is strongly summable and has sum zero. The termwise image also sums to zero, so preservation holds even though $|I| = \kappa$. One cannot replace the positive-family conclusion by an unconditional cardinality criterion for signed families. The exact criterion in general is that the missing-index sum in (8.3) vanish.

Corollary 8.3 (First failure cardinal [06, 07, 09]). *The least cardinality of a positive strongly summable family on which J fails to commute with strong summation is exactly κ .*

Proof. All smaller families are preserved, and the monomial family defining h_κ is a positive witness of size κ . $\square$

06 (its strong-additivity threshold) and 09 (the first part of its “Exact threshold”) prove the small-family half by enumerating the family by an ordinal $\mu < \kappa$: the transported family has domain $j(\mu) = \mu$ and term $J(x_\alpha)$ at each old index, and Lemma 2.6 turns the internal identity into the external one. There is no restriction on the support of any individual member. Both give the witness h_κ (06’s u_κ , 09’s Q_κ): the coefficient at exponent $-\kappa$ is one in $J(h_\kappa)$ and zero in $\sum_{\alpha < \kappa} J(\omega^{-\alpha})$.

There is no conflict between this theorem and the defect of an individual long-support x . A singleton family $\{x\}$ is always preserved by J . Its normal-form decomposition may have κ or more terms, and that is the different family on which J and H_j disagree.

8.1 An omnific monomial witness [09]

For every ordinal α put

$$q_\alpha = \omega^{-\alpha}, \quad \tau_\alpha = \omega^{q_\alpha} = \omega^{\omega^{-\alpha}}. \quad (8.4)$$

Both (q_α) and (τ_α) are strictly decreasing monomial families on every ordinal domain. Each q_α is positive, so every τ_α is larger than all positive real numbers. In particular, $\tau_\alpha \in \mathbf{Oz}$. For an ordinal θ , define

$$Y_\theta = \sum_{\alpha < \theta} \tau_\alpha. \quad (8.5)$$

It is a strong sum of length θ and is purely infinite whenever $\theta > 0$. Here “purely infinite” means that its normal form has only positive exponents, not that its sign must be positive for every possible choice of coefficients; Y_θ is positive because all its coefficients equal one.

Theorem 8.4 (Explicit threshold witnesses [09]). *The map J does not preserve all strongly summable families of cardinality κ . More explicitly,*

$$J(h_\kappa) = h_{j(\kappa)}, \quad H_j(h_\kappa) = h_\kappa, \quad D_j(h_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \omega^{-\beta}, \quad (8.6)$$

$$J(Y_\kappa) = Y_{j(\kappa)}, \quad H_j(Y_\kappa) = Y_\kappa, \quad D_j(Y_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \tau_\beta. \quad (8.7)$$

The leading discrepancies are respectively $\omega^{-\kappa}$ and $\tau_\kappa = \omega^{\omega^{-\kappa}}$. Every individual summand in the original κ -families is fixed by J .

Proof. The constructions (8.4) and (8.5) are uniform definable ordinal-indexed constructions. Hence $J(h_\kappa) = h_{j(\kappa)}$ and $J(Y_\kappa) = Y_{j(\kappa)}$. For each $\alpha < \kappa$ the ordinal α is fixed, and the omega identity gives $J(q_\alpha) = q_\alpha$ and $J(\tau_\alpha) = \tau_\alpha$. The strong lift therefore fixes both original sums. Subtraction leaves exactly the indicated tails, whose first indices are κ . $\square$

The row (8.6) is 07’s Theorem 7.1. Its discrepancy is infinitesimal even in absolute magnitude. The discrepancy in (8.7) is instead an infinite omnific integer. It is still infinitesimal *relative* to the common leading term $\tau_0 = \omega$. This supplies two different geometric manifestations of the same missing-index mechanism.

For a cardinal λ , let “ $< \lambda$ -strong” mean preservation of every strongly summable family indexed by a set of size less than λ . Theorem 8.1 and Theorem 8.4 state that J is $< \kappa$ -strong and is not $< \lambda$ -strong for any cardinal $\lambda > \kappa$. They do not claim that every individual family of size κ fails; many do not [09] (compare the signed family of the remark after Theorem 8.1).

Proposition 8.5 (Order continuity does not enforce strong additivity [09]). *Both J and H_j are continuous for the order topology on $\mathbf{No}$. Nevertheless, J is not strongly additive.*

Proof. The failure of strong additivity is Theorem 8.4. For continuity at zero, take any $\delta > 0$ and choose a positive ordinal $\beta > 1/\delta$. Elementary embeddings are nondecreasing on ordinal magnitudes: $j(\beta) \geq \beta$, as follows by induction from order preservation. The ordinal β has finite normal form, so $H_j(\beta) = J(\beta) = j(\beta)$. For either map $E \in \{J, H_j\}$ and $|x| < 1/\beta$, therefore,

$$|E(x)| < 1/E(\beta) \leq 1/\beta < \delta.$$

This proves continuity at zero; translation gives continuity everywhere. This is an interval-based statement about a class order, not an assertion that its topology is a set of open sets. $\square$

8.2 An intrinsic closure theorem for the image

The threshold can be stated without first specifying a family in the domain of J . This strengthens the noncommutation result to an actual failure of membership in the image.

Theorem 8.6 (Strong-sum closure and positive escape). *Let $(y_i)_{i \in I}$ be an external strongly summable family in $J(\mathbf{No})$, and let x_i be the unique preimage with $J(x_i) = y_i$. Then (x_i) is strongly summable and*

$$\sum_{i \in I} y_i \in J(\mathbf{No}) \iff \sum_{i \in I} y_i = J\left(\sum_{i \in I} x_i\right). \quad (8.8)$$

In particular, $J(\mathbf{No})$ is closed under all external strongly summable families of cardinality less than κ . If every $y_i > 0$, then

$$\sum_{i \in I} y_i \in J(\mathbf{No}) \iff |I| < \kappa. \quad (8.9)$$

In contrast, $H_j(\mathbf{No})$ is closed under every external set-indexed strong sum of its members.

Proof. Replacement with the embedding parameter gives the set-indexed preimage family. For every a , the normal-form defect theorem implies

$$[\omega^{J(a)}] y_i = [\omega^a] x_i.$$

Thus $J \text{ “ } \bigcup_i \text{supp}(x_i) \text{ is a subset of the reverse well-ordered union } \bigcup_i \text{supp}(y_i)$. Pulling this suborder back by the order embedding J proves reverse well ordering of $\bigcup_i \text{supp}(x_i)$. An exponent a occurs in x_i exactly when $J(a)$ occurs in y_i , so the coefficient fibers are finite as well. Hence the preimage family is strongly summable.

Let $x = \sum_i x_i$ and $y = \sum_i y_i$. At every exponent $J(a)$ the finite coefficient sums give

$$[\omega^{J(a)}] y = \sum_i [\omega^a] x_i = [\omega^a] x = [\omega^{J(a)}] J(x).$$

Therefore $J(x) - y$ has support disjoint from Γ_j . But every nonzero element of $J(\mathbf{No})$ has its leading exponent in Γ_j , by Corollary 5.2. If $y \in J(\mathbf{No})$, the difference $J(x) - y$ is in that field and must consequently be zero. This proves the forward implication in (8.8); the reverse implication is immediate.

For $|I| < \kappa$, Theorem 8.1 gives equality, without requiring positive terms. If all $y_i > 0$, then all $x_i > 0$, and that theorem makes equality equivalent to $|I| < \kappa$. Finally, a strong sum of series supported on Γ_j is again supported on Γ_j ; (4.1) proves the closure assertion for $H_j(\mathbf{No})$. $\square$

9 Descent of the companion to the target model [06]

The value $H_j(x)$ always exists in the ambient field. The question of this section, which is 06’s, is whether it exists in the target model. In a set-ultrapower the answer measures considerably more than the size of an arithmetic error: it detects closure under external sequences. In this section λ denotes the normal-form length of x , and by (5.1) and (5.2),

$$H_j(x) = \sum_{\alpha < \lambda} r_\alpha \omega^{J(a_\alpha)} = \sum_{\beta \in j \text{ “ } \lambda} j(r)(\beta) \omega^{j(a)(\beta)}, \quad (9.1)$$

the latter sum read in increasing order of the index β . It exists externally because j preserves the exponent ordering. Only J is *a priori* valued in $\mathbf{No}^M$.

9.1 Landing is equivalent to the seed

Theorem 9.1 (Universal landing criterion [06]). *Let x have normal-form length λ . Then*

$$H_j(x) \in \mathbf{No}^M \iff j \text{ “ } \lambda \in M. \quad (9.2)$$

In particular the answer depends only on the cardinality of the support, not on the actual coefficients or exponents. At a fixed support cardinality there are no exceptional normal forms that land when others do not.

Proof. Suppose first that $s = j \text{ `` } \lambda$ belongs to M . The transported arrays $j(r)$ and $j(a)$ are in M . Restrict them to s inside M , using its increasing enumeration to obtain an ordinal-indexed normal form. The value of this normal form belongs to $\mathbf{No}^M$ and, by Lemma 2.6, equals the external expression (9.1). It is $H_j(x)$.

Conversely suppose $y = H_j(x)$ belongs to $\mathbf{No}^M$. Its normal form computed in M is its external normal form. In particular its exponent support S belongs to M . The array $j(a)$ belongs to M and is injective on $j(\lambda)$. By Separation in M , the set

$$t = \{\beta < j(\lambda) : j(a)(\beta) \in S\}$$

belongs to M . Externally, the support formula (9.1) and injectivity of $j(a)$ identify t exactly with $j \text{ `` } \lambda$. This proves the converse, including the empty support case.

Finally, if $f : \mu \rightarrow \lambda$ is a bijection, then $j(f)$ maps $j \text{ `` } \mu$  bijectively onto  $j \text{ `` } \lambda$. Each seed is obtained from the other using $j(f)$ or its inverse, both belonging to M . Thus seed membership, and hence landing, is unchanged when the support length is replaced by its cardinality. $\square$

The forward implication is stronger than a statement about canonical 0–1 series alone. Even an elaborate normal form with arbitrary nonzero real coefficients cannot conceal a missing seed. Its exponents, compared with the transported original exponent array, recover that seed exactly [06].

Corollary 9.2 (Landing spectrum [06]). *Define the class of cardinals*

$$\mathcal{L}(j) = \{\mu : j \text{ `` } \mu \in M\}.$$

It is downward closed and contains every cardinal $\mu \leq \kappa$. Moreover

$$H_j^{-1}(\mathbf{No}^M) = \{x : |\text{supp}(x)| \in \mathcal{L}(j)\}$$

is a truncation-closed subfield of $\mathbf{No}^V$.

Proof. For ordinals $\mu \leq \lambda$, $j \text{ `` } \mu = j \text{ `` } \lambda \cap j(\mu)$, giving downward closure. Below κ the seed equals the ordinal, and $j \text{ `` } \kappa = \kappa$, so all these seeds belong to M . The displayed equality follows from Theorem 9.1. The inverse image of a subfield under a field embedding is a subfield. Truncation decreases the support cardinality. $\square$

9.2 Why the seed measures sequence closure in an ultrapower

The next lemma is a standard ultrapower fact, included with its proof because it explains precisely what is being imported from set theory. It is stated as Fact 7 in [12].

Lemma 9.3 (Seed–closure equivalence [06]). *Suppose $j : V \rightarrow M$ is the well-founded ultrapower by a countably complete ultrafilter on a set I . For any ordinal λ ,*

$$j \text{ `` } \lambda \in M \iff {}^\lambda M \cap V \subseteq M. \quad (9.3)$$

Here the right side means that every ambient set function with domain λ and values in M belongs to M .

Proof. If M is closed under such sequences, the sequence $\langle j(\alpha) : \alpha < \lambda \rangle$ belongs to M , so its range $j \text{ `` } \lambda$ does too. This direction does not require the ultrapower hypothesis.

For the other direction, let $s = j \text{ `` } \lambda \in M$ and take an ambient sequence $\langle x_\alpha : \alpha < \lambda \rangle$ of elements of M . Choose representatives $f_\alpha : I \rightarrow V$ with $x_\alpha = [f_\alpha]_U$, where brackets include the transitive collapse. This choice is legitimate for a set of indices: choose bounded-rank representatives first and then apply ordinary

Choice. Pack them into a single set function F with $F(\alpha, i) = f_\alpha(i)$. Let $d = [\text{id}_I]_U \in M$ be the usual ultrapower seed for I . The evaluation identity for ultrapowers gives

$$j(F)(j(\alpha), d) = x_\alpha.$$

The increasing enumeration $e : \lambda \rightarrow s$ belongs to M ; its order type is the actual ordinal λ , by absoluteness of well-orders. Inside M form

$$\alpha \mapsto j(F)(e(\alpha), d).$$

This is exactly the original ambient sequence. Hence it belongs to M . $\square$

Corollary 9.4 (One number tests all sequences [06]). *For a set-ultrapower embedding and a surreal x with support cardinal μ ,*

$$H_j(x) \in \mathbf{No}^M \iff {}^\mu M \cap V \subseteq M.$$

Thus descent of one such normal form implies descent of every surreal whose support has at most μ terms.

Proof. Combine Theorem 9.1, Lemma 9.3 and Corollary 9.2. $\square$

The word “one” refers to a test relative to the specified embedding and its target. It does not mean that a single pure-field formula with no access to j , M , or normal-form data defines supercompactness [06].

9.3 Normal measures: the exact three-way classification

Lemma 9.5 (Cover obstruction [06]). *Let $j : V \rightarrow M$ be a set-ultrapower on an index set I with critical point κ . Let $\lambda \geq \kappa$ be an ambient cardinal. Suppose that $s \in M$ satisfies*

$$j \text{ “ } \lambda \subseteq s \subseteq j(\lambda), \quad M \models |s| < j(\kappa).$$

Then

$$\lambda \leq |I| \cdot \kappa. \tag{9.4}$$

Proof. Represent s by a function $f : I \rightarrow V$. By the ultrapower theorem, the assertion $s \in j(P_\kappa(\lambda))$ implies that $f(i) \in P_\kappa(\lambda)$ on a measure-one set. Change f to the empty set outside that set. For every $\alpha < \lambda$, membership $j(\alpha) \in s$ says

$$\{i \in I : \alpha \in f(i)\} \in U.$$

In particular this set is nonempty. Consequently $\lambda = \bigcup_{i \in I} f(i)$. Each $f(i)$ has cardinality $< \kappa$, so the cardinality of the union is at most $|I| \cdot \kappa$. $\square$

This elementary counting argument is useful because it does not confuse an internal support bound with an external one. An internally short cover can be externally large, but not too large for its representing index set [06].

Lemma 9.6 (The image of the critical point is large [06]). *Under (2.3), $j(\kappa) > (2^\kappa)^V$.*

Proof. The ordinal $j(\kappa)$ is a strong-limit cardinal in M by Lemma 2.3 and elementarity. Let $\delta = (2^\kappa)^M$. The set $\mathcal{P}(\kappa)$ is the same in both models by Lemma 2.3, so a bijection in M from δ onto that power set is an actual bijection in V . Thus $\delta \geq (2^\kappa)^V$ as ordinals. Since M regards $\kappa < j(\kappa)$ and $j(\kappa)$ as strong limit, it has $\delta < j(\kappa)$. Combine these inequalities. $\square$

Theorem 9.7 (Normal-measure trichotomy [06]). *Suppose U is a normal nonprincipal κ -complete ultrafilter on κ and $j : V \rightarrow M = \text{Ult}(V, U)$ is its transitive ultrapower. For every surreal x ,*

$$H_j(x) \in \mathbf{No}^M \iff |\text{supp}(x)| \leq \kappa. \quad (9.5)$$

More precisely:

Support cardinal	Comparison	Descent of $H_j(x)$
$< \kappa$	$J(x) = H_j(x)$	belongs to $\mathbf{No}^M$
$= \kappa$	$J(x) \neq H_j(x)$	belongs to $\mathbf{No}^M$
$> \kappa$	$J(x) \neq H_j(x)$	does not belong to $\mathbf{No}^M$

In all three rows $J(x)$ belongs to $\mathbf{No}^M$.

Proof. By Corollary 9.2, every cardinal at most κ belongs to the landing spectrum. We prove that the ambient successor κ^+ does not. Suppose to the contrary that $s = j \text{ “ } \kappa^+ \in M$. As an ordered set of ordinals, s has order type κ^+ , and this order type is absolute to M . By Lemma 9.6,

$$\kappa^+ \leq (2^\kappa)^V < j(\kappa).$$

The increasing enumeration therefore witnesses in M that s has cardinality $< j(\kappa)$. Apply Lemma 9.5 with $I = \kappa$ and $\lambda = \kappa^+$. It gives $\kappa^+ \leq \kappa \cdot \kappa = \kappa$, a contradiction.

Downward closure now excludes every larger cardinal. The universal landing criterion proves (9.5), and the exact equalizer theorem gives the three rows. $\square$

Remark 9.8 (Slightly stronger scope [06]). Normality is not needed in the counting proof of Theorem 9.7. The same conclusion holds for a well-founded ultrapower on an index set of cardinality κ with critical point κ . The normal-measure formulation is retained because it also identifies the recovered measure in Section 10 with the original ultrafilter.

Corollary 9.9 (Landing domain versus equalizer [06]). *For a normal-measure ultrapower on κ , the maximal domain on which the Hahn lift lands in the target is exactly $\mathcal{S}_{\leq \kappa}$, whereas the equalizer of the two lifts is exactly $\mathcal{S}_{< \kappa}$. The target is closed under ambient κ -sequences but not under all ambient κ^+ -sequences.*

Proof. Use Theorem 9.7, Theorem 5.1 and Lemma 9.3. $\square$

No assertion that $j(\kappa)$ is inaccessible in V was used. It is inaccessible in M , and that is exactly the property needed for the internal support bound. These two assertions should not be interchanged [06].

10 Recovering a normal measure from one coefficient

10.1 Boolean masks at ordinal exponents

For each $A \subseteq \kappa$, define

$$h_A = \sum_{\alpha \in A} \omega^{-\alpha}. \quad (10.1)$$

The notation is unambiguous: the index set is ordered by the ordinal order, and zeros are omitted from the full κ -list. In particular, $h_\emptyset = 0$ and h_κ agrees with (7.1).

Theorem 10.1 (The measure-coefficient formula). *For every $A \subseteq \kappa$,*

$$H_j(h_A) = h_A, \quad (10.2)$$

$$J(h_A) = \sum_{\beta \in j(A)} \omega^{-\beta}, \quad (10.3)$$

$$[\omega^{-\kappa}] D_j(h_A) = \mathbf{1}_{\{\kappa \in j(A)\}}. \quad (10.4)$$

Consequently

$$U_j = \{A \subseteq \kappa : [\omega^{-\kappa}] D_j(h_A) = 1\} \quad (10.5)$$

is the nonprincipal normal κ -complete ultrafilter on κ derived from j .

Proof. The original exponents $-\alpha$, $\alpha < \kappa$, are fixed by J , so (10.2) follows from the companion formula. Transferring the whole mask construction gives (10.3). Since κ is not an original index, h_A has no coefficient at $\omega^{-\kappa}$. The coefficient in $J(h_A)$ is 1 precisely when $\kappa \in j(A)$. This proves (10.4) and identifies (10.5) with the classical derived measure.

For completeness, its properties can be read directly from this last identification. Since $\kappa \in j(\kappa)$, the whole space is in U_j and the empty set is not. Elementary transfer of complements and finite intersections makes it an ultrafilter. For $\alpha < \kappa$, $j(\{\alpha\}) = \{\alpha\}$ does not contain κ , so it is nonprincipal. If $\mu < \kappa$ and $A_\xi \in U_j$ for $\xi < \mu$, then $j(\mu) = \mu$ and

$$\kappa \in \bigcap_{\xi < \mu} j(A_\xi) = j\left(\bigcap_{\xi < \mu} A_\xi\right),$$

which proves κ -completeness.

Finally, suppose $A \in U_j$ and $f : A \rightarrow \kappa$ is regressive, meaning $f(\alpha) < \alpha$ for all nonzero $\alpha \in A$. Then $j(f)(\kappa) < \kappa$, so $j(f)(\kappa) = \eta = j(\eta)$ for some $\eta < \kappa$. Therefore

$$\kappa \in j(\{\alpha \in A : f(\alpha) = \eta\}).$$

The indicated fiber is in U_j , proving normality. $\square$

The ultrafilter properties just proved are standard properties of derived measures [6]. The new information is their simultaneous realization by (10.4) within one explicitly given surreal coefficient space. No real-valued probability measure on all surreal sets is being introduced.

All three sources recover the derived normal measure from one coefficient, with the same proof of its ultrafilter, completeness and normality properties (06 in Lemma 2.3 and its recovery theorem, 09 in its “Normal-measure coefficient formula”). They use three different codes. 07’s h_A has coefficients 0, 1 and is bounded. 06’s code has coefficients 1, 2, so that its support has exactly κ terms for every A ; 09’s code is omnific from the start. We print the two further codes with the computations that differ.

Theorem 10.2 (A bounded code of constant length [06]). *For $A \subseteq \kappa$ put*

$$\tilde{h}_A = h_\kappa + h_A = \sum_{\alpha < \kappa} (1 + \mathbf{1}_A(\alpha)) \omega^{-\alpha}. \quad (10.6)$$

Every coefficient is one or two, so the support has exactly κ terms, independently of A , and $1 < \tilde{h}_A < 3$. For any embedding satisfying (2.3), U_j is a normal measure on κ , $H_j(\tilde{h}_A) = \tilde{h}_A$, and

$$[\omega^{-\kappa}] D_j(\tilde{h}_A) = 1 + \mathbf{1}_{U_j}(A), \quad (10.7)$$

$$\text{st}(\omega^\kappa D_j(\tilde{h}_A)) = 1 + \mathbf{1}_{U_j}(A), \quad (10.8)$$

where st is the standard part of a finite surreal. If j is the ultrapower by a normal measure U on κ , then $U_j = U$.

Proof. The bounds hold because the tail after the constant term is a nonzero positive infinitesimal and the constant term is at most two. The measure properties are those proved in Theorem 10.1 (06 proves them in Lemma 2.3 and by the same regressive-function argument). All exponents $-\alpha$ with $\alpha < \kappa$ and all coefficients in (10.6) are fixed by J , so its Hahn lift is $\tilde{h}_A$ itself. The transported whole code has normal form

$$J(\tilde{h}_A) = \sum_{\beta < j(\kappa)} (1 + \mathbf{1}_{j(A)}(\beta)) \omega^{-\beta}.$$

The first κ terms are exactly $\tilde{h}_A$. Its first remaining coefficient, at $-\kappa$, is two if $\kappa \in j(A)$ and one otherwise. This proves (10.7). Multiplication by ω^κ shifts that exponent to zero; all later exponents become negative. The standard part therefore extracts the same coefficient, proving (10.8).

For a normal ultrapower on κ , the class of the identity function is the ordinal κ . Indeed every smaller ultrapower ordinal has a regressive representative on a large set and hence is constant modulo U , whereas every constant below κ is smaller than $[\text{id}]_U$. Thus $\kappa \in j(A)$ if and only if $A \in U$, which gives $U_j = U$. $\square$

The same seed calculation is the step “the measure derived at the seed κ is the original measure” in the proof of Corollary 10.4; 09 cites it from [17].

Theorem 10.3 (An omnific code [09]). *For $A \subseteq \kappa$ define the omnific mask*

$$Y_A = \sum_{\alpha \in A} \omega^{q_\alpha} = \sum_{\alpha \in A} \tau_\alpha, \quad (10.9)$$

with q_α and τ_α as in (8.4). Its support is $\{q_\alpha : \alpha \in A\}$ and its coefficients are all one; for $A = \kappa$ it is Y_κ of (8.5). Then

$$[\omega^{q_\kappa}] J(Y_A) = [\omega^{q_\kappa}] D_j(Y_A) = \begin{cases} 1, & A \in U_j, \\ 0, & A \notin U_j, \end{cases} \quad (10.10)$$

where $[\omega^{q_\kappa}] x$ is the coefficient of $\tau_\kappa = \omega^{q_\kappa}$ in x .

Proof. Uniform definability of the mask construction and Lemma 2.6 give

$$J(Y_A) = \sum_{\beta \in j(A)} \tau_\beta.$$

For $\alpha < \kappa$ all the exponents q_α are fixed by J , so $H_j(Y_A) = Y_A$. Since Y_A has no term τ_κ , subtracting it does not change the coefficient at τ_κ . That coefficient is one exactly when $\kappa \in j(A)$. $\square$

This gives a faithful numerical presentation of a normal measure once the set-induced embedding is available. It is not a proof that an arbitrary field embedding produces a normal measure. The set-theoretic origin and its action on every subset code are indispensable hypotheses [09].

Corollary 10.4 (Distinguishing normal measures [06, 07]). *Suppose U and W are distinct normal measures on the same κ , with normal ultrapower embeddings j_U and j_W . Their coefficient functionals*

$$A \mapsto [\omega^{-\kappa}] D_{j_U}(h_A), \quad A \mapsto [\omega^{-\kappa}] D_{j_W}(h_A)$$

are distinct. Thus the first failure cardinal does not determine the coefficient-level defect data.

Proof. Choose A in exactly one of U, W . In the normal ultrapower construction, the measure derived at the seed κ is the original measure. Apply Theorem 10.1. $\square$

06 proves the same with its bounded codes, and adds that the Hahn lifts fix those codes pointwise.

Corollary 10.5 (Faithful bounded coding [06]). *Distinct normal measures on the same κ induce different transport maps on the set $\{\tilde{h}_A : A \subseteq \kappa\} \subset (1, 3)$. Every associated Hahn lift fixes that entire set pointwise.*

Proof. Choose A which belongs to one measure and not to the other. The coefficient in (10.7), and hence the coefficient of the transported code at $-\kappa$, differs. The fixed-point assertion was proved in Theorem 10.2. $\square$

This recovers the ultrafilter from a coefficient observable rather than by recovering all of set theory from birthday data. The coefficient extraction and the monomial scale are specified additional structure; neither is claimed definable in the pure field language [06].

Remark 10.6 (A zero probe does not mean a zero defect [07, 09]). If $A \subseteq \kappa$ has cardinality κ but does not belong to U_j , then $D_j(h_A) \neq 0$ by the exact support theorem, although its coefficient at $\omega^{-\kappa}$ is zero. Such sets exist: partition κ into two sets of size κ , at most one of which is in the ultrafilter. The first missing *normal-form index* is always κ for a long support, but its associated *exponent* depends on the exponent list. The fixed mask probe need not be the leading nonzero term of every mask defect. 09 makes the same warning for its masks: if A has size κ and $A \notin U_j$, then $[\omega^{q_\kappa}] D_j(Y_A) = 0$ but $D_j(Y_A) \neq 0$; other additional exponents remain, and the distinguished coefficient records measure membership, not the entirety of the support discrepancy.

10.2 Real-valued masks and the Hadamard product

For $f : \kappa \rightarrow \mathbb{R}$, set

$$h_f = \sum_{\alpha < \kappa} f(\alpha) \omega^{-\alpha}. \quad (10.11)$$

This includes zero coefficients with the convention that they are suppressed in the canonical normal form. Let $\mathcal{B}_\kappa$ denote the set of all these masks. Give it its coefficientwise, or Hadamard, product

$$h_f \odot h_g = h_{fg}, \quad (fg)(\alpha) = f(\alpha)g(\alpha). \quad (10.12)$$

Together with ordinary addition and real scalar multiplication, this makes $\mathcal{B}_\kappa$ an $\mathbb{R}$ -algebra isomorphic to $\mathbb{R}^\kappa$. Its multiplicative unit is $h_\kappa = h_1$, *not* the surreal number 1. The map from scalars into this algebra is $r \mapsto rh_\kappa$.

Theorem 10.7 (Ultrafilter evaluation as a defect coefficient [07, 09]). *The functional*

$$\chi_j(h_f) = [\omega^{-\kappa}] (J(h_f) - H_j(h_f)) \quad (10.13)$$

satisfies

$$\chi_j(h_f) = j(f)(\kappa). \quad (10.14)$$

It is a unital $\mathbb{R}$ -algebra homomorphism $(\mathcal{B}_\kappa, +, \odot) \rightarrow \mathbb{R}$. Its value is the unique $r \in \mathbb{R}$ for which $f^{-1}(\{r\}) \in U_j$.

Proof. All original exponents are fixed, so $H_j(h_f) = h_f$. The whole transformed mask is

$$J(h_f) = \sum_{\beta < j(\kappa)} j(f)(\beta) \omega^{-\beta}.$$

Reading its coefficient at $-\kappa$ proves (10.14). The value lies in $\mathbb{R}$, since $j(f)$ takes values in $j(\mathbb{R}) = \mathbb{R}$. If $r = j(f)(\kappa)$, then $j(r) = r$ and

$$\kappa \in j(f^{-1}(\{r\})),$$

which gives the asserted fiber. Two distinct fibers cannot both belong to an ultrafilter. Transferring pointwise addition and multiplication gives

$$j(f + g)(\kappa) = j(f)(\kappa) + j(g)(\kappa), \quad j(fg)(\kappa) = j(f)(\kappa)j(g)(\kappa).$$

Constant functions take their constant real value. These are exactly the claimed algebra-homomorphism laws under (10.12). $\square$

09 proves the same evaluation for weighted omnific masks, with a different proof that the fiber exists.

Proposition 10.8 (Weighted omnific evaluation [09]). *For $f : \kappa \rightarrow \mathbb{R}$ put*

$$Y_f = \sum_{\alpha < \kappa} f(\alpha)\tau_\alpha, \tag{10.15}$$

with zero coefficients omitted. The map $f \mapsto Y_f$ is an injective $\mathbb{R}$ -linear map into the purely infinite part of $\mathbf{Oz}$, and

$$[\omega^{q\kappa}] J(Y_f) = j(f)(\kappa). \tag{10.16}$$

There is a unique $r \in \mathbb{R}$ such that $f^{-1}(\{r\}) \in U_j$, and the right side of (10.16) equals that r . Hence this coefficient is the character of $\mathbb{R}^\kappa$ associated with U_j , when $\mathbb{R}^\kappa$ has pointwise addition and multiplication.

Proof. The normal-form coefficient at τ_κ after applying j to (10.15) is exactly $j(f)(\kappa)$, including the possibility that it is zero. Since κ is measurable (by Lemma 2.3, or by Theorem 10.1), $|\mathbb{R}| < \kappa$. The partition of κ into the nonempty fibers of f has fewer than κ parts. If no fiber belonged to U_j , the intersection of their complements would belong to U_j by completeness and would be empty. At most one fiber can belong to an ultrafilter, proving existence and uniqueness. Applying j to that fiber identity gives $j(f)(\kappa) = j(r) = r$. Values of sums and pointwise products are therefore the sums and products of these selected real values, either directly by j or by intersecting the corresponding large fibers. $\square$

The image of $\mathbb{R}^\kappa$ in (10.15) is naturally an algebra under coefficientwise, or Hadamard, multiplication $Y_f \odot Y_g = Y_{fg}$. It is *not* a subalgebra under ordinary surreal multiplication. For masks, $Y_A \odot Y_B = Y_{A \cap B}$, but in general $Y_A Y_B \neq Y_{A \cap B}$: the latter product adds exponents and creates terms $\omega^{q\alpha + q\beta}$. Thus the ultrafilter interpretation uses explicitly specified coefficient operations; it cannot be smuggled into the ordinary ring language by treating masks as idempotents. A field has no nontrivial ordinary idempotents [09].

Product warning. The character theorem concerns the Hadamard product $\odot$. It does not say that coefficient extraction at $\omega^{-\kappa}$ is multiplicative for ordinary surreal multiplication. Ordinary multiplication convolves exponents; Hadamard multiplication multiplies coefficients at matching exponents.

Infinite direct-product homomorphisms and their ultrafilter factorizations are a classical subject; see, for example, Bergman [7]. The next result records an exact large-cardinal criterion in this particular mask presentation. Its underlying ultrafilter characterization is not claimed to be new.

Theorem 10.9 (A mask criterion for measurability). *Let κ be an uncountable cardinal, without initially assuming an embedding j . The following are equivalent.*

- (i) κ is measurable.

- (ii) *There is a unital $\mathbb{R}$ -algebra homomorphism $\chi : (\mathcal{B}_\kappa, +, \odot) \rightarrow \mathbb{R}$ such that $\chi(h_{\{\alpha\}}) = 0$ for every $\alpha < \kappa$, and for every pairwise disjoint family $(A_\xi)_{\xi < \mu}$ with $\mu < \kappa$,*

$$\chi \left(h_{\bigcup_{\xi < \mu} A_\xi} \right) = \sum_{\xi < \mu} \chi(h_{A_\xi}). \quad (10.17)$$

The scalar sum in (10.17) has at most one nonzero term, so no infinite real summation convention is needed. Under measurability, χ can be chosen to be the defect functional (10.13) of a normal ultrapower embedding.

Proof. Assume (i) and choose a normal measure and its ultrapower. The functional from Theorem 10.7 vanishes on singleton masks. Its value on each indicator mask is 0 or 1 according to membership in the measure. For fewer than κ disjoint sets, their union belongs to a κ -complete ultrafilter exactly when one of them belongs to it: otherwise all their complements have intersection in the ultrafilter. This proves (10.17).

Conversely, assume (ii). Each h_A is a Hadamard idempotent, so $\chi(h_A)^2 = \chi(h_A)$ and hence $\chi(h_A) \in \{0, 1\}$. Define $U = \{A \subseteq \kappa : \chi(h_A) = 1\}$. The identities

$$h_A \odot h_B = h_{A \cap B}, \quad h_{\kappa \setminus A} = h_\kappa - h_A$$

show that U is an ultrafilter. For example, upward closure follows from $h_A \odot h_B = h_A$ when $A \subseteq B$, and exactly one of a set and its complement has value 1. The singleton hypothesis makes it nonprincipal.

To verify κ -completeness, let $A_\xi \in U$ for $\xi < \mu < \kappa$. Set $B_\xi = \kappa \setminus A_\xi$, and disjointify the family by

$$C_\xi = B_\xi \setminus \bigcup_{\eta < \xi} B_\eta.$$

Each $C_\xi \subseteq B_\xi$ has value zero: multiplying its idempotent by h_{B_ξ} leaves it unchanged. By (10.17), $\bigcup_{\xi < \mu} C_\xi = \bigcup_{\xi < \mu} B_\xi$ also has value zero. Its complement $\bigcap_{\xi < \mu} A_\xi$ is therefore in U . Thus U is a nonprincipal κ -complete ultrafilter on κ , which is the usual measure characterization of measurability. $\square$

This criterion detects measurability only because the mask algebra and its small-union condition explicitly retain a κ -indexed Boolean algebra. It is not a characterization of measurability in the pure ordered-field language of **No**.

10.3 A compatible-field-embedding criterion [09]

09 gives a second criterion, of a different form: its witness is an ordered field embedding of all of **No**, not a functional on a mask algebra, and its extra operation is a Hadamard product on all surreals. For arbitrary surreals define

$$x \odot y = \sum_{\alpha \in \text{supp}(x) \cap \text{supp}(y)} [\omega^\alpha] x [\omega^\alpha] y \omega^\alpha.$$

The intersection is reverse well-ordered, so this operation is always defined. It restricts to the operation on the masks h_f and Y_f introduced above.

Theorem 10.10 (Compatible-field-embedding criterion for measurability [09]). *Fix an uncountable cardinal κ . In a class framework with Separation and Replacement for the maps in question, the following are equivalent.*

- (i) *The cardinal κ is measurable.*
- (ii) *There is an ordered field embedding $E : \mathbf{No} \rightarrow \mathbf{No}$ fixing $\mathbb{R}$ and commuting with Conway's omega-map such that:*

- (a) $E(\tau_\alpha) = \tau_\alpha$ for every $\alpha < \kappa$;
- (b) E preserves every strongly summable family of cardinality less than κ ;
- (c) for all $A, B \subseteq \kappa$, $E(Y_{A \cap B}) = E(Y_A) \odot E(Y_B)$;
- (d) $[\omega^{q_\kappa}] E(Y_\kappa) = 1$.

In the forward direction the witness E can be chosen definable from a normal measure. In the reverse direction no elementary embedding of set universes is assumed.

Proof. Assume measurability, choose a normal measure and take $E = J$. Theorem 8.1 and Theorem 8.4 give (a) and (b), and (8.7) gives (d). Hadamard multiplication is definable from normal forms and real coefficient multiplication. Its value for old inputs is absolute by Lemma 2.6; thus J preserves it. Since $Y_A \odot Y_B = Y_{A \cap B}$, condition (c) follows.

Conversely, let E satisfy (ii) and put

$$\chi_E(A) = [\omega^{q_\kappa}] E(Y_A), \quad U_E = \{A \subseteq \kappa : \chi_E(A) = 1\}.$$

Separation for the graph of E makes U_E a set. Taking $A = B$ in (c) shows $\chi_E(A)^2 = \chi_E(A)$. As a real idempotent, $\chi_E(A)$ is zero or one. Condition (d) gives $\chi_E(\kappa) = 1$, and additivity of the field embedding, applied to $Y_A + Y_{\kappa \setminus A} = Y_\kappa$, gives $\chi_E(A) + \chi_E(\kappa \setminus A) = 1$. Condition (c) further gives $\chi_E(A \cap B) = \chi_E(A)\chi_E(B)$. These identities show that U_E is an ultrafilter. Its singleton values are zero by (a), since τ_α has no coefficient at τ_κ for $\alpha < \kappa$. Thus U_E is nonprincipal.

To prove completeness, take $\nu < \kappa$ and sets $A_\xi \in U_E$ for $\xi < \nu$. Put $B_\xi = \kappa \setminus A_\xi$ and disjointify them:

$$C_\xi = B_\xi \setminus \bigcup_{\eta < \xi} B_\eta.$$

Each $C_\xi \subseteq B_\xi$ has $\chi_E(C_\xi) = 0$ by the ultrafilter identities. The family $(Y_{C_\xi})_{\xi < \nu}$ is strongly summable: its support union is a subset of the fixed decreasing scale $(q_\alpha)_{\alpha < \kappa}$ and each exponent occurs at most once. Its sum is Y_B , where $B = \bigcup_{\xi < \nu} B_\xi$. By (b) its image family is strongly summable and sums to $E(Y_B)$. Taking the coefficient at τ_κ gives zero, since every summand has zero there and coefficient extraction of a strong sum is a finite coefficient calculation. Consequently $\chi_E(B) = 0$, so $\bigcap_{\xi < \nu} A_\xi \in U_E$.

We have produced a nonprincipal κ -complete ultrafilter on κ , which is exactly measurability. The extra ordered-field and omega-map compatibility in (ii) was not needed for this converse, but the forward construction shows that it can be required simultaneously. $\square$

The implication from the coefficient identities to a measure is the classical ultrafilter criterion expressed in normal-form coordinates. The simultaneous compatibility with ordinary surreal field arithmetic, the omega-map, and all small strong sums is supplied by the large-cardinal construction. The theorem is an infinitary criterion with a specified coefficient operation and all subset masks, not a first-order criterion in the bare field or omnific-ring language. Determining which of these extra requirements can be removed is Question 21.22 [09]. [merge] Theorem 10.9 and Theorem 10.10 share the ultrafilter argument (idempotents, disjointification, small unions); they differ in what carries it, a character of an abstract mask algebra (07) or an ordered field embedding of **No** compatible with $\odot$ on masks (09). Neither removes the coefficient algebra.

11 Omnific integers and a purely infinite measure code

11.1 The ring and its short-support part

The omnific integers form the discretely ordered subring

$$\mathbf{Oz} = \left\{ \sum_a r_a \omega^a : \text{supp}(x) \subseteq \mathbf{No}^{\geq 0}, \quad r_0 \in \mathbb{Z} \right\}. \quad (11.1)$$

A missing zero exponent has coefficient zero. Nonzero coefficients at positive exponents are arbitrary reals, not necessarily ordinary integers. This is Conway’s normal-form description [1, 2]. We use it rather than an informal criterion such as “having no fractional part,” which can hide a convention at exponent zero. Put

$$\mathcal{J}_{<\kappa} = \mathbf{Oz} \cap \mathcal{S}_{<\kappa}.$$

Theorem 11.1 (Omnific equalizer, intersection, and fractions [06, 07, 09]). *The maps J and H_j preserve and reflect membership in $\mathbf{Oz}$, and*

$$\{z \in \mathbf{Oz} : J(z) = H_j(z)\} = \mathcal{J}_{<\kappa}, \quad (11.2)$$

$$J(\mathbf{Oz}) \cap H_j(\mathbf{Oz}) = J(\mathcal{J}_{<\kappa}) = H_j(\mathcal{J}_{<\kappa}), \quad (11.3)$$

$$\text{Frac}(\mathcal{J}_{<\kappa}) = \mathcal{S}_{<\kappa}, \quad (11.4)$$

$$\text{Frac}(J(\mathbf{Oz}) \cap H_j(\mathbf{Oz})) = C_j. \quad (11.5)$$

Also $\text{ct}(J(z)) = \text{ct}(H_j(z)) = \text{ct}(z)$ for every $z \in \mathbf{Oz}$.

Proof. For H_j this preservation and reflection is part of Lemma 4.1. For J , transfer the canonical normal form and the condition in (11.1). Nonnegativity of all exponents and the integer condition on the constant coefficient are absolute; equivalently, apply elementarity to the normal-form description and then Lemma 2.6. A coefficient at exponent zero is fixed, and a new missing-image exponent cannot equal zero because $0 \in \Gamma_j$. The same reasoning gives the constant-term identity for both maps.

The equalizer formula is Theorem 5.1 restricted to $\mathbf{Oz}$. If $y \in J(\mathbf{Oz}) \cap H_j(\mathbf{Oz})$, write $y = J(z)$ with $z \in \mathbf{Oz}$. The field intersection theorem forces $z \in \mathcal{S}_{<\kappa}$, so $z \in \mathcal{J}_{<\kappa}$. Conversely, both maps have the same value on any $z \in \mathcal{J}_{<\kappa}$. This proves (11.3).

To prove (11.4), take nonzero $x \in \mathcal{S}_{<\kappa}$. Its support is a set, so choose $d > 0$ such that $d > -a$ for every $a \in \text{supp}(x)$. The monomial $q = \omega^d$ and the shifted series $p = \omega^d x$ have only strictly positive exponents. Their constant coefficients vanish, and both are in $\mathbf{Oz}$. Their supports have cardinality less than κ , so $p, q \in \mathcal{J}_{<\kappa}$ and $x = p/q$. The reverse inclusion follows because $\mathcal{S}_{<\kappa}$ is a field containing $\mathcal{J}_{<\kappa}$. Applying the field embedding J gives (11.5). $\square$

Preservation of $\mathbf{Oz}$ and the equalizer (11.2) are also proved by 06 (Theorem 11.4) and 09 (its “Restricted equalizers”). The constant-term identity is 09’s Proposition “Constant coefficients survive”, stated for all $x \in \mathbf{No}$: $[\omega^0] J(x) = [\omega^0] H_j(x) = [\omega^0] x$, so $D_j(z)$ has constant coefficient zero for every $z \in \mathbf{Oz}$. For J , coefficient extraction is absolute by Lemma 2.6 and definable, so its value is the image of the fixed real $[\omega^0] x$; for H_j , injectivity of the exponent map and $J(0) = 0$ show that only the old exponent zero can contribute [09]. It is also the case $a = 0$ of Proposition 14.1. For arbitrary surreals the constant-coefficient operation is not a field homomorphism: positive and negative exponents can multiply to zero. On $\mathbf{Oz}$, whose exponents are nonnegative, it is a ring homomorphism to $\mathbb{Z}$. The large discrepancies here therefore lie in its purely infinite kernel. This does not conflict with claims about maps to set-sized rings in the repository: the embeddings in this article have proper-class codomains [09].

11.2 A defect with no finite residue

For $A \subseteq \kappa$, define

$$z_A = \omega^\kappa h_A = \sum_{\alpha \in A} \omega^{\kappa - \alpha}. \quad (11.6)$$

Every exponent is strictly positive, since $\alpha < \kappa$. Thus z_A is a purely infinite omnific integer when A is nonempty, and is zero otherwise. Here and below $\kappa - \alpha$ is the difference in the surreal ordered field.

Theorem 11.2 (Omnific recovery of the measure). *For $A \subseteq \kappa$,*

$$D_j(z_A) = \omega^{j(\kappa)} D_j(h_A), \quad (11.7)$$

$$[\omega^{j(\kappa) - \kappa}] D_j(z_A) = \mathbf{1}_{\{A \in U_j\}}, \quad (11.8)$$

$$\text{ct}(D_j(z_A)) = 0. \quad (11.9)$$

In particular, for $z_\kappa = \omega^\kappa h_\kappa$,

$$D_j(z_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \omega^{j(\kappa) - \beta} > 0, \quad \text{LT}(D_j(z_\kappa)) = \omega^{j(\kappa) - \kappa}.$$

Every nonzero term of every $D_j(z_A)$ is purely infinite.

Proof. The two embeddings agree on ω^κ , with common value $\omega^{j(\kappa)}$, and both are multiplicative. This proves (11.7). Multiplication by that monomial shifts exponents by $j(\kappa)$, so (11.8) follows from (10.4). Moreover $j(A) \cap \kappa = A$, so the defect of h_A has its possible indices only in $j(A) \setminus A \subseteq [\kappa, j(\kappa))$. After shifting, all exponents $j(\kappa) - \beta$ are strictly positive, which proves (11.9) and the last assertion. Taking $A = \kappa$ gives the explicit nonempty series and its leading term. $\square$

The constant-term map therefore loses this large-cardinal information, even though one specified infinite-scale coefficient recovers it. This is compatible with the repository's emphasis on constant-term factorizations in omnific arithmetic [11]; it does not supply a new factorization theorem for set-sized quotient rings.

06 shifts its bounded code the same way. 09's code Y_A of Theorem 10.3 is omnific from the start, with exponents $q_\alpha \in (0, 1]$ instead of $\kappa - \alpha$.

Corollary 11.3 (Omnific measure codes [06]). *The elements*

$$\tilde{z}_A = \omega^\kappa \tilde{h}_A = z_\kappa + z_A = \sum_{\alpha < \kappa} (1 + \mathbf{1}_A(\alpha)) \omega^{\kappa - \alpha} \quad (11.10)$$

are purely infinite omnific integers. Their transport defects satisfy

$$[\omega^{j(\kappa) - \kappa}] D_j(\tilde{z}_A) = 1 + \mathbf{1}_{U_j}(A). \quad (11.11)$$

Thus the same measure is recoverable from the two induced maps on $\mathbf{Oz}$.

Proof. For $\alpha < \kappa$, the surreal difference $\kappa - \alpha$ is positive. This proves membership in $\mathbf{Oz}$. Both embeddings send ω^κ to $\omega^{j(\kappa)}$. Multiplicativity gives $D_j(\tilde{z}_A) = \omega^{j(\kappa)} D_j(\tilde{h}_A)$, and (11.11) follows by shifting the coefficient in (10.7). Preservation of the ring by both embeddings is Theorem 11.1. $\square$

Theorem 11.4 (Omnific restrictions [06]). *Both J and H_j preserve the ring $\mathbf{Oz}$ of omnific integers and restrict to injective unital ring maps. Their surcomplex extensions preserve $\mathbf{Oz}[i]$. The equalizer and landing criteria remain exact on each of these rings, with support computed in its real or complex normal form. For a normal-measure ultrapower, all three support regimes of Theorem 9.7 occur already in $\mathbf{Oz}$.*

Proof. In the normal-form description (11.1), J carries the transported array of positive exponents to an array of positive exponents by elementarity and absoluteness. The unique constant coefficient, when present, is an ordinary integer and is fixed. Thus $J(x)$ is omnific. The same argument for H_j is termwise: positivity of exponents and the integer constant term are preserved. Ring preservation and injectivity follow from the field-embedding theorem.

For $\mathbf{Oz}[i]$ apply the real result to the two coordinates. The exact criteria follow by restriction of the real and complex theorems (Theorem 5.1, Theorem 9.1 and Theorem 12.3). For the three regimes, the omnific normal forms $\zeta_\mu = \omega^\mu h_\mu = \sum_{\alpha < \mu} \omega^{\mu-\alpha}$ have support cardinality $|\mu|$. Take respectively a nonzero finite cardinal, κ , and κ^+ . $\square$

Remark 11.5 (Preserving is not fixing [06]). Preserving $\mathbf{Oz}$ does not mean preserving its elements pointwise. Nor is this a theorem that all endomorphisms of the omnific integers come from large cardinals. It constructs and classifies the behavior of a specified pair arising from an elementary universe embedding. The distinction is important in comparison with the repository's independent work on omnific automorphisms and set-sized quotients.

Remark 11.6 (An $\mathbf{Oz}$ -reflecting embedding that is not strong [merge]). The omnific-preserving-automorphisms report asks whether every field embedding $f : \mathbf{No} \hookrightarrow \mathbf{No}$ with $f^{-1}(\mathbf{Oz}) = \mathbf{Oz}$ must be strongly additive, and what changes if also $\text{ct} \circ f = \text{ct}$ (`opa:as:q:embeddingstrong` in `docs/surreal/omnific-preserving-automorphisms`, where J is written $\hat{f}$). If there is a measurable cardinal, or more generally an embedding as in (1.1) available as a class, then J answers both clauses negatively. It is an ordered field embedding fixing $\mathbb{R}$ (Proposition 2.10); it preserves and reflects $\mathbf{Oz}$, and preserves constant terms (Theorem 11.1); it carries every strongly summable family to a strongly summable family, but the family $(\omega^{-\alpha})_{\alpha < \kappa}$ has image sum $h_\kappa \neq J(h_\kappa) = h_{j(\kappa)}$ (Theorem 8.1, Theorem 7.1; the omnific family $(\tau_\alpha)_{\alpha < \kappa}$ of Theorem 8.4 does the same inside $\mathbf{Oz}$). So J is not strongly additive. It is not onto (Proposition 7.4), so there is no conflict with that report's theorem that every ring automorphism of $\mathbf{Oz}$ extends uniquely to a strongly $\mathbb{R}$ -linear automorphism of $\mathbf{No}$ (`opa:as:thm:main`). A positive answer to the first clause is therefore not provable in the base theory unless that theory with a measurable cardinal is inconsistent. Whether a counterexample exists without large cardinals is not addressed here, and nothing here bears on that question's clauses about dense images or images that are neither closed nor discrete. The pair (J, H_j) also shows that preservation and reflection of $\mathbf{Oz}$ together with agreement on all monomials does not determine an embedding: H_j has the same properties and is strong. *Batch 32.* Manuscript 10 sharpens the large-cardinal-free case for coefficient-fixing embeddings: if there is no measurable cardinal, every countably strong coefficient-fixing field embedding preserving $\mathbf{Oz}$ is strong (Theorem 26.2), so such a counterexample would have to fail a countable sum. Embeddings that move real coefficients are outside that theorem, and the existence question itself stays open (Question 26.8).

12 Surcomplex numbers and the Gaussian omnific lattice

Let $\mathbf{SC} = \mathbf{No}(i)$ with a fixed named element i satisfying $i^2 = -1$, and write conjugation as $\overline{x + iy} = x - iy$. A surcomplex normal form has complex coefficients and a reverse well-ordered set of surreal exponents. Its support is the union of the supports of its real and imaginary coordinates: a complex coefficient vanishes exactly when both real coordinates vanish. Define

$$J_{\mathbb{C}}(x + iy) = J(x) + iJ(y), \quad H_{j,\mathbb{C}}(x + iy) = H_j(x) + iH_j(y).$$

These definitions fix the chosen orientation i ; they are not a statement about arbitrary automorphisms of the pure surcomplex field.

Theorem 12.1 (Surcomplex transfer [06, 07, 09]). *The maps $J_{\mathbb{C}}, H_{j,\mathbb{C}} : \mathbf{SC} \rightarrow \mathbf{SC}$ are field embeddings fixing $\mathbb{C}$ and commuting with conjugation. The second is strongly $\mathbb{C}$ -linear. They are elementary in the pure field language, and*

$$\text{Eq}(J_{\mathbb{C}}, H_{j,\mathbb{C}}) = \mathcal{S}_{<\kappa}(i), \quad (12.1)$$

$$J_{\mathbb{C}}(\mathbf{SC}) \cap H_{j,\mathbb{C}}(\mathbf{SC}) = C_j(i). \quad (12.2)$$

Both fields on the right are algebraically closed. The image fields on the left of (12.2) are incomparable. Their intersection with the Gaussian omnific lattice satisfies

$$J_{\mathbb{C}}(\mathbf{Oz}[i]) \cap H_{j,\mathbb{C}}(\mathbf{Oz}[i]) = J(\mathcal{J}_{<\kappa})[i]. \quad (12.3)$$

Proof. The quadratic-extension multiplication law proves the field-homomorphism claims coordinatewise. Uniqueness of real and imaginary parts gives injectivity and commutation with conjugation. The strong summation claim is the real statement on both coordinates. Since $\mathbf{No}$ is real closed, $\mathbf{SC}$ and both image fields are algebraically closed of characteristic zero. Quantifier elimination for ACF_0 gives elementarity, formula by formula, as before.

Equality of $J_{\mathbb{C}}(x + iy)$ and $H_{j,\mathbb{C}}(x + iy)$ is equivalent to both $J(x) = H_j(x)$ and $J(y) = H_j(y)$. This is precisely $x, y \in \mathcal{S}_{<\kappa}$. Equivalently, the union of the two supports has size less than κ : a finite union of sets of size less than an infinite cardinal still has size less than that cardinal. This proves (12.1). The same coordinate comparison and Theorem 6.1 prove (12.2). The fields $\mathcal{S}_{<\kappa}$ and C_j are real closed, so adjoining i makes them algebraically closed.

The real witnesses h_κ and $h_{j(\kappa)}$ still witness incomparability. If a real element is the image of $x + iy$, its imaginary coordinate is zero, forcing $y = 0$; allowing a complex preimage cannot repair the failure of a real preimage. Finally apply Theorem 11.1 separately to the two integer coordinates to obtain (12.3). $\square$

The exact defect formula also holds directly with complex coefficients: for $z = \sum_{\alpha < \theta} c_\alpha \omega^{a_\alpha}$, set $c^* = j(c)$ after identifying $\mathbb{C}$ with $\mathbb{R}^2$. Then

$$J_{\mathbb{C}}(z) - H_{j,\mathbb{C}}(z) = \sum_{\beta \in P_\theta} c_\beta^* \omega^{a_\beta^*}.$$

Nonzero complex coefficients at distinct exponents cannot cancel. Thus long complex support, not a sign or order on complex numbers, is what forces disagreement. The Boolean measure masks from Section 10 already lie on the real axis and require no complex choice beyond the named inclusion.

Corollary 12.2 (Conjugation and modulus). *If $|z| = (z\bar{z})^{1/2} \in \mathbf{No}^{\geq 0}$, then*

$$|J_{\mathbb{C}}(z)| = J(|z|), \quad |H_{j,\mathbb{C}}(z)| = H_j(|z|).$$

Every surcomplex number in either image outside $C_j(i)$ is transcendental over $C_j(i)$.

Proof. Both maps preserve conjugation and the unique nonnegative square root. The transcendence assertion follows because $C_j(i)$ is algebraically closed. $\square$

06 and 09 prove the transfer and the equalizer $\mathcal{S}_{<\kappa}(i)$ as well; 09 notes that the same κ -sized real omnific family (τ_α) witnesses the failure of full strong additivity for $J_{\mathbb{C}}$, and 06 that properness is witnessed by the real elements of Proposition 7.4: a nonreal preimage of a real value is impossible, since the injective real-coordinate map would have to send a nonzero imaginary part to zero. 06 adds descent.

Theorem 12.3 (Surcomplex descent [06]). *For every $z \in \mathbf{SC}$ of complex normal-form length λ (the support being the union of the supports of the two coordinates),*

$$H_{j,\mathbb{C}}(z) \in \mathbf{SC}^M \iff j \restriction \lambda \in M.$$

The missing-index formula and the normal-measure trichotomy (Theorem 9.7) hold with complex coefficients. In a set-ultrapower, descent is again equivalent to closure under sequences of the support cardinality.

Proof. Apply j to the packed complex coefficient and real exponent arrays. The transformed coefficient array remains nonzero at every index. The termwise image retains exactly the indices $j \restriction \lambda$, so normal-form uniqueness proves the equalizer and defect formulas. For descent, if the complex number belongs to $\mathbf{SC}^M$, its two coordinates and their normal forms belong to M , hence their combined complex support does as well. The preimage under $j(a)$ used in Theorem 9.1 recovers $j \restriction \lambda$. Conversely an internal seed permits evaluation of both coordinates inside M . The sequence and normal-measure conclusions follow unchanged, by Lemma 9.3 and the proof of Theorem 9.7. $\square$

The local analytic conclusion of Proposition 15.2 therefore applies to complex formal power series at surcomplex infinitesimals. No choice of a global surcomplex logarithm or exponential at arbitrary infinite arguments is needed. In particular no assertion about such a global function is hidden in the notation $\mathbf{SC}$ [06]. For surcomplex numbers the field-theoretic transfer is automatic once the real result is known, but descent still depends on the real exponent support and on the same seed [06].

13 Algebraic independence over the short-support field [09]

This section does not assume a measurable cardinal or an elementary embedding. Fix any uncountable cardinal κ . The exponent scale $q_\alpha = \omega^{-\alpha}$ of (8.4) has two useful features: it is decreasing, and its members are linearly independent over $\mathbb{Q}$. The latter follows immediately from uniqueness of finite Conway normal forms, since the q_α are distinct monomials. The omnific masks Y_A are those of (10.9).

Definition 13.1 (Almost disjoint families [09]). A family $\mathcal{A} \subseteq [\kappa]^\kappa$ is κ -almost disjoint if $|A \cap B| < \kappa$ whenever $A, B \in \mathcal{A}$ are distinct.

13.1 A coordinate separation lemma

Lemma 13.2 (Avoiding all coefficient directions [09]). *Let $G_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}q_\alpha$ and let $\Xi \subseteq (\mathbf{No}, +)$ be a rational vector subspace of cardinality less than κ . There is a set $B \subseteq \kappa$ of size less than κ such that, for every $\alpha \notin B$, the coordinate map*

$$G_\kappa \longrightarrow \mathbb{Q}, \quad \sum_{\xi} c_\xi q_\xi \longmapsto c_\alpha$$

extends to a linear functional $\ell_\alpha : \Xi + G_\kappa \rightarrow \mathbb{Q}$ that vanishes on Ξ .

Proof. Choose a rational basis of $\Xi \cap G_\kappa$. Every vector in this basis has finite support in the displayed basis of G_κ . Let B be the union of these finite supports. The number of basis vectors is at most $|\Xi|$, so $|B| < \kappa$, with the finite cases interpreted in the evident way. This estimate does not require κ to be regular.

For $\alpha \notin B$ the α -coordinate vanishes on $\Xi \cap G_\kappa$. Define $\ell_\alpha(\xi + v)$ to be that coordinate of v for $\xi \in \Xi$ and $v \in G_\kappa$. If $\xi + v = \xi' + v'$, then $v - v' \in \Xi \cap G_\kappa$, so the coordinate is independent of the representation. The resulting map is linear and kills Ξ . $\square$

The lemma is what allows arbitrary short-support surreal coefficients, not just coefficients whose exponents already lie in G_κ [09].

13.2 The independence theorem

Theorem 13.3 (Almost-disjoint omnific independence [09]). *For every uncountable cardinal κ and every κ -almost-disjoint family $\mathcal{A} \subseteq [\kappa]^\kappa$, the family*

$$\{Y_A : A \in \mathcal{A}\} \quad (13.1)$$

*is algebraically independent over the entire field $\mathcal{S}_{<\kappa}$ inside **No**. Its members are positive purely infinite omnific integers of normal-form length κ .*

Proof. Choose distinct $A_1, \dots, A_n \in \mathcal{A}$ and a nonzero polynomial

$$\Phi(X_1, \dots, X_n) = \sum_{\mathbf{m}} c_{\mathbf{m}} X_1^{m_1} \cdots X_n^{m_n} \in \mathcal{S}_{<\kappa}[X_1, \dots, X_n]. \quad (13.2)$$

Only finitely many coefficients are nonzero. Let S be the union of their normal-form supports and set $\Xi = \text{span}_{\mathbb{Q}}(S)$. Then $|\Xi| < \kappa$, exactly as in the proof of Theorem 3.1. Obtain B and the coordinate functionals from Lemma 13.2.

For each i choose

$$\alpha_i \in A_i \setminus \left(B \cup \bigcup_{k \neq i} A_k \right). \quad (13.3)$$

Such a choice is possible because the removed part of A_i has cardinality less than κ : it is a finite union of small intersections and the small set B . The indices α_i are distinct. More importantly, q_{α_i} occurs in the support of Y_{A_i} and in none of the other selected masks.

Choose a multi-index $\mathbf{m} = (m_1, \dots, m_n)$ with nonzero coefficient whose total degree $d = m_1 + \dots + m_n$ is maximum among the nonzero terms of Φ . Choose one exponent $d_0 \in \text{supp}(c_{\mathbf{m}})$, and let $\varrho \neq 0$ be its real coefficient. We claim that the exponent

$$\gamma = d_0 + \sum_{i=1}^n m_i q_{\alpha_i} \quad (13.4)$$

has coefficient exactly ϱ in $\Phi(Y_{A_1}, \dots, Y_{A_n})$.

A contribution to this coefficient from a term with multi-index $\mathbf{n}$ consists of an exponent $d' \in \text{supp}(c_{\mathbf{n}})$ and, for each of its n_i factors of Y_{A_i} , a choice of one exponent q_{β} with $\beta \in A_i$. The total exponent is d' plus all these chosen exponents. It belongs to the set-sized group $\Xi + G_\kappa$. Apply the n functionals ℓ_{α_i} to a putative equality of this total exponent with (13.4). Each functional kills d' and d_0 . It counts exactly the number of selected factors whose exponent is q_{α_i} . That count must be m_i .

The counts sum to d , so at least d factors have already been used. On the other hand the maximality of d gives $\sum_i n_i \leq d$. Hence exactly d factors occur, all of them at the specified private indices. Because α_i belongs only to A_i among the selected sets, the equality forces $n_i = m_i$ for every i . There are no further factors, and the exponent equation now forces $d' = d_0$.

Within the powers $Y_{A_i}^{m_i}$ all m_i factors must select the same term τ_{α_i} . The coefficient of that choice is one; there is no further multinomial multiplicity. Thus the only contribution to the coefficient at γ is ϱ , as claimed. All convolutions used here are legitimate finite coefficient convolutions by Hahn summability. The argument

also covers $d = 0$, in which case the polynomial is a nonzero constant. We have proved that every nonzero polynomial (13.2) evaluates nontrivially.

Each mask has positive exponents and coefficients one, so is a positive purely infinite omnific integer. A subset of κ of cardinality κ has inherited order type κ : its order type is at most κ and cannot be a smaller ordinal of the same cardinality. Therefore its normal-form length is κ . $\square$

Remark 13.4 (Why cancellation cannot be hidden in the coefficients [09]). A coefficient in $\mathcal{S}_{<\kappa}$ can contain exponents unrelated to any q_α . Such exponents can create complicated additive relations. The proof does not assume they are absent. It puts their entire rational span into Ξ and chooses private coordinates annihilating Ξ . The maximal *total* degree then exhausts every factor, leaving exactly one nonzero coefficient. Using only distinct leading exponents of the masks would not establish this stronger coefficient-field assertion.

13.3 Maximal cardinality inside a fixed Hahn field

Theorem 13.5 (A 2^κ family [09]). *Suppose κ is uncountable and $2^{<\kappa} = \kappa$. Let*

$$G_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}q_\alpha, \quad K_\kappa = \mathbb{R}((\omega^{G_\kappa})).$$

Then K_κ contains 2^κ positive purely infinite omnific integers, algebraically independent over $\mathcal{S}_{<\kappa}$. Moreover

$$|K_\kappa| = 2^\kappa, \quad \text{trdeg}(\mathcal{S}_{<\kappa}K_\kappa/\mathcal{S}_{<\kappa}) = 2^\kappa. \quad (13.5)$$

*The compositum in (13.5) is the subfield of **No** generated by $\mathcal{S}_{<\kappa}$ and K_κ .*

Proof. The binary tree $2^{<\kappa} = \bigcup_{\alpha < \kappa} 2^\alpha$ has cardinality κ . Fix a bijection $\rho : 2^{<\kappa} \rightarrow \kappa$. For each branch $b \in 2^\kappa$ put

$$A_b = \{\rho(b \upharpoonright \alpha) : \alpha < \kappa\}.$$

Each A_b has size κ . Distinct branches share only their prefixes through their first disagreement; their sets therefore intersect in fewer than κ points. The family $(A_b)_{b \in 2^\kappa}$ is κ -almost disjoint, so Theorem 13.3 applies. Every resulting Y_{A_b} is supported in G_κ , hence belongs to K_κ . No order-preserving property of ρ is needed: any subset of the ordinal κ has a well-ordered increasing enumeration.

The group G_κ has cardinality κ , and $|\mathbb{R}| \leq \kappa$ follows from $2^{<\kappa} = \kappa$. A Hahn series is specified by a coefficient function on that group, so

$$|K_\kappa| \leq |\mathbb{R}|^\kappa \leq \kappa^\kappa = 2^\kappa.$$

The constructed masks already give the reverse inequality. They provide the lower bound for the transcendence degree in (13.5); the set K_κ , which generates the extension, gives the upper bound.

Although the base field is a proper class, this last use of transcendence degree causes no size problem. Algebraic independence of a subset of the set K_κ is a definable set-theoretic condition: each polynomial involves only finitely many surreal coefficients. Zorn's lemma on subsets of K_κ gives a maximal independent set over $\mathcal{S}_{<\kappa}$. Maximality makes the set-generated extension algebraic over that independent set. The usual exchange argument is finitary and applies unchanged. Thus the transcendence basis under discussion is a set. $\square$

Corollary 13.6 (Complex coefficients and measurable cardinals [09]). *The families in Theorems 13.3 and 13.5 are also algebraically independent over $\mathcal{S}_{<\kappa}(i)$ inside **SC**. If κ is measurable, Theorem 13.5 applies, and its 2^κ omnific integers all lie outside the equalizer of J and H_j .*

Proof. Split a putative polynomial relation with coefficients in $\mathcal{S}_{<\kappa}(i)$ into real and imaginary parts. At least one part is a nonzero polynomial over $\mathcal{S}_{<\kappa}$, contradicting real algebraic independence. A measurable cardinal is strongly inaccessible, hence $2^{<\kappa} = \kappa$ [18]. Every constructed mask has support of size κ , so Theorem 5.1 gives the disagreement assertion. $\square$

The independence theorem itself is therefore not a large-cardinal theorem. For example, it applies at ω_1 under CH. The measurable-cardinal hypothesis adds the two embeddings whose disagreement it witnesses [09]. [merge] This is transcendence over the *domain* equalizer $\mathcal{S}_{<\kappa}$; it does not decide 07’s Question 21.6 on the joint algebraic dependence of the two *image* fields over C_j .

14 Defect algebra and a logarithmic homomorphism

14.1 What all old-image coefficients fail to see

Proposition 14.1 (Coefficient invisibility). *For every $a, x \in \mathbf{No}$,*

$$[\omega^{J(a)}] J(x) = [\omega^a] x = [\omega^{J(a)}] H_j(x). \quad (14.1)$$

In particular, J and H_j preserve the coefficient at exponent zero. They preserve the finite-surreal valuation ring and its infinitesimal ideal, and induce the identity on its residue field $\mathbb{R}$.

Proof. If a is in the support of x , its unique original index gives the coefficient in (5.2). A missing-image index cannot supply exponent $J(a)$ by Theorem 5.1. If a is not in the support, the same range-reflection argument used in that theorem excludes $J(a)$ from the support of $J(x)$. The identity for H_j is its definition. Setting $a = 0$ gives the constant coefficient. The valuation identities in Corollary 5.2, together with $J(0) = 0$ and order preservation, give preservation of the valuation ring and its maximal ideal. On a finite surreal, its residue is its coefficient at zero. $\square$

The constant-coefficient case $a = 0$ is also 09’s (see the note after Theorem 11.1). Thus (10.4) detects genuinely new exponent positions. No coefficient extraction at an exponent in $J(\mathbf{No})$ can see the defect. Both embeddings nevertheless have the same value-group action and the same residue action.

Proposition 14.2 (Twisted product law [06, 07, 09]). *The additive map D_j satisfies*

$$D_j(xy) = J(x)D_j(y) + D_j(x)H_j(y). \quad (14.2)$$

For $c \in \mathcal{S}_{<\kappa}$ one has $D_j(cx) = J(c)D_j(x)$.

Proof. Expand the right side as $J(x)J(y) - J(x)H_j(y) + J(x)H_j(y) - H_j(x)H_j(y)$. The middle terms cancel and multiplicativity gives the left side. The scalar identity follows because $D_j(c) = 0$. $\square$

Equation (14.2) is a twisted derivation identity, not an ordinary Leibniz rule for a derivation on $\mathbf{No}$. No claim is made that D_j is a Berarducci–Mantova derivation or a strongly additive derivation.

06 (Proposition “Algebra of the defect”) and 09 (Proposition “Defect identities”) state the same law in the equivalent symmetric form

$$D_j(xy) = H_j(x)D_j(y) + H_j(y)D_j(x) + D_j(x)D_j(y), \quad (14.3)$$

obtained by substituting $J = H_j + D_j$ in $J(xy) = J(x)J(y)$. 09 adds that D_j is additive and $\mathbb{R}$ -linear with kernel $\mathcal{S}_{<\kappa}$, and that for $x \neq 0$

$$D_j(x^{-1}) = -\frac{D_j(x)}{J(x)H_j(x)},$$

the difference of $J(x)^{-1}$ and $H_j(x)^{-1}$; the quadratic last term in (14.3) cannot be discarded [09]. 06 shows that D_j is not a derivation relative to H_j : if the ordinary H_j -Leibniz identity held, then putting $x = y = h_\kappa$ would force $D_j(h_\kappa)^2 = 0$, contradicting $D_j(h_\kappa) \neq 0$ (Theorem 7.1) in a field [06].

14.2 Passing from a multiplicative discrepancy to an additive one

For $x \neq 0$, let

$$\eta_j(x) = \frac{J(x)}{H_j(x)}. \quad (14.4)$$

By Corollary 5.2, this ratio lies in $1 + \mathfrak{m}$. For $u \in \mathfrak{m}$, the formal series

$$\log(1 + u) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} u^n \quad (14.5)$$

is strongly summable by the Hahn–Neumann lemma. The usual formal power-series identities are legitimate after such infinitesimal substitution; in particular, this logarithm takes products in $1 + \mathfrak{m}$ to sums [2, 3]. Only this near-one logarithm is needed, not a preservation theorem for the global surreal exponential.

Theorem 14.3 (Logarithmic defect). *The map*

$$L_j : \mathbf{No}^\times \longrightarrow (\mathfrak{m}, +), \quad L_j(x) = \log \eta_j(x)$$

is a group homomorphism with kernel $\mathcal{S}_{<\kappa}^\times$. If x has normal form (2.1) of length $\theta \geq \kappa$, then

$$\text{LT}(L_j(x)) = \frac{r_\kappa^*}{r_0} \omega^{a_\kappa^* - J(a_0)}. \quad (14.6)$$

Thus a nonzero additive infinitesimal invariant records every departure from the equalizer, after multiplicative normalization.

Proof. Because both embeddings are multiplicative, $\eta_j(xy) = \eta_j(x)\eta_j(y)$. The logarithm identity gives $L_j(xy) = L_j(x) + L_j(y)$. If $\eta_j(x) = 1 + u$ with $u \neq 0$, then the first term of (14.5) is u , while all higher powers have strictly larger valuation. Hence the logarithm is nonzero and has leading term $\text{LT}(u)$. It follows that $L_j(x) = 0$ precisely when $J(x) = H_j(x)$, proving the kernel assertion.

Finally $u = D_j(x)/H_j(x)$. Its leading term is the quotient of the leading term in (5.5) by the common leading term from Corollary 5.2, namely (14.6). Its exponent is negative since $a_\kappa^* < a_0^* = J(a_0)$, so it is infinitesimal as required. $\square$

For the basic witness,

$$\text{LT}(L_j(h_\kappa)) = \omega^{-\kappa}.$$

The homomorphism formulation avoids introducing a quotient whose individual equivalence classes might be proper classes. We use its kernel as a class predicate; no extra axiom for forming hyperclasses or class-sized quotient objects is needed.

15 An exact exponential compatibility locus

This section adds the canonical Gonshor exponential. It is independent of the proof of the main defect and image-intersection theorems, but the same support calculation yields a second exact boundary.

Write every surreal uniquely as

$$x = p(x) + r(x) + \varepsilon(x),$$

where $p(x)$ has only positive exponents, $r(x) \in \mathbb{R}$, and $\varepsilon(x) \in \mathfrak{m}$. Thus $p(x)$ is its purely infinite part. We use the classical exponential facts that $\exp(p)$ is a Conway monomial for purely infinite p , that the exponential on infinitesimals is its strong power series, and that the inverse logarithm of a monomial is purely infinite [2, Chapter 10][3, Section 5].

Lemma 15.1 (Exponential transfer [06, 07]). *The elementary-induced embedding satisfies*

$$J(\exp x) = \exp J(x), \quad J(\log y) = \log J(y) \quad (y > 0).$$

Proof. Gonshor's exponential is defined by a uniform recursive cut on the canonical options of its argument, using exponentials of simpler options, finite Taylor polynomials, field operations, and their inequalities. The cut absoluteness argument from Lemma 2.6 applies to this recursion as well. Therefore the exponential computed in M on an old argument equals the one in V . Elementary transfer of the defining recursion gives the first identity. The second follows from the uniqueness of the inverse of the strictly increasing bijection $\exp$. $\square$

06 includes the exponential in its absoluteness lemma, with the same recursion argument (the note after Lemma 2.6). 06 also proves the infinitesimal part of the next theorem as a separate proposition, for both maps and for power series in several variables.

Proposition 15.2 (Formal evaluation [06]). *Let $A \in \mathbb{R}[[T_1, \dots, T_n]]$ and let $\varepsilon_1, \dots, \varepsilon_n$ be infinitesimal surreals. With strong formal evaluation,*

$$\begin{aligned} H_j(A(\varepsilon_1, \dots, \varepsilon_n)) &= A(H_j(\varepsilon_1), \dots, H_j(\varepsilon_n)), \\ J(A(\varepsilon_1, \dots, \varepsilon_n)) &= A(J(\varepsilon_1), \dots, J(\varepsilon_n)). \end{aligned}$$

The same statements hold with complex coefficients and surcomplex infinitesimals after the extension of Section 12.

Proof. The Neumann lemma makes the family of evaluated monomial terms strongly summable. It is countable, since there are finitely many variables. Both embeddings fix every coefficient and preserve finite products. For H_j use strong additivity; for J use Theorem 8.1, since $\aleph_0 < \kappa$. Ordered-field embeddings fixing the reals preserve the condition of being infinitesimal. The coefficientwise proof over $\mathbb{C}$ is identical. $\square$

This includes the Taylor exponential on $\mathfrak{m}$ and the logarithm on $1 + \mathfrak{m}$. Arbitrarily large supports of the individual infinitesimals cause no problem: the *number of Taylor terms* is still countable. This is another distinction between support size and summation-family size [06].

Theorem 15.3 (Exact exponential defect [06, 07]). *For every $x \in \mathbf{No}$,*

$$\frac{H_j(\exp x)}{\exp(H_j(x))} = \exp(D_j(p(x))). \quad (15.1)$$

In particular,

$$H_j(\exp x) = \exp(H_j(x)) \iff |\text{supp}(p(x))| < \kappa. \quad (15.2)$$

The companion commutes with the exponential on every finite surreal, including finite surreals with arbitrarily long infinitesimal support. For a positive $y = r\omega^a(1+u)$, with $r > 0$ real and $u \in \mathfrak{m}$,

$$\log(H_j(y)) - H_j(\log y) = D_j(\log(\omega^a)). \quad (15.3)$$

Proof. The companion preserves the purely infinite, real, and infinitesimal parts separately, because it preserves the sign of exponents. Since $\exp(p(x))$ is a single monomial,

$$H_j(\exp(p(x))) = J(\exp(p(x))) = \exp(J(p(x))).$$

Strong linearity and infinitesimal power-series evaluation give $H_j(\exp(\varepsilon(x))) = \exp(H_j(\varepsilon(x)))$, while real exponential values are fixed. Applying H_j to $\exp x = \exp(p(x))e^{r(x)}\exp(\varepsilon(x))$ and dividing by $\exp(H_j(x))$ proves (15.1). Injectivity of the exponential and Theorem 5.1 give (15.2). For finite x , its purely infinite part is zero.

Finally use $\log y = \log r + \log(\omega^a) + \log(1+u)$. The near-one logarithm commutes with H_j by its strong power-series formula. Also $\log(H_j(\omega^a)) = \log(J(\omega^a)) = J(\log(\omega^a))$. Subtracting the two decompositions gives (15.3). $\square$

06 proves (15.1) and (15.2) with the same factorization, using Proposition 15.2 for the infinitesimal factor, and records the intermediate formula

$$H_j(\exp x) = \exp(J(p(x)) + r(x) + H_j(\varepsilon(x))). \quad (15.4)$$

It notes that a purely infinite series may have either sign: “purely infinite” describes its support, not its positivity; and it cites the classical facts that the exponential maps the group of purely infinite series onto the monomial group and agrees with the real exponential and the infinitesimal Taylor exponential on the other two parts [2, 13]. The identity (15.2) fails at every purely infinite input with at least κ nonzero terms [06]. The logarithmic identity (15.3) is 07’s.

Corollary 15.4 (A scale-sized exponential discrepancy [06]). *The ratio in (15.1) is always a monomial. It is either exactly one, a positive infinitesimal, or larger than every positive real number. In particular*

$$H_j(\exp x) \sim \exp(H_j x) \iff H_j(\exp x) = \exp(H_j x),$$

where $y \sim w$ means $y/w \in 1 + \mathfrak{m}$ for positive y, w (06 writes $p \sim q$).

Proof. Both $J(p(x))$ and $H_j(p(x))$ are purely infinite. Their difference is therefore purely infinite, possibly zero. Gonshor exponentiation maps this additive group to the monomial group. A nonidentity positive monomial is either infinitesimal or infinitely large relative to the reals. Apply (15.1). $\square$

The two criteria of Corollary 5.6 and (15.2) are different. For example, h_κ has κ terms but is finite, so the Hahn lift fails to commute with ω^x at h_κ , while commuting with $\exp x$ there. For the purely infinite omnific integer

$$\zeta_\kappa = z_\kappa = \omega^\kappa h_\kappa = \sum_{\alpha < \kappa} \omega^{\kappa - \alpha},$$

both identities fail [06].

Corollary 15.5 (Exponential closure without logarithmic closure). *The fields $\mathcal{S}_{<\kappa}$ and C_j are closed under the Gonshor exponential but not under logarithms of positive elements. With this induced exponential, neither is an elementary substructure of $(\mathbf{No}, \exp)$, although both are elementary ordered subfields of $\mathbf{No}$.*

Proof. If $x \in \mathcal{S}_{<\kappa}$, then $p(x) \in \mathcal{S}_{<\kappa}$. By the preceding theorem and Lemma 15.1,

$$H_j(\exp x) = \exp(H_j(x)) = \exp(J(x)) = J(\exp x).$$

The equalizer theorem yields $\exp x \in \mathcal{S}_{<\kappa}$. The same closure for $C_j = J(\mathcal{S}_{<\kappa})$ follows from exponential transfer.

The purely infinite element z_κ of Section 11 has κ outer terms, so it is not in $\mathcal{S}_{<\kappa}$. But $\exp(z_\kappa)$ is a positive monomial and hence belongs to $\mathcal{S}_{<\kappa}$. Its unique logarithm in $\mathbf{No}$ is z_κ , which is absent. Similarly $J(\exp(z_\kappa))$ lies in C_j , whereas its logarithm $J(z_\kappa)$ does not, by injectivity of J and the description of C_j . Thus the sentence asserting that every positive element has an exponential preimage holds in $(\mathbf{No}, \exp)$ and fails in both smaller exponential structures. Their ordered-field elementarity follows from real closedness. $\square$

[merge] *Related (batch 32).* For $\mathcal{S}_{<\kappa}$ the first- κ report proves the first assertion without the embedding, for every uncountable κ , regular or singular, with the same witness z_κ (`fkc: sb: thm: expclosure`), and the failure of exponential elementarity with the same sentence (`fkc: sb: cor: expsep`). It also shows that no ordered-group isomorphism $(\mathcal{S}_{<\kappa}, +, <) \rightarrow (\mathcal{S}_{<\kappa}^{>0}, \cdot, <)$ exists at all, so no choice of exponential on $\mathcal{S}_{<\kappa}$ is surjective (`fkc: sb: thm: noexp`). The field C_j is not treated there.

Corollary 15.6 (One image is exponentially closed and the other is not). *The image $J(\mathbf{No})$ is closed under both exponential and logarithm. The image $H_j(\mathbf{No})$ is not closed under exponential.*

Proof. The first assertion follows immediately from Lemma 15.1. For the second, use the classical order-preserving bijection $g : \mathbf{No}^{>0} \rightarrow \mathbf{No}$ specified by

$$\exp(\omega^b) = \omega^{\omega^{g(b)}} \quad (b > 0).$$

The purely infinite exponential formula says that $\exp(\sum r_\alpha \omega^{b_\alpha}) = \omega^{\sum r_\alpha \omega^{g(b_\alpha)}}$ when all $b_\alpha > 0$; see [2, Theorem 10.13] and [3, Lemma 5.2, proof]. Define

$$p_\kappa = \sum_{\alpha < \kappa} \omega^{g^{-1}(-\alpha)}.$$

The exponents are strictly decreasing and positive, so this is a purely infinite normal form and $\exp(p_\kappa) = \omega^{h_\kappa}$. The definition of g , together with transfer of exponential and the omega-map, shows that $J(g(b)) = g(J(b))$ and hence that J commutes with g^{-1} . Since $-\alpha$ is fixed for $\alpha < \kappa$, all the displayed exponents are fixed. Therefore $H_j(p_\kappa) = p_\kappa$, so $p_\kappa \in H_j(\mathbf{No})$.

But $h_\kappa \notin J(\mathbf{No}) = \Gamma_j$ by Theorem 7.1. The monomial ω^{h_κ} consequently has its exponent outside Γ_j and is not in $H_j(\mathbf{No})$ by (4.1). This proves the failure of exponential closure. $\square$

Theorem 15.7 (Two canonical closures of the common field). *Among class subfields of $\mathbf{No}$ containing C_j , the least one closed under logarithms of positive elements is $J(\mathbf{No})$, and the least one closed under external set-indexed strong sums is $H_j(\mathbf{No})$. More explicitly, every element of $J(\mathbf{No})$ has the form*

$$\frac{\log m_1}{m_2}, \quad m_1, m_2 \in \{\omega^{J(a)} : a \in \mathbf{No}\} \subseteq C_j, \quad (15.5)$$

and every element of $H_j(\mathbf{No})$ is a strong sum of elements of C_j . These two closures are incomparable and intersect in exactly C_j . In contrast, both the logarithmic field closure and the strong-sum closure of the domain equalizer $\mathcal{S}_{<\kappa}$ are all of $\mathbf{No}$.

Proof. The field $J(\mathbf{No})$ contains C_j and is logarithmically closed by Lemma 15.1. Take $J(x) \in J(\mathbf{No})$. Choose $d > 0$ so that $z = \omega^d x$ is purely infinite, exactly as in Theorem 11.1. Then $\exp z$ is a monomial. Both $\exp z$ and ω^d have one-term support and hence lie in $\mathcal{S}_{<\kappa}$. Set $m_1 = J(\exp z)$ and $m_2 = J(\omega^d)$; they are monomials in C_j . Exponential transfer gives

$$J(x) = \frac{J(z)}{J(\omega^d)} = \frac{\log m_1}{m_2}.$$

Any logarithmically closed class subfield containing C_j must therefore contain every $J(x)$, proving minimality and (15.5).

The field $H_j(\mathbf{No})$ is strongly closed by Theorem 8.6. Every normal-form term $r\omega^{J(a)}$ in one of its members already lies in C_j , so strong closure of any class containing C_j forces that member to be included. This proves the other minimality assertion. Incomparability and the intersection were proved earlier.

For $\mathcal{S}_{<\kappa}$, the identical shift argument without J expresses every $x \in \mathbf{No}$ as a quotient of the logarithm of a monomial in $\mathcal{S}_{<\kappa}$ by another monomial in $\mathcal{S}_{<\kappa}$. Also every surreal is a strong sum of real multiples of monomials, all of which lie in $\mathcal{S}_{<\kappa}$. This proves both final assertions. The minimality statements are explicit universal properties; they do not require forming an intersection of a collection of all classes. $\square$

The two image fields are therefore canonically isomorphic as ordered valued fields over C_j , but their actual placement in $\mathbf{No}$ has different compatibility with the canonical exponential. Neither their shared residue field and value group nor their common monomials detect this difference. The support criterion (15.2) is different from the whole-support equalizer criterion: the long-support finite number h_κ lies in the former locus and not the latter.

16 Supercompactness as a seed-coefficient system

The first support threshold depends only on the critical point. Thus it cannot by itself distinguish a measurable embedding from a stronger embedding with the same critical point. A richer family of coefficient probes does retain the classical seed information used for supercompactness.

16.1 Coding an arbitrary index set [09]

The measure coefficient used the special seed $\kappa \in j(\kappa)$. A seed can instead be an arbitrary element of an embedded set. Let I be a nonempty set, fix a bijection $e : \theta \rightarrow I$ from an ordinal, and put

$$Y_{A,e} = \sum_{\substack{\alpha < \theta \\ e(\alpha) \in A}} \tau_\alpha \quad (A \subseteq I). \quad (16.1)$$

For $s \in j(I)$ let $\beta_s = (j(e))^{-1}(s) < j(\theta)$.

Theorem 16.1 (Seed coefficient formula [09]). *For every $A \subseteq I$,*

$$[\omega^{q\beta_s}] J(Y_{A,e}) = 1 \iff s \in j(A). \quad (16.2)$$

The coefficient is always zero or one. The resulting collection $U_s = \{A \subseteq I : s \in j(A)\}$ is a κ -complete ultrafilter on I . It may be principal for some choices of s .

Proof. Applying j to the defining mask gives

$$J(Y_{A,e}) = \sum_{\substack{\beta < j(\theta) \\ j(e)(\beta) \in j(A)}} \tau_\beta.$$

At the unique position β_s its coefficient is one precisely when $s \in j(A)$. Complements and finite intersections give an ultrafilter, and intersections indexed by a fixed ordinal below κ give κ -completeness, as in Theorem 10.1. If $s = j(a)$ for some $a \in I$, then the resulting ultrafilter is principal at a , explaining the qualification. $\square$

Changing the enumeration e merely changes which coefficient position carries the same seed. It does not change the derived ultrafilter. At the same time, a single derived ultrafilter need not determine the whole embedding j or its target M ; the theorem recovers exactly the ultrafilter data stated, not an arbitrary ambient embedding from one bit [09]. When s is not of the form $j(a)$, the index β_s is a missing-image index and the coefficient (16.2) is a coefficient of the defect $D_j(Y_{A,e})$, as in the next theorem.

16.2 The supercompact seed [07]

Assume $\lambda \geq \kappa$, $j(\kappa) > \lambda$, and M is closed under λ -sequences from V . Let

$$I = P_\kappa(\lambda) = \{a \subseteq \lambda : |a| < \kappa\}, \quad s = j \text{ `` } \lambda.$$

Closure gives $s \in M$, and in M it has size at most $\lambda < j(\kappa)$, so $s \in j(I)$. These are the standard hypotheses and seed for a λ -supercompactness measure [6]. Choose a bijection $e : \theta \rightarrow I$. Let $\beta_s < j(\theta)$ be the unique index satisfying

$$j(e)(\beta_s) = s. \tag{16.3}$$

For $A \subseteq I$, define the mask

$$m_A = \sum_{\substack{\alpha < \theta \\ e(\alpha) \in A}} \omega^{-\alpha}.$$

The construction depends on the enumeration e , which is explicitly part of the data.

Theorem 16.2 (Seed extraction). *The index β_s is a missing-image index: $\beta_s \notin j \text{ `` } \theta$. Moreover*

$$[\omega^{-\beta_s}] D_j(m_A) = \mathbf{1}_{\{s \in j(A)\}} \quad (A \subseteq I). \tag{16.4}$$

Consequently the seed coefficient recovers the fine normal κ -complete ultrafilter

$$U_s = \{A \subseteq P_\kappa(\lambda) : s \in j(A)\}.$$

Proof. If $\beta_s = j(\alpha)$, then $s = j(e(\alpha)) = j(a)$ for some $a \in P_\kappa(\lambda)$. Since $|a| < \kappa$, Lemma 2.5 gives $j(a) = j \text{ `` } a$ . But  $s = j \text{ `` } \lambda$, so injectivity of j implies $a = \lambda$, a contradiction. Thus β_s is missing from the pointwise index image.

The transformed mask has coefficient 1 at $-\beta_s$ exactly when $j(e)(\beta_s) = s$ belongs to $j(A)$. The companion has its exponents only at positions $-j(\alpha)$, so it contributes zero there. This gives (16.4).

Ultrafilter and κ -completeness follow by the same Boolean and short-index argument as in Theorem 10.1, with s replacing κ . For each $\xi < \lambda$, the seed contains $j(\xi)$, so

$$\{a \in P_\kappa(\lambda) : \xi \in a\} \in U_s;$$

this is fineness. If $A \in U_s$ and $f : A \rightarrow \lambda$ satisfies $f(a) \in a$, then $j(f)(s) \in s = j \text{ `` } \lambda$. Write $j(f)(s) = j(\xi)$ for some $\xi < \lambda$. The fiber where $f(a) = \xi$ then belongs to U_s , proving normality. $\square$

Fine normal κ -complete ultrafilters on $P_\kappa(\lambda)$ are the classical ultrafilter characterization of supercompactness, due to Solovay; see [6]. We have not discovered a new large-cardinal equivalence. The contribution is that the seed can be read as a coefficient of the same defect construction, and that its index is provably outside the termwise image. An intrinsic, enumeration-free coefficient geometry for these seed systems is a further research question, not a result claimed here.

16.3 Covering seeds and exact seeds [09]

For a cardinal $\lambda \geq \kappa$, an ultrafilter U on $P_\kappa(\lambda)$ is *fine* if $\{a : \xi \in a\} \in U$ for every $\xi < \lambda$. It is *normal* if, whenever $A \in U$ and $f : A \rightarrow \lambda$ satisfies $f(a) \in a$, there is a $\xi < \lambda$ such that $\{a \in A : f(a) = \xi\} \in U$. These are the standard ultrafilter notions [18, 19].

Proposition 16.3 (Covering seeds and exact seeds [09]). *Let $I = P_\kappa(\lambda)$ and suppose $s \in j(I)$.*

- (i) *If $j \text{ `` } \lambda \subseteq s$, then U_s is fine and κ -complete.*
- (ii) *If $s = j \text{ `` } \lambda$ belongs to M and is an element of $j(I)$, then U_s is normal and fine.*
- (iii) *For a fine ultrafilter on $P_\kappa(\lambda)$, the ultrapower seed represented by the identity map contains every $j(\xi)$ for $\xi < \lambda$. If the ultrafilter is normal, that seed is exactly $j \text{ `` } \lambda$.*

Proof. For each $\xi < \lambda$, let $A_\xi = \{a \in I : \xi \in a\}$. Its image is the set of $b \in j(I)$ containing $j(\xi)$. A covering seed belongs to each $j(A_\xi)$, proving fineness; completeness is Theorem 16.1.

For normality in (ii), apply j to $f(a) \in a$ on a U_s -large set. Then $j(f)(s) \in s = j \text{ `` } \lambda$. Thus $j(f)(s) = j(\xi)$ for some $\xi < \lambda$, and s lies in the image of the corresponding constant fiber. That fiber belongs to U_s .

For (iii), write the seed as $[\text{id}]_U$ before transitive collapse. Fineness says $[\text{const}_\xi]_U \in [\text{id}]_U$, which gives one inclusion after collapse. Conversely, any element of this seed is represented by a function f for which $f(a) \in a$ on a large set. Normality makes f constant there, so its collapsed class is some $j(\xi)$. This proves equality for a normal fine ultrafilter. $\square$

Part (ii) is the ultrafilter half of Theorem 16.2, proved there by the same argument; 09 derives the membership $s \in j(I)$ the same way: a supercompactness embedding with $j(\kappa) > \lambda$ and $M^\lambda \subseteq M$ has $j \text{ `` } \lambda \in M$, and in M this set has order type and cardinality $\lambda < j(\kappa)$, because its external increasing enumeration has order type λ and the order type of a well-ordered set in a transitive model is absolute.

The usual ultrafilter characterization of strong compactness is the existence, for every $\lambda \geq \kappa$, of a fine κ -complete ultrafilter on $P_\kappa(\lambda)$. Supercompactness has the corresponding normal-fine characterization [18]. Through Theorem 16.1, these existing characterizations become coefficient-selection principles on the specified omnific masks. The difference between a seed *covering* $j \text{ `` } \lambda$  and an available seed *equal* to  $j \text{ `` } \lambda$ is retained explicitly [09].

Warning 16.4 (Translation is not a new consistency equivalence [09]). The preceding paragraph still quantifies over set embeddings, seeds and subset codes. It is a numerical presentation of established large-cardinal mechanisms, not a characterization in the bare ordered-field language of **No** or the bare ring language of **Oz**. Eliminating that extra structure is a substantive further question.

17 Strong compactness as a support cover; supercompactness as exact descent [06]

17.1 The set-theoretic principles and their witnesses

Fix an uncountable cardinal κ and a cardinal $\lambda \geq \kappa$, and use the fine and normal ultrafilters on $P_\kappa(\lambda)$ of the previous section. To avoid ambiguity about local terminology, in this section the (κ, λ) *strong-compactness principle* means existence of a fine κ -complete ultrafilter on $P_\kappa(\lambda)$. The corresponding *supercompactness principle* requires the ultrafilter to be normal as well. Requiring the respective principles for every cardinal $\lambda \geq \kappa$ gives the usual global large cardinal properties. The proofs below also identify the usual sequence-closed elementary embedding characterization of λ -supercompactness. The seed construction is standard [6, 12]; 06 gives the argument to make the exact translation to surreal supports explicit. The letter W denotes a fine ultrafilter in this section only.

Lemma 17.1 (Cover and exact-seed characterizations [06]). *The (κ, λ) strong-compactness principle is equivalent to existence of a set-ultrapower embedding $j : V \rightarrow M$ of critical point κ and a set $s \in M$ such that*

$$j \text{ “ } \lambda \subseteq s \subseteq j(\lambda), \quad M \models |s| < j(\kappa). \quad (17.1)$$

Such an embedding necessarily has $j(\kappa) > \lambda$. The supercompactness principle is equivalent to existence of such a set-ultrapower embedding with $j(\kappa) > \lambda$ and $j \text{ “ } \lambda \in M$, or equivalently with ${}^\lambda M \cap V \subseteq M$.

Proof. Start with a fine κ -complete ultrafilter W on $P_\kappa(\lambda)$. Its ultrapower is well-founded: a putative infinite descending membership chain would, by countable completeness, yield a pointwise descending membership chain on a nonempty common measure-one set, contradicting Foundation. Take its transitive collapse. The seed $s = [\text{id}]_W$ is an element of $j(P_\kappa(\lambda))$, so it is a subset of $j(\lambda)$ of internal size $< j(\kappa)$. Fineness says that it contains every $j(\alpha)$ for $\alpha < \lambda$.

Completeness fixes all ordinals below κ : any function into such an ordinal is constant modulo W , since otherwise the intersection of the complements of all its fibers would be large and empty. Induction on the ordinal gives the assertion in the transitive collapse. On the other hand, if $j(\kappa) = \kappa$, then an internal enumeration of s of length $< \kappa$ would be an external enumeration of a set containing $\lambda \geq \kappa$ distinct elements, which is impossible. Hence the critical point is exactly κ . More generally, an internal cardinal bound $\nu < j(\kappa)$ for s gives an external injection of λ into ν . Since λ is an ambient cardinal, $\lambda \leq \nu < j(\kappa)$.

Conversely, suppose an embedding and s as in (17.1) are given, and let $U_s = \{X \subseteq P_\kappa(\lambda) : s \in j(X)\}$ (06’s W_s). Because $s \in j(P_\kappa(\lambda))$, this is an ultrafilter. Preservation of intersections indexed below the critical point gives κ -completeness. Containment of $j \text{ “ } \lambda$ in s gives fineness. This proves the cover characterization.

If the initial fine ultrafilter is normal, every member of $[\text{id}]_W$ is represented by a selector $f(a) \in a$ on a large set. Normality makes it equal to a constant $j(\alpha)$. Fineness gives the converse inclusion, so $[\text{id}]_W = j \text{ “ } \lambda \in M$. Conversely, if $s = j \text{ “ } \lambda$ and $j(\kappa) > \lambda$, then its increasing enumeration in M has length $\lambda < j(\kappa)$, so it satisfies (17.1). For a selector f , the value $j(f)(s) \in s$ equals $j(\alpha)$ for some $\alpha < \lambda$. The corresponding fiber belongs to U_s , proving normality. Finally apply Lemma 9.3 to obtain the sequence-closure formulation. A sequence-closed embedding not initially presented as an ultrapower also yields this normal ultrafilter by the same exact-seed argument, and hence yields a set-ultrapower witness. $\square$

17.2 Binary normal forms and coefficientwise covers

For $t \subseteq \theta$ a set of ordinals, put

$$B(t) = \sum_{\beta \in t} \omega^{-\beta},$$

with its indices in increasing order. A *binary series below* θ has this form. On this family define

$$B(t) \preccurlyeq_{\text{coeff}} B(t') \iff t \subseteq t'.$$

Equivalently every coefficient of the difference $B(t') - B(t)$ is nonnegative. This coefficientwise order is not the ordinary numerical order on $\mathbf{No}$. In particular a small leading coefficient inequality cannot replace a support inclusion. For the canonical detector h_λ of (7.1),

$$J(h_\lambda) = h_{j(\lambda)}, \quad H_j(h_\lambda) = B(j \text{ `` } \lambda). \quad (17.2)$$

The second expression is always defined externally, even when it fails to belong to M .

Theorem 17.2 (Support-cover characterization of strong compactness [06]). *The (κ, λ) strong-compactness principle holds if and only if there is a set-ultrapower embedding $j : V \rightarrow M$ of critical point κ and a binary series $\widehat{B} \in \mathbf{No}^M$ below $j(\lambda)$ such that*

$$H_j(h_\lambda) \preccurlyeq_{\text{coeff}} \widehat{B} \preccurlyeq_{\text{coeff}} J(h_\lambda), \quad M \models |\text{supp}(\widehat{B})| < j(\kappa). \quad (17.3)$$

Proof. If s is the cover seed in Lemma 17.1, let $\widehat{B} = B(s)$, computed in M . Absoluteness identifies it with the ambient binary series, and the containments of sets in (17.1) become the coefficientwise inequalities in (17.3). The support map $\beta \mapsto -\beta$ is a bijection in M , so the internal cardinal bounds are identical.

Conversely, take the normal form of $\widehat{B}$ inside M and negate its exponents to obtain $s \subseteq j(\lambda)$ in M . The inequalities in (17.3) say exactly that $j \text{ `` } \lambda \subseteq s$. The internal bound on the support gives the bound on s . Apply Lemma 17.1. $\square$

Theorem 17.3 (Single-number characterization of supercompactness [06]). *The (κ, λ) supercompactness principle holds if and only if there is a set-ultrapower embedding $j : V \rightarrow M$ with $\text{crit}(j) = \kappa$, $j(\kappa) > \lambda$, and*

$$H_j(h_\lambda) \in \mathbf{No}^M. \quad (17.4)$$

For this equivalence h_λ may be replaced by any fixed surreal of support cardinal λ . It may in particular be replaced by the omnific integer

$$\zeta_\lambda = \omega^\lambda h_\lambda = \sum_{\alpha < \lambda} \omega^{\lambda - \alpha}. \quad (17.5)$$

Proof. By Theorem 9.1, (17.4) is exactly the assertion $j \text{ `` } \lambda \in M$. Now use Lemma 17.1. The landing criterion depends only on support cardinality, proving the replacement claim. Every exponent in (17.5) is positive, so ζ_λ is omnific. Alternatively, use the identity $H_j(\zeta_\lambda) = \omega^{j(\lambda)} H_j(h_\lambda)$ and the fact that the nonzero multiplying monomial and its inverse both belong to $\mathbf{No}^M$. $\square$

Corollary 17.4 (Omnific support covers [06]). *The criterion (17.3) is equivalent to existence of a purely infinite binary-coefficient omnific integer $\widehat{B}^+ \in \mathbf{Oz}^M$ whose support lies among the exponents $\{j(\lambda) - \beta : \beta < j(\lambda)\}$ and such that*

$$H_j(\zeta_\lambda) \preccurlyeq_{\text{coeff}} \widehat{B}^+ \preccurlyeq_{\text{coeff}} J(\zeta_\lambda), \quad M \models |\text{supp}(\widehat{B}^+)| < j(\kappa).$$

Proof. Multiply all three series in (17.3) by $\omega^{j(\lambda)}$. This translates the support bijectively, preserves coefficientwise inclusion and internal cardinality, and makes every exponent positive. Dividing by that monomial reverses the construction. $\square$

[merge] In this corollary $\preccurlyeq_{\text{coeff}}$ is read on the shifted binary series, whose exponents are $j(\lambda) - \beta$; 06 uses the same symbol.

17.3 What the distinction says, and what it does not say

Strong compactness provides an internally short support *containing* the transported old indices. Supercompactness makes the exact set of old indices an internal support. The cover may include extra monomials; exact descent permits none. This is the mathematical distinction behind the two criteria, rather than a change of terminology for numerical approximation.

At $\lambda = \kappa$ a normal measure supplies both tests. At $\lambda = \kappa^+$, an ultrapower indexed just by κ cannot supply even the cover, by Lemma 9.5. Stronger witnesses, when they exist, must use sufficiently large index sets. One must not conclude that a measurable cardinal fails a stronger large-cardinal property merely because its particular normal-measure ultrapower has the short landing spectrum from Theorem 9.7.

These are equivalences with classical principles, not new consistency proofs. For example, the global existence of exact-descent witnesses for every λ is equivalent to supercompactness, and hence has exactly that hypothesis and relative consistency strength. No assertion that any of these large cardinals exists is made in ZFC alone [06]. For omnific integers the statements are not merely about an ambient field: the measure and compactness detectors can themselves be taken to be omnific integers [06]. [merge] 07's Theorem 16.2 reads the same normal seed as a defect coefficient; 06's theorems read it as membership of one value in M ; 09's Proposition 16.3 records the covering and exact seeds as ultrafilters. All three are translations of the classical fine and normal-fine characterizations.

18 Weak compactness as an omnific prefix principle [09]

There is a second large-cardinal mechanism that does not start with a measure: compactness of trees. 09 gives its explicit normal-form version, while retaining the exact topology and the known set-theoretic content.

18.1 Length-preserving binary codes

For $\sigma \in 2^\alpha$ put

$$w_\sigma = \sum_{\xi < \alpha} (1 + \sigma(\xi)) \tau_\xi. \quad (18.1)$$

Every coefficient is one or two, so the length of the normal form is exactly α . This convention is important. If one used coefficients zero and one without also recording the length, appending a zero bit would not change the number and the coding of tree nodes would not be injective. For $\beta \leq \alpha$, the normal-form truncation of w_σ to length β is $w_{\sigma \upharpoonright \beta}$. Thus the prefix order of binary strings is literally normal-form truncation on these omnific codes.

Definition 18.1 (Omnific prefix compactness at κ [09]). Let κ be strongly inaccessible. The principle $\text{OPC}(\kappa)$ says: whenever $\mathcal{T}$ is a set of numbers of the form w_σ with $\sigma \in 2^{<\kappa}$, is closed under normal-form truncation, and has a node of every length $\alpha < \kappa$, there is a number

$$w_b = \sum_{\xi < \kappa} (1 + b(\xi)) \tau_\xi, \quad b \in 2^\kappa,$$

all of whose proper normal-form truncations belong to $\mathcal{T}$.

Theorem 18.2 (A transported weak-compactness criterion [09]). *For a strongly inaccessible cardinal κ ,*

$$\text{OPC}(\kappa) \iff \kappa \text{ has the tree property} \iff \kappa \text{ is weakly compact.}$$

Proof. The last equivalence is the standard inaccessible-cardinal tree characterization of weak compactness [18]. For the first, transport $\mathcal{T}$ through (18.1). At level $\alpha < \kappa$ there are at most $2^{|\alpha|} < \kappa$ nodes. Hence it is a κ -tree, and the tree property yields the desired branch.

Conversely, binary prefix trees suffice; this avoids an implicit assumption about unique limit nodes in an arbitrary tree. Let $\mathcal{R}$ be a κ -tree, with each level $\mathcal{R}_\alpha$ of size less than κ . For each α choose a nonzero ordinal $\delta_\alpha < \kappa$ and an injection $\iota_\alpha : \mathcal{R}_\alpha \rightarrow 2^{\delta_\alpha}$. Such a choice is possible, for example by taking δ_α large enough to code a well-ordering of that level. Define $\Delta_\alpha = \sum_{\beta \leq \alpha} \delta_\beta$ by ordinal addition. Regularity gives $\Delta_\alpha < \kappa$, and these lengths are cofinal in κ .

For $t \in \mathcal{R}_\alpha$, concatenate the labels $\iota_\beta(t \upharpoonright \beta)$ of its predecessors at all levels $\beta \leq \alpha$, including the label of t itself. Here $t \upharpoonright \alpha = t$, and each earlier restriction denotes the unique predecessor at that height. Take the binary tree of all prefixes of all such concatenated codes. Its height is κ , and each level has fewer than κ members by the strong-limit property. Applying OPC(κ) through (18.1) yields a binary branch.

At every completed block Δ_α , that branch determines a unique node of $\mathcal{R}_\alpha$, since ι_α is injective. Any sufficiently long code witnessing the branch prefix confirms that the nodes determined at two completed blocks are in the predecessor relation. They therefore form a cofinal branch of $\mathcal{R}$, as required. $\square$

This is an exact coding of a classical tree principle. It is not a new proof of the relative consistency of weak compactness. Its value is to show which additional organization of omnific numbers carries the compactness information [09].

18.2 The branch is not an order-topological limit

The topology naturally associated with (18.1) is the prefix or truncation topology: a basic neighborhood fixes an initial normal-form segment. It is not the order topology inherited from **No**. Indeed, **Oz** is discrete in the inherited order topology because

$$(z - 1/2, z + 1/2) \cap \mathbf{Oz} = \{z\} \quad (z \in \mathbf{Oz}).$$

Thus the branch theorem cannot be rephrased as ordinary convergence of the omnific partial sums. Moreover every proper prefix has support smaller than κ , whereas the completed branch has support exactly κ . Demanding a branch *inside* $\mathcal{S}_{<\kappa}$ would fail even for the full binary tree, regardless of weak compactness.

In generalized analysis, Carl–Galeotti–Löwe relate a specifically *weak* κ -Bolzano–Weierstrass principle to the tree property for suitable ordered fields, including their surreal-based field $\mathbb{R}_\kappa$ [21]. Their hypotheses and notion of weak convergence must not be replaced by the usual real Bolzano–Weierstrass theorem. The prefix principle above makes a related distinction in a different, explicitly combinatorial setting [09].

19 What large cardinals change, and what they do not

19.1 No change to finite algebraic truth

Both J and H_j are elementary ordered-field embeddings. Their complex lifts are elementary embeddings of algebraically closed fields. Consequently they preserve every first-order formula in those respective field languages. The distinction made in this article is in additional structure: normal forms, external supports, indexed strong sums, and their coefficient probes. It would be wrong to describe (7.4) as a failure of an ordinary field axiom.

Likewise, a measurable cardinal does not alter the formula for adding two specified old surreal sign sequences. It supplies a nontrivial elementary embedding of the surrounding set universe, and that map moves certain *entire set-indexed presentations*. The finite identities still transfer; the pointwise index set does not exhaust the transferred one.

19.2 Why there is no Kunen contradiction

The set-theoretic map is $j : V \rightarrow M$, not an elementary self-embedding $V \rightarrow V$. After forming surreal numbers in M , we compose with the identity inclusion into the ambient $\mathbf{No}^V$. That inclusion is elementary in the *ordered-field reduct*, by real closedness. It is not being claimed elementary in a structure that recovers all of the surrounding set theory from birthdays or other codes.

This matters when comparing with the repository’s birthday-enriched structures [10]. An order-field embedding can preserve specific operations and a birthday identity on its image without making the full ambient birthday expansion an elementary extension. The extra quantifiers of that expansion can range over new sign codes. Accordingly, our construction is not an exception to Kunen-type obstructions for a set-theoretically faithful elementary self-embedding.

09 makes the same point [09]. By quantifier elimination for real closed fields, both J and H_j are elementary in the ordered-field language; similarly their complexifications are elementary in the language of algebraically closed fields of characteristic zero [20]. For these definable class structures, the assertion is interpreted formula by formula, or syntactically via quantifier elimination; it is not an assertion that a set-sized satisfaction relation for all classes has been constructed. The birthday function, the omega-map, normal-form truncation and an entire summable family are additional data, and pure field elementarity does not force their preservation. In particular, composing $j : V \rightarrow M$ with the inclusion $M \subseteq V$ does not yield an elementary set-theoretic self-embedding of V . The two field embeddings do not evade the usual restrictions on such self-embeddings. The enriched structures considered in the repository and the announced Chen–Hamkins–Yang work concern a stronger language [10, 22].

[merge] The birthday-cutoffs report proves the relevant obstruction for the full birthday-expanded class: every class elementary self-embedding of it is the identity, by transfer of Kunen’s inconsistency (`hset : thm : kunen` in `docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets`). Its correspondence between elementary embeddings of H_κ and of the birthday-bounded surreal structures (`hset : thm : embeddings`) acts on sign sequences exactly as J does, and the remark following it warns, as Lemma 2.5 does here, that $j(A)$ must not be replaced by $j \text{ “ } A$: the transported sign sequence has domain $j(b(x))$ and may have coordinates outside $j \text{ “ } b(x)$. Neither result is used above, and neither implies the defect formula.

19.3 A hierarchy of information

The first failure cardinal of positive strong summation is κ . The first missing coefficient on the masks h_A recovers the normal measure U_j . At a supercompact seed, the wider mask family m_A recovers the associated fine normal measure. These are three different levels of data:

$$\text{critical point} \quad \longleftarrow \quad \text{normal-measure coefficient functional} \quad \longleftarrow \quad \text{seed-coefficient systems.}$$

The arrows mean forgetting information, not an implication that every normal measure extends to every seed system. In particular, two distinct normal measures can have the same critical cardinal and the same equalizer field $\mathcal{S}_{<\kappa}$, while their defect functionals differ by Corollary 10.4.

19.4 Choice and the scope of set formation

The proof does use Choice, already present in ZFC, to enumerate an arbitrary set in Lemma 2.5 and to use the conventional measure characterizations. It uses Replacement with the embedding parameter to form pointwise images and inverse images of *sets*. It does not choose a basis for the entire proper class **No**, does not appeal to a class version of Zorn’s lemma, and does not form the collection of all class endomorphisms as a set or class of its own. A definable normal ultrapower provides all the specific maps needed. [merge] The merged material keeps this scope. 09’s Theorem 13.5 uses Zorn’s lemma only on subsets of the set K_κ ; 06’s Lemma 9.3 uses ordinary Choice for a set of representatives.

19.5 What is visible at each level of structure [09]

Proposition 19.1 (Pure field sentences do not detect large cardinals [09]). *Every parameter-free sentence about **No** in the ordered-field language has the truth value prescribed by the complete theory of real closed fields. Every parameter-free field sentence about **SC** has the truth value prescribed by algebraically closed fields of characteristic zero. Accordingly, an additional large-cardinal axiom cannot change any of these truth values, provided the same ordinary axioms of surreal arithmetic are being used.*

Proof. Real closed fields and algebraically closed fields of fixed characteristic have complete theories, as follows from quantifier elimination [20]. The canonical surreal structures satisfy the respective theories. The same syntactic decision of each sentence therefore applies in every ambient model of the stated foundations. $\square$

This standard observation rules out an overly simple research program: one cannot discover a parameter-free polynomial-field sentence whose truth in the surreals switches exactly when a measurable cardinal exists. The positive theorems instead concern class embeddings, normal-form coefficients, families of prescribed cardinality and tree branches [09].

Data retained	Information obtained	Important qualification
Pure ordered field	Real closed field theory	No large-cardinal detection by parameter-free sentences
Normal forms and a support cutoff	$\mathcal{S}_{<\kappa}$ and long-support independence	Available for uncountable κ in ZFC
A measurable-cardinal embedding	Exact equalizer and additional-term formula	Conditional on the embedding’s set-theoretic origin
Omnific masks and a coefficient seed	A normal or derived ultrafilter	Uses all subset codes and coefficient operations
Covering or exact seeds on $P_\kappa(\lambda)$	Fine or normal-fine measures	Standard strong/supercompactness mechanisms
Prefix-coded omnific trees	Weak compactness at an inaccessible	Truncation topology, not the inherited order topology

(09’s table.) A birthday predicate is another way to recover substantial set-theoretic information, as the repository and the Chen–Hamkins–Yang announcement emphasize [10, 22]. 09 does not need to resolve the exact expressive strength of that predicate. Its main comparison uses a more concrete invariant: the set of additional indices in one transformed normal form.

Why this is not simply a change of notation [09]. Several individual ingredients are standard. Nevertheless the exact equalizer puts them together in a restrictive way. The two maps have identical actions on every monomial, identical leading coefficients, the same induced action on values, and infinitesimally close relative outputs. Both are elementary as pure ordered-field embeddings and both preserve the omnific ring. Yet their agreement locus is a sharply defined, proper-class real closed subfield, and they disagree on an explicitly maximal-cardinality independent family inside one fixed set-sized Hahn field. The discrepancy is not an arbitrary error term. Its support is the exact complement $j(\theta) \setminus j''\theta$ in an internally transformed ordinal index set. Thus the theorem converts a specific set-theoretic failure of pointwise surjectivity into a full normal-form identity, which contains more information than preservation or failure of one infinite sum alone.

19.6 What has actually been established [06]

The central conclusions are exact rather than asymptotic. The first comparison threshold is strict support size $< \kappa$; the normal-measure landing threshold is nonstrict support size $\leq \kappa$. The defect is the actual series on the omitted indices, not an error estimate. Every nonzero coefficient survives. The landing criterion holds for each individual input and is insensitive to coefficients. These features distinguish the results from a generic construction of an embedding or a general assertion that large cardinals produce nonstandard algebraic structures.

The exponential theorem identifies a different threshold, on only the positive-exponent part. This supplies a concrete separation between preserving all strong sums and preserving global exponential structure: the Hahn lift has the first property and fails the second on exactly the specified inputs. The set-transport map has the opposite global pattern: it preserves Gonshor exponentiation, but its strong summation law stops at the critical cardinal.

19.7 Hypotheses which cannot be silently removed [06]

The target must be transitive and the canonical constructions must be identified through the absoluteness interface. The seed–closure converse uses a *set-ultrapower* presentation; it was not proved for an arbitrary elementary embedding into an arbitrary inner model. The no-landing result above κ uses an index set of cardinality κ , not merely that the critical point is κ .

The support sizes in the comparison theorems are external cardinalities. The bound in the compactness cover theorem is an internal cardinal bound in M . The parameter λ in the compactness theorems is an ambient cardinal. These distinctions prevent several tempting but incorrect shortcuts in the proofs.

No automorphism classification, Kunen inconsistency result, new large-cardinal consistency theorem, or first-order characterization in the pure surreal field is established. In particular, one should not infer a large cardinal from the existence of an arbitrary non-strong field map; such behavior already has constructions unrelated to the universe embeddings used here. [merge] *Re-scoped in batch 32*. For maps that preserve every countable strong sum the inference is valid: by Theorem 24.1 the first failure of a countably strong $\mathbb{R}$ -linear map is a measurable cardinal. The known construction without large cardinals, the Kaplan–Krapp–Serra automorphism, fails a countable sum (Remark 26.3).

20 Position relative to the literature and the supplied repository

20.1 Classical ingredients and proposed contributions

The following separation is part of the claim, not merely an editorial qualification. Conway normal forms, the real closedness of $\mathbf{No}$, the algebraic closedness of $\mathbf{No}(i)$, Hahn reindexing, and Gonshor’s exponential are established background. Normal measures derived from embeddings, the ultrapower construction, and the fine-normal-measure characterization of supercompactness are also established background. The article uses these results to obtain an explicit comparison inside one ambient surreal field.

Item	Status in this article	Relevant precedent
Conway normal forms and strong summation	Classical input; conventions and absoluteness requirements made explicit.	Conway, Gonshor, van den Dries–Ehrlich, Ehrlich [1, 2, 3, 4].
Monomial reindexing	Classical Hahn construction; elementary proof supplied.	General Hahn-field theory and the strong maps studied in [5].
Critical-index defect	Candidate original exact formula, including its first term and exclusion of image exponents.	Standard distinction $j(I) \neq j \text{ “ } I$ supplies the set-theoretic mechanism.
Equalizer and intersection	Candidate original package: exact short-support equalizer, common-image formula, incomparability, and coefficient projection.	Real-closed-field facts are classical; the present pair is constructed from the embedding.
Strong-sum escape	Candidate original intrinsic closure threshold for $J(\mathbf{No})$, not only a non-commutation example.	Strong summability itself is classical.
Measure coefficient	Candidate original arithmetic realization. The derived ultrafilter and its properties are not new.	Normal ultrapowers and seed measures [6].
Mask criterion	Classical ultrafilter criterion transported to an explicitly specified Hadamard algebra.	Direct-product homomorphisms and ultrafilters [7].
Exponential boundary	Candidate original compatibility locus for this companion and its two-closure description.	Gonshor’s exponential normal-form formula [2, 3].
Omnific and surcomplex transfer	Consequences proved here from the real construction; no separate large-cardinal assumption.	Canonical integer parts and quadratic extensions are classical.

The phrase “candidate original” means that this formulation and proof package was not located in the targeted material inspected for this article. It does not assert that no specialist has previously observed some or all of the consequences. In particular, elementary embeddings and Hahn series are mature subjects; the mechanism is simple once the correct two images are compared.

06 draws its own line between established ingredients and proposed contributions [06]:

Ingredient or statement	Status in manuscript 06
Conway normal forms; Hahn arithmetic; infinitesimal formal evaluation	Classical prerequisites, principally [1, 2].
Termwise exponent lift; strong versus ordinary additivity	Established machinery; compare [5].
Ultrafilter seeds; fine and normal ultrafilters; seed-closure lemma	Classical set theory, with proofs included where used [6, 12].
Exact equalizer and missing-index defect; precise exponential ratio	Proposed comparison results proved there.
Universal individual-input landing criterion and normal-measure trichotomy	Principal proposed contributions, derived from the preceding machinery.
Coefficient recovery of the normal measure; omnific witness versions	Proposed explicit constructions, not a new definition of measurability.
Support covers and exact descent for compactness principles	Exact reformulations; no extra consistency-strength claim.

Its literature check was targeted, not exhaustive. It included the primary sources in the bibliography and the repository's relevant indices and scope statements. A proof being short once two theories are placed together does not establish priority. Conversely, a result can be a useful new formulation without settling a named conjecture [06]. 06 adds that Kaplan–Krapp–Serra already construct a non-strongly additive automorphism, so mere existence of such a map is not a contribution; ultrapower seeds, normal measures and sequence-closure criteria are established set theory [6, 12]; the novelty 06 proposes lies in the exact *interaction* of these constructions, especially the universal landing criterion for each individual normal form.

09's comparison [09]. Hahn lifting and strong additivity are established subjects. Kuhlmann and Serra study coefficient and exponent lifts and strongly additive Hahn-field automorphisms [16]. Kaplan, Krapp and Serra apply related ideas to the surreals and distinguish strong additivity from preservation of additional exponential structure [5]. The maps here are generally *embeddings*, not automorphisms; 09's proposed addition is the exact critical-point equalizer and its term-by-term discrepancy formula. The field $\mathcal{S}_{<\kappa}$ belongs to the cardinal-bounded Hahn-field tradition [15]; 09 does not claim the short-support field construction as new, and its proof includes singular uncountable cardinals and separates this cardinal condition from birthday cutoffs, which are studied classically in [3]. Connections between large cardinals and surreal-based generalized analysis already occur in Carl–Galeotti–Löwe [21]; their weak Bolzano–Weierstrass principle is important precedent for Section 18, and no claim to have initiated that connection is made. 09's dependency ledger marks the exact discrepancy and the almost-disjoint independence theorem as its main proposed contributions; the absoluteness lemma, the short-support field and the strong lift as standard constructions with proofs included; the normal and seed coefficient formulas as explicit coding results with the seed theory credited; the numerical measurability criterion as an infinitary criterion, not a bare-field equivalence; and the strong compactness, supercompactness and weak compactness statements as established set theory transported to the codes.

20.2 The repository snapshot actually inspected

The repository was read at commit

0865f043aec113c14c69ef45006bbc7546a4e75a.

The targeted inspection comprised the root `README.md`, the `docs/README.md` report index, the READMEs for *Surreal Fields Across Set-Theoretic Universes* and *Birthday Cutoffs and Hereditary Sets*, and the former article’s LaTeX preamble [8, 9, 10]. The style here follows that report’s restrained theorem-centered layout. This was not a line-by-line audit of the whole repository or its formalization ledger.

The cross-universe report describes omitted cuts, fresh sign sequences, external saturation, first new birthdays, and Prikry examples. The birthday report describes hereditary-set interpretations and the recovery of set-theoretic information from birthday-enriched structures. Its README explicitly attributes the announced full-class bi-interpretation to Chen–Hamkins–Yang. Those earlier themes are relevant context but are not used as unproved premises of our theorems.

The present article takes a different route. It does not add birthday coding to recover the entire set universe, and it does not change the universe by forcing. It compares the pointwise elementary action and its strong monomial extension inside a fixed universe. The output is an arithmetic defect whose support and coefficients can be calculated exactly. The constant-term discussion of omnific integers is also kept separate from the repository’s proposed set-sized quotient theorems [11].

The root README reports existing formal Hahn infrastructure, including summable families and exponent-embedding maps. That makes a substantial part of the algebraic proof a plausible formalization target. It does *not* mean the large-cardinal construction, the absoluteness lemma, or any theorem specific to this article is already checked in Lean. The repository itself warns that research reports, finite tests, and formal proof coverage are different kinds of evidence.

06 and 09 inspected the same snapshot [06, 09]. 06 read the root README, the documentation index, directory trees, and the READMEs of the birthday-cutoff and universes reports, and records the identifier as a “tree object”, not asserted to be a commit (it is in fact the commit `0865f04`, “Merge reviewed documentation after angular multiplicity validation”; Appendix D). 09 read the main documentation and the introductions and scope statements of the universes, birthday-cutoffs and bounded-support reports. Both describe those reports as 06 and 07 do: birthday interpretations, elementary-embedding transfer and forcing comparisons belong to the project and are not advertised as new; none of the new proofs requires an unreviewed repository theorem. 09 adds that the repository’s *Bounded-Support Hahn Arithmetic* concerns support bounded in the *order* of a specified exponent group [23], while its independence theorem concerns support bounded in *cardinality*, with a coefficient field containing short-support surreals with arbitrary exponents outside the selected set-sized Hahn field; these are genuinely different base fields. 09 also retains the repository’s separation of written arguments, finite tests and checked Lean declarations. 06’s Remark after Theorem 11.4 keeps its results apart from the repository’s work on omnific automorphisms and set-sized quotients.

20.3 The collection at the placement commit [merge]

The sources’ repository statements were checked at the placement commit `21375f8` and at the time of writing. The reports they cite are unchanged since the snapshot, except the set-sized-quotients report, which has since been extended (07 cites only its catalogue entry). The statements stand: the universes report proves absoluteness of the arithmetic and of cuts (`univ:prop:absolute`), not of normal forms; the bounded-support report (`bst:`) bounds supports in order, not in cardinality; the root README describes Lean infrastructure in which strictly increasing additive exponent embeddings preserve Hahn sums, and no statement of this report is formalized. The relations to the neighbouring reports are:

- `docs/foundations-and-computation/definable-surreals-and-omnific-integers (dsn:)`. Lemma 2.6 supplies the omega-map and normal-form absoluteness that its `dsn:q:support` leaves unproved, for same-ordinal inner models with the same reals (Remark 2.9); that report now carries a reciprocal remark. The question itself (classify the intermediate support fields) stays open.

- `docs/surreal/omnific-preserving-automorphisms (opa:)`. J gives a negative answer, under a measurable cardinal, to the first two clauses of `opa:as:q:embeddingstrong` (Remark 11.6); that report records this, writing J as $\hat{J}$. [merge] Batch 32: Theorem 26.2 shows that without measurable cardinals a coefficient-fixing counterexample to those clauses must fail a countable sum, and Theorem 26.4 gives the explicit unrepresentable target test ω^κ for `opa:as:q:targettests`, whose conditional negative answer already followed from `opa:as:thm:adjoint` and the non-strongness of J .
- `docs/surcomplex/first-kappa-coefficients (fk:;)` written concurrently in batch 32, now cited by label). It studies Hahn fields of series with fewer than κ nonzero terms; over all surreal monomials that field is $\mathcal{S}_{<\kappa}$ here. By Theorem 24.4 a $<\kappa$ -strong linear map agrees with its canonical strong part on $\mathcal{S}_{<\kappa}$. Its Part II proves, for $\mathcal{S}_{<\kappa}$ and without large cardinals, the results listed after Proposition 3.5 and Corollary 15.5.
- `docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (hset:)`. Its embedding correspondence `hset:thm:embeddings` and Kunen transfer `hset:thm:kunen` are the birthday-enriched side of Section 19; its remark on $j(A)$ versus $j \text{ “ } A$ is the same distinction as Lemma 2.5.
- `docs/foundations-and-computation/surreal-fields-across-universes (univ:)`. Its `univ:prop:absolute` is the arithmetic part of Lemma 2.6. Its forcing and saturation themes are the subject of Questions 21.11 and 21.20; none of its named questions is answered here.
- `docs/surreal/transcendence-over-bounded-support (bst:)`. Maximal transcendence over order-bounded supports there; Theorem 13.5 is over cardinality-bounded supports. Neither implies the other. [merge] Batch 32: its Galois part asks (`bst:gr:q:cardinal`) which of its constructions survive in fields of Hahn series with fewer than κ terms, at regular and singular κ . Theorem 3.1 supplies the ambient real closed field $\mathcal{S}_{<\kappa}$ for every uncountable κ , singular included; the Galois constructions are not treated here, and the question is not answered.
- `docs/surreal/set-sized-quotients-of-omnific-integers (osq:)`. The constant-term discussion of Section 11 is kept apart from its quotient theorems, as all three sources say.
- `docs/surreal/independent-surreal-copies (isc:)`, written concurrently as part of batch 31 (781b19e) and now cited by label. It identifies H_j as its strong exponent lift for the exponent map J (`isc:thm:lift`), and relates its linear-disjointness theorem for full Hahn fields (`isc:thm:sliceddisjoint`) to Question 21.6 without answering it (see the note after that question): $J(\mathbf{No})$ is not a full Hahn field (Theorem 8.6), and C_j is a proper class. The question stays open here.

20.4 Relation to current automorphism work

Kaplan–Krapp–Serra [5] provide a contemporary framework for strongly additive maps, monomial actions, valuation structure, and additional surreal operations. We use their definitions and general Hahn context, not a particular non-strong automorphism construction from that paper. Our maps are nonsurjective embeddings, and their valuations are transported by J rather than fixed pointwise. Results that assume bijectivity or pointwise preservation of the value group must not be applied to this pair without checking those hypotheses.

No claim is made to settle one of that paper’s numbered questions. Rather, our construction answers a new precisely stated family of structural questions under an explicit large-cardinal hypothesis. A specialist comparison with the unpublished details of related projects would be appropriate before submitting a priority claim. [merge] Manuscript 10’s lower bound is consistent with the non-strong automorphism of

[5, Example 4.4]: that automorphism fixes $\mathbb{R}$ and every monomial and fails a countable sum, so it is not countably strong (Remark 26.3).

21 Further research questions and concrete next steps

The following questions are proposed by this article. They are not presented as an established list of famous open problems. Each isolates something not proved above and indicates what a meaningful answer would require.

Question 21.1 (Optimal consistency strength). Suppose two ordered-field embeddings $P, Q : \mathbf{No} \rightarrow \mathbf{No}$ fix $\mathbb{R}$, agree on all ordinals and monomials, Q is strongly linear, and their equalizer and image intersection satisfy the formulas of Theorem 6.1 for an uncountable κ . Suppose also that P fixes every ordinal below κ , moves κ , and commutes with the omega-map. Must κ be measurable, or can such a pair exist without an ambient elementary embedding of set theory?

The obstruction to a converse is precise: the first defect coefficient is not automatically a character for the Hadamard product merely because P is multiplicative for ordinary surreal multiplication. Theorem 10.9 solves the converse only when the coefficient-algebra structure and its small-union law are available. [merge] *Status (batch 32): partly answered.* If P is in addition countably strong, then $Q = P_{\text{at}}$ and $\text{fail}(P)$ is a measurable cardinal at most κ ; if P is $<\kappa$ -strong, κ itself is measurable (Proposition 26.7 (a), from manuscript 10's Theorem 24.1). So for countably strong P such a pair cannot exist without a measurable cardinal, hence without an ultrapower embedding of set theory. Whether κ itself must be measurable for countably strong P , and the case of a P failing a countable sum, stay open.

Question 21.2 (Classification of strong companions). For an arbitrary real-fixing omega-map-compatible ordered-field embedding P , classify the equalizer of P with the Hahn reindexing map T_P . Which equalizers are cardinal support bounds, and which depend on the shape, cofinality, or iterated structure of the exponent support?

The construction T_P is always available by Lemma 4.1. What is special to the present large-cardinal setting is the exact missing-index formula and the absence of cancellation in it. [merge] *Status (batch 32): partial progress.* For countably strong P , T_P is the canonical strong part P_{at} , and either $P = T_P$ or the equalizer contains $\mathcal{S}_{<\text{fail}(P)}$ with $\text{fail}(P)$ measurable (Proposition 26.7 (b)); without measurable cardinals every countably strong such P equals T_P . The classification itself stays open.

Question 21.3 (Recovering an embedding from fewer coefficient probes). What is the least family of masks and coefficient positions that determines j on a specified hereditary-size universe H_λ ? Can normal-form probes of bounded complexity recover more than the derived normal measure at the critical point?

Using all subsets of all ordinals with all coefficients would retain far more information than a single normal measure. The meaningful task is to specify a restricted family and prove a sharp reconstruction or nonreconstruction theorem.

Question 21.4 (Measure order from defect data). Can the Mitchell ordering, or another comparison relation between normal measures on κ , be recovered from a natural algebraic relation between their coefficient functionals χ_j , without naming the ambient ultrapower models as extra predicates?

Corollary 10.4 gives injective detection of individual measures. It gives no criterion for their relative position in an ultrapower hierarchy.

Question 21.5 (Intrinsic supercompact seed systems). Can the seed construction of Theorem 16.2 be formulated without choosing an enumeration of $P_\kappa(\lambda)$? Is there a natural coordinate-free compatibility law for the resulting systems as λ varies that detects the appropriate closure of M ?

Any proposal must preserve fineness and normality and must distinguish an actual seed in $j(P_\kappa(\lambda))$ from an arbitrary missing-image index. Merely having many extra coefficients does not imply supercompactness.

Question 21.6 (Joint algebraic dependence of the images). Are $J(\mathbf{No})$ and $H_j(\mathbf{No})$ linearly disjoint over C_j ? More modestly, for a finite tuple x , can one decide the transcendence degree of $H_j(\mathbf{No})(J(x))$ over $H_j(\mathbf{No})$ from the missing-image supports?

Every single element outside the intersection is transcendental over the opposite real-closed image field. That does not decide joint algebraic independence of a tuple, and the exact intersection theorem is not a substitute for a disjointness proof. *Related result* (batch 31). The independent-copies report docs/surreal/independent-surreal-copies proves that full Hahn fields on two subgroups G_1, G_2 of one ordered group, set or class, meet in the full Hahn field on $G_1 \cap G_2$ and are linearly disjoint over it (isc:thm:sliceddisjoint). This does not answer the question. The field $H_j(\mathbf{No}) = \mathcal{H}(\Gamma_j)$ is the full Hahn field on the subgroup Γ_j by (4.1), but $J(\mathbf{No})$ is not a full Hahn field on any subgroup: it is not closed under positive strong sums of κ terms by (8.9), whereas a full Hahn field on a subgroup is closed under every set-indexed strong sum of its members. Nor is C_j such a field: it contains the monomials $\omega^{-\alpha}$, $\alpha < \kappa$, but not their strong sum, which has κ terms (Theorem 6.1). The question stays open.

Question 21.7 (Alternating the two closures). Starting with C_j , what field is obtained by alternating logarithmic field closure and external strong-sum closure? Do finitely many alternations stabilize, and can the result be characterized directly from the exponent image $J(\mathbf{No})$?

Theorem 15.7 determines each first closure exactly. Their incomparability makes the mixed closure a genuinely different question. One should distinguish closure under sums from closure under all normal-form subseries or the omega-map.

Question 21.8 (Differential compatibility). For a specified canonical surreal derivation, what is the exact locus on which J or H_j commutes with it? Can a differential analogue of (15.2) be expressed using support or the iterated logarithmic structure of the exponents?

The exponential locus is settled here. A derivation may depend on further normal-form and logarithmic data, so its preservation is not a consequence of being an ordered-field embedding. No differential claim should be imported without verifying the actual chosen derivation.

Question 21.9 (Definability in weaker signatures). Which of $\mathcal{S}_{<\kappa}$, C_j , and the coefficient projection Π_j are definable in natural reducts of the structure with the maps J, H_j ? Can any of the large-cardinal information survive after the embedding symbols and the Hadamard product are forgotten?

With both maps named, $\mathcal{S}_{<\kappa}$ is the immediate equalizer formula. The question concerns losing symbols, not that trivial definition. Pure field elementarity alone cannot recover the external strong-sum threshold.

Question 21.10 (Extending the coefficient-projection isomorphism). Under what explicit additional hypotheses does the isomorphism $\Pi_j : J(\mathbf{No}) \rightarrow H_j(\mathbf{No})$ extend to an automorphism of a larger specified surreal subfield, or of the full ordered field? Which extension requirements on monomials, simplicity, or exponentiation obstruct it?

The given isomorphism fixes the common field and all common monomials. It is not globally multiplicative on $\mathbf{No}$, so its existing definition cannot simply be reused as the desired extension.

Question 21.11 (Forcing and lifted embeddings). If an embedding j lifts through a forcing extension, which of its old coefficient defects are preserved and which new supports appear? Can the lifting criterion be expressed through a commuting square of strong companions and coefficient projections?

This would connect the present arithmetic construction to the repository’s cross-universe saturation work. One must first specify the ground and extension, verify the lift, and distinguish new index sets from changed cardinalities; no automatic preservation is asserted.

Question 21.12 (A reusable formal theorem package). Can the exact defect, image intersection, and closure theorems be formalized in a way that separates abstract ordered Hahn algebra from the set-theoretic embedding interface? What is the smallest verified interface sufficient for the elementary-embedding application?

The proposed architecture in Appendix A treats the critical-point and absoluteness obligations as explicit theorems to be proved, not as silently trusted axioms. This would make the genuinely new algebraic consequences independently reusable and auditable.

[merge] The twelve questions above are 07’s. The nine questions of 06 and the thirteen of 09 follow, each set with its source’s own framing; all 34 are kept, and where two sources ask related questions the overlap is noted rather than merged, since their formulations differ. One question is partly answered inside this report: the exponential half of Question 21.30 (09) is settled by 06’s and 07’s Theorem 15.3. All others remain open here. *Batch 32*. Manuscript 10 (Sections 22–26) answers Question 21.22 for the criterion (Corollary 26.5), meets the ultrafilter demand of Question 21.13, and makes partial progress on Questions 21.1, 21.2 and 21.23; the status notes at those questions give the scope. Its own twelve questions are in Section 26, for 46 in all.

21.1 Questions of manuscript 06 [06]

06’s questions are proposed directions, not assertions that each is a previously published open problem. They are designed to distinguish new mathematical content from immediate reformulations of the theorems already proved.

Question 21.13 (An intrinsic converse without a universe embedding). Can a suitably enriched surreal structure characterize which pairs of maps arise as (J, H_j) from an elementary universe embedding?

The coefficient formula recovers a candidate measure from a map on the codes $\tilde{h}_A$, but field multiplication alone does not give coefficientwise intersection of binary supports. Determine a minimal additional operation or relation—for example controlled coefficientwise multiplication on a designated coding family—which makes the ultrafilter laws follow intrinsically. A genuinely new converse would have to construct the ultrafilter or embedding, not assume it in the language from the outset. (Compare Question 21.1 and Question 21.22; the criteria Theorem 10.9 and Theorem 10.10 add exactly such a coefficientwise operation.) [merge] *Status (batch 32): partly answered*. Manuscript 10 constructs the ultrafilter instead of assuming it, with summation as the only extra structure: for any countably strong $\mathbb{R}$ -linear map that is not strong, the null ideal of one output coefficient on a least failing support carries a nonprincipal complete ultrafilter on the first failing cardinal (Theorem 24.1, Proposition 26.7 (c)); no coefficientwise multiplication is used. Which pairs arise as (J, H_j) from a universe embedding is not characterized.

Question 21.14 (Which landing spectra occur?). For a fixed critical point κ , classify the downward-closed classes $\mathcal{L}(j)$ realized by set-ultrapower embeddings. How does this spectrum constrain the least possible cardinality of an index set?

The normal-measure answer is exactly the cardinals at most κ . The index-size obstruction supplies a necessary bound whenever an internally short cover exists. Beyond these statements, investigate which

successor, limit, or singular endpoints can occur and whether two embeddings with the same landing spectrum can have substantially different support-cover spectra.

Question 21.15 (Extender targets and failure of the seed–closure converse). For elementary embeddings not generated by one set-ultrapower seed, what additional hypothesis makes $j \restriction \lambda \in M$ imply closure under all external λ -sequences of elements of M ?

The surreal landing theorem still reduces to seed membership, but the proof of sequence closure uses a single evaluation parameter $[\text{id}_I]$. An extender presentation may require a family of parameters. A useful theorem would identify the necessary seed support and express it as a strengthened descent condition for a family of normal forms.

Question 21.16 (Higher coefficients and derived measures). What structural information about an embedding is recoverable from the whole coefficient profile of $J(\tilde{h}_A)$ as A varies?

The explicit normal form already gives $[\omega^{-\beta}] J(\tilde{h}_A) = 1 + \mathbf{1}_{j(A)}(\beta)$ for every $\beta < j(\kappa)$. Thus it records the ultrafilters derived from all such seeds. The nontrivial next problem is to recover relations among these ultrafilters—factor maps, equivalences, or iteration data—from relations among coefficient observables, rather than merely restating the membership formula. (Related: Questions 21.3 and 21.25.)

Question 21.17 (Iteration and multilevel transport defects). Develop a coherent comparison theory for iterated elementary embeddings and successive Hahn lifts, keeping track of which intermediate values actually belong to the next domain.

A formal composition of termwise lifts is straightforward in a common ambient universe, but iteration through inner models introduces descent conditions. Determine whether a hierarchy of first missing indices gives an invariant of the iteration independent of the chosen presentations. Any proposed cocycle identity must specify these domains before performing algebra with the defects. (Related: Question 21.24.)

Question 21.18 (Canonical derivations and transport). How do J and H_j interact with the Berarducci–Mantova derivation or other specified strongly additive surreal derivations?

First prove the necessary absoluteness or naturality theorem for the chosen derivation; it is not a consequence of field elementarity alone. Then compare differentiation of the missing-index formula with transport of derivatives. The goal is an exact commutation locus comparable to (15.2), not a generic statement about differentiating formal sums. (Related: Questions 21.8 and 21.30.)

Question 21.19 (Birthday cost of a critical-support detector). For a fixed κ , what is the least birthday of a surreal with κ nonzero normal-form terms? What is the least birthday of an omnific witness to failure of exponential compatibility?

06's theorems use the explicit h_κ and ζ_κ , but do not optimize their birthdays. Sharp answers would connect this support-based theory to the repository's birthday investigations without confusing the two invariants. Normal-form length, support cardinal, and sign length must be tracked separately throughout.

Question 21.20 (Forcing and lifted coefficient codes). When an elementary embedding lifts through a forcing extension, how do the code families $\tilde{h}_A$ and their defect coefficients change for new subsets $A \subseteq \kappa$?

For old subsets the original measure code is fixed data, while new subsets require an extension of the measure. Formulate a preservation theorem distinguishing persistence of old coefficients, extension of the coding domain, and changes to the landing spectrum. This could connect the present theory to forcing across surreal universes without assuming that ground-model and extension normal-form families coincide. (Related: Question 21.11.)

Question 21.21 (Local restrictions and omega-preserving automorphisms). Which set-sized restrictions of J extend to automorphisms of a surreal structure containing the monomial map, or both the monomial map and Gonshor exponentiation?

The full J cannot be such an automorphism by Proposition 7.4. A local extension theorem, or a concrete local obstruction, would nevertheless bear on the richer automorphism questions in [5]. Homogeneity of the pure ordered field does not by itself settle the question for the monomial-enriched structure.

21.2 Questions of manuscript 09 [09]

09's questions are proposed by that manuscript. Except where a published source is explicitly named, no claim is made that each is a previously formulated open problem. Their purpose is to specify what a next theorem should say and which extra hypotheses would have to be controlled.

Question 21.22 (Removing coefficientwise multiplication). How far can the extra coefficient algebra be removed from a numerical criterion for measurability? In the compatible-field-embedding criterion of Theorem 10.10, is preservation of Hadamard multiplication on all masks implied by weaker assumptions about order, the omega-map, truncations, and $< \kappa$ -strong additivity?

A useful first target is preservation only of disjointness of mask supports, together with binary-valued coefficients at the designated probe. Any proposed converse should explicitly explain how it yields a *set* of subsets of κ and why that set is closed under fewer than κ intersections. Bare field elementarity cannot supply this by itself. [merge] *Status (batch 32): answered for the criterion.* By manuscript 10's Theorem 24.1, the whole coefficient algebra can be removed: κ is measurable if and only if some $\mathbb{R}$ -linear map $E : \mathbf{No} \rightarrow \mathbf{No}$ satisfies (a), (b) and (d) of Theorem 10.10, with no order, multiplicativity, omega-map or Hadamard condition (Corollary 26.5); (a) may even be weakened to $[\omega^{q_\kappa}] E(\tau_\alpha) = 0$. The set of subsets is the ultrafilter of a null ideal (Theorem 23.7), and its closure under fewer than κ intersections comes from $< \kappa$ -strong additivity. The mask criterion can likewise drop the Hadamard product (Corollary 26.6). $\mathbb{R}$ -linearity is used and is not dropped. Whether condition (c) actually follows from the other conditions of Theorem 10.10 is not decided, and is no longer needed for the criterion.

Question 21.23 (The bare exact-equalizer principle). For which uncountable cardinals κ can there exist ordered field embeddings $E, H : \mathbf{No} \rightarrow \mathbf{No}$ agreeing on every monomial, with H fully strong and $\text{Eq}(E, H) = \mathcal{S}_{<\kappa}$, without assuming that E comes from a set embedding? What is the consistency strength of adding omega compatibility for E ?

The current theorem proves one implication from measurability. It does not prove that the equalizer statement alone implies measurability, nor that it is impossible at an ordinary uncountable cardinal. Separating the algebraic condition from an encoded ultrafilter would clarify the genuine large-cardinal content of the main construction. (Related: Question 21.1, which adds more conditions on the pair.) [merge] *Status (batch 32): partial progress.* If E fixes $\mathbb{R}$ and is countably strong, and H is strongly $\mathbb{R}$ -linear, then $H = E_{\text{at}}$ and $\text{fail}(E)$ is a measurable cardinal at most κ ; if E is also $< \kappa$ -strong, κ is measurable (Proposition 26.7 (a)). Embeddings failing a countable sum, embeddings moving reals, and the consistency strength of omega compatibility stay open.

Question 21.24 (Iterated discrepancies). Given an iteration of normal ultrapowers, can its successive missing-index sets be described by a normal-form calculus that is functorial under composition? Which terms distinguish the pointwise image of a pointwise image from the full iterated image of an ordinal index set?

A concrete starting problem is a two-step iteration with critical points κ and $j(\kappa)$ inside the appropriate models. Each step has its own formula (5.3); the difficulty is comparing the strong lift of the composite with the composite of the two strong lifts when their domains and internal series differ. Statements about all maps must specify the universe in which each summability assertion is evaluated.

Question 21.25 (What one coefficient forgets). Which additional families of seed coefficients determine the seed hull or an extender associated with j , rather than only one normal measure? Can the passage from one seed to several seeds be expressed by explicitly compatible operations on the corresponding omnific masks?

A single normal measure does not in general specify every possible embedding that derives it. A successful reconstruction theorem should state exactly which factor of the embedding is recovered. Compatibility under projections between finite tuples of seeds is a natural first milestone.

Question 21.26 (Mitchell-order information). Can comparison or iteration of measures be recognized in a system of coefficient probes that is stable under the induced surreal embeddings? In particular, can membership of one measure in another ultrapower be encoded without merely reinstating the full membership relation as an extra predicate?

This is deliberately stronger than coding each ultrafilter separately. The problem is to represent the relation between the codes in a useful surreal language, not simply to assign another number to another set. (Related: Question 21.4.)

Question 21.27 (Beyond $2^{<\kappa} = \kappa$). When $2^{<\kappa} > \kappa$, what is the largest family of explicitly constructed positive omnific masks algebraically independent over $\mathcal{S}_{<\kappa}$? Can one still attain 2^κ in K_κ by replacing the almost-disjoint branch construction?

Theorem 13.3 applies to every available κ -almost-disjoint family, so one route is combinatorial. Another would replace private coordinates by a stronger support-separation argument. A complete answer should distinguish limitations of the particular masks from limitations on the transcendence degree itself.

Question 21.28 (Singular support cutoffs). For singular uncountable κ , which analogues of the equalizer and summability-threshold results can be realized by abstract Hahn embeddings? Can cofinality and support cardinality produce different equality loci?

The field $\mathcal{S}_{<\kappa}$ is already real closed for singular κ (Theorem 3.1). An ordinary nontrivial elementary embedding of the universe cannot have such a critical point under the present hypotheses. Thus an affirmative algebraic construction at singular κ would necessarily use a mechanism different from the normal-ultrapower one.

Question 21.29 (Other coefficient fields and ordered groups). How much of Theorem 13.3 survives for Hahn fields with coefficients of positive characteristic, nondiscrete valuation on the coefficients, or exponent groups without the displayed free rational subspace?

The selected coefficient in the proof has multiplicity one, so it avoids one familiar source of cancellation in positive characteristic. The main remaining issue is to formulate the correct small-support coefficient base and to supply enough independent exponent coordinates. This offers a plausible extension beyond surreal numbers without changing the core separation argument.

Question 21.30 (Exponential and differential refinements). How do the pair (J, H_j) and the discrepancy interact with the Gonshor exponential and with a specified surreal derivation? Can one calculate an exact equality locus for either operation analogous to Corollary 5.6?

09 proves no compatibility with these operations, and adds that an approach through their recursive definitions must verify absoluteness, and an approach through transseries must verify all relevant summability conditions. The questions about enriched automorphisms in [5] provide context, but those automorphisms and the generally nonsurjective embeddings here should not be identified. [merge] *Status*. For the exponential this is answered by Theorem 15.3 (06, 07): the exact locus of $H_j \circ \exp = \exp \circ H_j$ is $|\text{supp}(p(x))| < \kappa$, and J commutes with $\exp$ everywhere (Lemma 15.1), with the absoluteness of the recursion verified there. The derivation half is open, as in Questions 21.8 and 21.18.

Question 21.31 (Set-sized restrictions). For which pairs of birthday cutoffs can the main pair of embeddings be restricted to set-sized real closed surreal fields, retaining the same support equalizer on the common domain? What are the optimal birthday bounds on the images and the discrepancy?

The identity $b(J(x)) = j(b(x))$ is a starting bound for one map. The strong lift may need a different support-to-birthday estimate. A complete theorem should name its cutoffs and closure properties rather than appeal to the proper-class theorem informally. (Related: Question 21.19.)

Question 21.32 (Omnific arithmetic without masks). Can any part of the measure-recovery mechanism be expressed using only ordinary omnific addition, multiplication, order and a small number of named maps or parameters? In particular, how much of normal-form coefficient extraction is definable in such a structure?

The lack of nontrivial idempotents in an integral domain makes a naive identification with a Boolean algebra impossible. One could instead seek interpreted quotients or definable families. This direction should be compared with the repository’s omnific definability and quotient projects, not assumed to follow from them.

Question 21.33 (Higher compactness in truncation spaces). Can stronger branch principles on two-parameter or seed-indexed families of omnific normal forms distinguish weak compactness, strong compactness and supercompactness using one coherent truncation-space framework?

The distinction between a covering seed and an exact seed suggests a specific invariant to retain. Any topological formulation must specify the topology and the allowed cardinalities of covers or branches; the ordinary inherited order topology on $\mathbf{Oz}$ is too discrete for the prefix argument. (06’s support covers and exact descent, Section 17, are one such pair of invariants, stated with embeddings rather than branches.)

Question 21.34 (A checked exact-equalizer theorem). Can the main theorem be formalized with an abstract elementary-embedding interface whose only large-cardinal inputs are the critical-point and normal-form transport properties, while separately proving the bridge from a genuine ultrapower?

A practical first deliverable would formalize Hahn subseries partition, strong exponent lifts, and the private-coordinate independence argument in a set-sized Hahn field. The second deliverable would formalize Lemma 2.5 and the exact comparison theorem for an ordinal-indexed normal-form transport. Only the third would connect that interface to actual surreal sign sequences and ultrapowers. This ordering would keep the algebraic verification independent of the largest foundational infrastructure task. (Related: Question 21.12.)

22 Measurable first failures: source, conventions and headline [10]

[merge] Sections 22–26 print manuscript 10 of this report: manuscript 05 of batch 32, delivered as `surreal_measurable_spectrum.zip`, a 24-page article titled *Measurable Cardinals and the First Failure of Surreal Summation: Exact failure spectra, canonical strong parts, and omnific-preserving additive*

symmetries, prepared for Vladimir Reshetnikov with ChatGPT on 23 September 2026 and pinned to repository commit 7af8a30. It was added after the three-source merge and is placed after the questions of Section 21, so that no earlier number changes. The preceding sections construct an obstruction from a measurable cardinal; manuscript 10 proves a converse. For coefficient-linear maps that preserve every *countable* Hahn sum, the first failure of a larger sum is always a measurable cardinal, and every measurable cardinal occurs. This is the converse that Section 1.4 disclaims, obtained for countably strong maps only (Remark 26.3). In Sections 22–26 every passage is 10’s unless it is marked [merge]; results that this report already proves are printed once, with their sources. The conventions of these five sections are fixed in Section 22.4 before they are used.

22.1 The problem and the headline statement

An infinite normal form has two distinct features: the family of its terms is admissible, and a proposed map commutes with their sum. For finite sums these issues disappear. For arbitrary Hahn sums they must be kept separate. The main result of manuscript 10 is that, once countable sums are respected, *the first remaining obstruction is exactly measurable-cardinal sized*.

For the introductory statement regard **No** as the Hahn field $\mathbb{R}((\omega^{\mathbf{No}}))$ of Conway normal forms, with reverse well-ordered *set* supports, as in (2.1). The exponent of a monomial is itself a surreal; this is notation for Conway normal forms, not the introduction of proper-class supports. Definitions and foundations follow in Section 23.

Theorem 22.1 (Exact surreal failure spectrum [10]). *Let $\Psi : \mathbf{No} \rightarrow \mathbf{No}$ be $\mathbb{R}$ -linear and preserve every countable strongly summable family and its sum. Then it preserves the summability of every set-indexed strongly summable family. If it does not commute with every such sum, define*

$$\text{fail}(\Psi) = \min \left\{ |I| : \Psi \left(\sum_{i \in I} x_i \right) \neq \sum_{i \in I} \Psi(x_i), \quad (x_i)_{i \in I} \text{ strongly summable} \right\}. \quad (22.1)$$

The cardinal $\text{fail}(\Psi)$ is measurable.

*Conversely, for every measurable cardinal κ and every set $Z \subseteq \mathbf{No}$, there is an $\mathbb{R}$ -linear additive automorphism Θ of **No** with $\text{fail}(\Theta) = \kappa$. It preserves order and leading terms, stabilizes **Oz**, fixes Z pointwise, and fixes every Conway monomial. It is not a field homomorphism. Both Θ and Θ^{-1} preserve all strong sums indexed by sets of size less than κ .*

Proof pointers: the first assertion is Corollary 23.4; measurability of $\text{fail}(\Psi)$ is Theorem 24.1; the converse, with all the additional preservation properties, is Theorem 25.2 together with Theorem 25.4. The distinction between an *additive* automorphism and a *field* automorphism is indispensable. In particular, the theorem is compatible with the omnific-preserving report’s theorem that every field automorphism of **No** preserving **Oz** is strongly additive (opa:as:thm:main in docs/surreal/omnific-preserving-automorphisms).

22.2 Why the result is more than an example

A measurable cardinal has long supplied ultrapower embeddings, and the preceding sections of this report turn those embeddings into explicit non-strong surreal field embeddings. An example by itself gives an upper bound on the set-theoretic strength of an obstruction. The present theorem supplies a converse under a precise hypothesis: *every coefficient-linear map that passes all countable strong-sum tests but fails a larger one yields an actual measurable cardinal*. It also identifies the first failing cardinal, not merely the existence of some measurable cardinal somewhere below a large support.

The proof is local to one well-ordered support and one output coefficient. It does not assume elementarity, multiplicativity, order preservation, valuation compatibility, or an omnific predicate. These stronger structures enter only in the applications and sharp examples.

22.3 Prior work and proposed contributions

Conway normal forms and their Hahn arithmetic are classical [1, 2]. The distinction between strong and unrestricted surreal automorphisms belongs to an existing literature, including Kaplan–Krapp–Serra [5]. Generalized power series also fit the finiteness-space framework of Blute–Cockett–Jacqmin–Scott [27]. The filter-theoretic engine below is the finite complete-ultrafilter factorization mechanism treated by Bergman [25], with related linear product-map results in [26]; 10 gives the specialized proof needed here, rather than claim a new general theorem about maps on products.

At its pin, 10 found in the omnific-preserving report automatic strongness for **Oz**-automorphisms, constant-term adjoints, a proper-class dual obstruction, and proper embeddings [30], and its question `opa:as:q:targettests` on representing pulled-back target tests for coefficient-fixing omnific-preserving embeddings. It found in this report (then 07’s placed text) the elementary embedding J , its strong companion H_j and their support defects. None of those results is claimed anew.

The proposed additions are the general least-failure theorem, its complete linear localization, the canonical strong-part construction for all countably strong linear maps, the exact full-Hahn workspace criterion and size bound, and the monomial-fixing ordered additive transvections. The field-embedding equivalence and target-test obstruction combine these results with the existing elementary-embedding construction. An audit of this boundary is Section 26.4.

22.4 Conventions, notation and the pin [merge]

Exponent convention. Manuscript 10 writes Hahn series in a variable t with $t^\gamma = \omega^{-\gamma}$ and *well-ordered* supports, and writes $[t^\gamma]x$ for the coefficient of $\omega^{-\gamma}$. This report writes ω^a with *reverse well-ordered* supports and $[\omega^a]x$ for the coefficient of ω^a ((2.1), Definition 2.1). Every statement of 10 is printed here after the substitution $t^\gamma \mapsto \omega^{-\gamma}$, so every exponent changes sign: 10’s $[t^\gamma]x$ is $[\omega^{-\gamma}]x$ here, “well ordered” becomes “reverse well ordered”, 10’s negative support is positive support here, and 10’s detector exponents $d - \omega^{-\alpha}$ with $d \leq -1$ become $\delta + q_\alpha$ with $\delta = -d \geq 1$. For an abstract ordered group Γ the two fields are identified by $t^\gamma \mapsto \omega^{-\gamma}$; no other normalization changes. The valuation is unaffected: 10’s $v(x) = \min \text{supp}_t(x)$ is this report’s $v(x) = -(\text{leading exponent})$ of (2.1).

Renamed symbols. The following symbols of 10 collide with symbols of Sections 1–21 and are renamed in Sections 22–26; 10’s delivered proof-status and audit files keep 10’s names.

10’s symbol	Here	Reason and meaning
$t^\gamma, [t^\gamma]x, k((t^\Gamma))$	$\omega^{-\gamma}, [\omega^{-\gamma}]x, k((\omega^\Gamma))$	the exponent convention above
$L = k((t^\Delta))$	$K' = k((\omega^{\Gamma'}))$	L_j is the logarithmic defect
$T, T_{\text{at}}, T_{\text{gh}}$	$\Psi, \Psi_{\text{at}}, \Psi_{\text{gh}}$	T_f, T_P are reindexing maps and 09’s T is H_j
G (a ghost map)	Υ	G_κ is 09’s exponent group
F (a field embedding)	E	F is 07’s letter for a family $F : I \rightarrow \mathbf{No}$; E is also 09’s field embedding in Theorem 10.10
D (a derivation)	∂	D_j is the defect $J - H_j$
θ in “ $< \theta$ -strong”	ν	θ is a normal-form length in (2.1)

10's symbol	Here	Reason and meaning
ν of Lemma 4.4 (least incomplete partition)	$\mu_{\mathcal{U}}$	freed for the previous line
U, U_j (ultrafilters), index j	$\mathcal{U}, \mathcal{U}_p$, index p	U_j is the derived measure, j the embedding
$\ell_{S,\eta}, \ell$	$\psi_{X,b}, \psi$	ℓ_α is 09's coordinate functional (Lemma 13.2)
support $S, K[X]$	$X, K[X]$	$\mathcal{S}_{<\kappa}$ is the short-support field
output exponent η	b	η is an ordinal index throughout
null ideal $\mathcal{I}_\ell$; atoms	$\mathcal{N}_\psi$; cells $I_1, \dots, I_m$	$\mathcal{S}_{<\kappa}$ is the short omnific ring; A is a subset of κ
$A_1, \dots, A_m$		
vectors $a \in k^I$; series f_i	$f \in k^I; x_i$	a is an exponent; f as in h_f (10.11)
$d \leq -1, s_\alpha = d - \omega^{-\alpha}, S_d$	$\delta \geq 1$, the exponent	sign change; $q_\alpha = \omega^{-\alpha}$ is 09's scale (8.4)
	$\delta + q_\alpha, X_\delta$	
$a = t^d$	ω^δ	a is an exponent
$\lambda_U, N_U, A_{c,U}$	$\vartheta_{\mathcal{U}}, N_{\mathcal{U}}, \Theta_{c,\mathcal{U}}$	λ is a length or cardinal; A a subset of κ
masks $F_B = \sum_{\alpha \in B} t^{s_\alpha}, B \subseteq \kappa$;	$\omega^\delta Y_A, A \subseteq \kappa; \omega^\delta Y_\kappa$	09's omnific mask (10.9) shifted by ω^δ ; 09's F_κ is $\mathcal{S}_{<\kappa}$
$F = F_\kappa$		
$B = \sum_{\alpha < \kappa} t^\alpha$	h_κ	07's (7.1)
Λ	Λ_j	it depends on j
parameter set P , union of supports E	Z, E_Z	P_θ is the missing-index set
Γ_κ, K_κ of Theorem 8.2	G_κ, K_κ	the same objects as 09's (Theorem 13.5); see Theorem 25.7
Π_k, m_k	$\mathfrak{P}_K, m_K$	Π_j is the coefficient projection
multiplication operator M_f	mul_y	M is the target model
$\mathbf{No}[i], \mathbf{Oz}[i], \langle x, y \rangle_0$	$\mathbf{SC}, \mathbf{Oz}[i], \text{ct}(xy)$	this report's notation

Kept: $k \in \{\mathbb{R}, \mathbb{C}\}$, Γ (an abstract exponent group, not $\Gamma_j = J(\mathbf{No})$), $K = k((\omega^\Gamma))$ (a generic Hahn field, not 09's K_κ except in Theorem 25.7), $\text{ev}_{\mathcal{U}}$, fail, the subscripts at and gh, ct, c for scalars, m for a finite count (not 07's supercompact masks m_A), and $O_\Gamma = \mathbb{Z} \oplus \mathfrak{P}_K$.

Tempting false readings. The detector functional $\vartheta_{\mathcal{U}}$ of Section 25 is an ultrafilter limit of κ coefficients of its *input*; it involves no embedding. The functional χ_j of (10.13) reads one coefficient of the *image* $J(h_f)$. Both evaluate along a κ -complete ultrafilter, on different supports (X_δ and $\{-\alpha : \alpha < \kappa\}$). The masks $\omega^\delta Y_A$, 09's Y_A and 07's $z_A = \omega^\kappa h_A$ are three different numbers. The pointwise product on k^X is not Hahn multiplication: a Boolean mask is not an idempotent of the field. “Countably strong” is a summation condition, not continuity in the valuation topology. The transvections $\Theta_{c,\mathcal{U}}$ fix every monomial but are not the identity; fixing every monomial as an element is not compatibility with the omega-map as a function. An additive automorphism is not a field automorphism. Reversing the exponent convention without changing the signs would destroy well-ordering: the exponents $\delta + q_\alpha$ decrease in α .

Hypotheses. Set-sized statements are proved in ZFC; class-sized statements are read in NBG with global choice, with each map an individual class function and all summation families sets (Section 23.1). A measurable cardinal is assumed exactly where stated. Section 26 uses the normal-measure ultrapower instance of (1.1) and imports this report's absoluteness package (Lemma 2.6); the earlier sections do not.

The pin. 10 was pinned to 7af8a30, which precedes the three-source merge de45cee. At the pin this report was 07's placed text alone, and the omnific-preserving report lacked its source 14. 10 therefore cites this

report as “source 07” and its “Sections 2–4” (now Sections 2, 4 and 5), and did not see 06’s and 09’s material, in particular 09’s criterion Theorem 10.10 and the questions 21.13 and 21.22, which it answers or bears on (Section 26.3). Its claim to give “a concrete conditional negative answer” to `opa:as:q:targettests` is stale against the tree; the negative answer already followed from results recorded at the pin, and 10’s contribution there is the explicit test (the note after Theorem 26.4). Nothing else 10 says about the repository is false at the time of this merge.

23 Countable detection and complete-ultrafilter functionals [10]

23.1 Fields, supports, and class size

Unless explicitly varied, k is $\mathbb{R}$ or $\mathbb{C}$. For an ordered abelian group Γ , write

$$K = k((\omega^\Gamma)) = \left\{ \sum_{a \in X} c_a \omega^a : X \subseteq \Gamma \text{ reverse well ordered, } c_a \in k \right\}.$$

Zero coefficients are omitted from the support. For set-sized Γ this is an ordinary Hahn field. We also allow $\Gamma = \mathbf{No}$, but every individual support must still be a set. This gives **No** for $k = \mathbb{R}$ and **SC** for $k = \mathbb{C}$ by separating real and imaginary coefficients. The valuation is $v(x) = -\max \text{supp}(x)$ for $x \neq 0$ and $v(0) = \infty$. For real coefficients the sign is the sign of the leading coefficient.

Set-sized statements are proved in ZFC. Class-sized statements may be read in NBG with global choice, with each map an individual class function and all quantified summation families sets. The proofs never form a class whose members are all proper-class maps. Assertions about all such maps are class-quantified statements, not an assertion that their collection is a set or an NBG class. Restrictions of a class map to set-sized products are set functions. This is precisely what the localization arguments need.

23.2 Hahn summability

Definition 23.1 (Hahn summability [10]). A set-indexed family $(x_i)_{i \in I}$ in K is *Hahn-summable* if

- (i) $X = \bigcup_{i \in I} \text{supp}(x_i)$ is reverse well ordered;
- (ii) for every $a \in \Gamma$, the set $\{i : [\omega^a] x_i \neq 0\}$ is finite.

Its sum has coefficient $\sum_i [\omega^a] x_i$ at ω^a , a finite sum at each exponent. The empty family has sum zero.

[merge] For $K = \mathbf{No}$ this is Definition 2.1; “Hahn-summable” and “strongly summable” are used interchangeably below.

A family of ordinary scalars, regarded as constant Hahn series, is Hahn-summable exactly when only finitely many members are nonzero. Thus nothing below replaces an infinite numerical sum by a conditionally convergent or analytically convergent one.

Definition 23.2 (Strong, countably strong, $<\nu$ -strong [10]). A k -linear map between Hahn fields is *strong* if it sends every Hahn-summable set family to a Hahn-summable family and commutes with its sum. It is *countably strong* if this holds for countable families, and *$<\nu$ -strong* if it holds for families of cardinality less than an uncountable cardinal ν .

Here countably strong is the same as $<\aleph_1$ -strong. Linearity is over the coefficient field, not over the whole Hahn field. The phrase “countably strong” is a summation condition; it is not shorthand for continuity in the

valuation topology. A summable Hahn family need not have valuation-topological partial sums converging to its Hahn sum. [merge] For $\mathbb{R}$ -linear maps of **No**, “strong” is “strongly $\mathbb{R}$ -linear” of Definition 2.1, and $<\nu$ -strong is the notion introduced after Theorem 8.4; compare Remark 2.2.

23.3 The omnific subring

Put

$$\mathfrak{P}_K = \{x : \text{supp}(x) \subseteq \Gamma^{>0}\}, \quad \mathfrak{m}_K = \{x : \text{supp}(x) \subseteq \Gamma^{<0}\}.$$

Both include zero. For $k = \mathbb{R}$ the canonical omnific-type ring is $\mathcal{O}_\Gamma = \mathbb{Z} \oplus \mathfrak{P}_K$. For $\Gamma = \mathbf{No}$ this is Conway’s **Oz** of (11.1), $\mathfrak{P}_{\mathbf{No}}$ is the class of purely infinite parts $p(x)$ and $\mathfrak{m}_{\mathbf{No}} = \mathfrak{m}$. The surcomplex analogue is $\mathbb{Z}[i] \oplus \mathfrak{P}_{\text{SC}} = \mathbf{Oz}[i]$. The direct decomposition

$$K = \mathfrak{P}_K \oplus k \oplus \mathfrak{m}_K$$

is additive. Constant extraction is denoted by ct . On the whole field it is linear, not multiplicative; on $k \oplus \mathfrak{P}_K$ it is a ring homomorphism. This distinction will prevent several tempting but incorrect shortcuts.

23.4 Products on a fixed support are only coefficient products

For a reverse well-ordered set $X \subseteq \Gamma$, define

$$K[X] = \{x \in K : \text{supp}(x) \subseteq X\}.$$

The map $f \mapsto \sum_{s \in X} f(s)\omega^s$ identifies $K[X]$ with the *vector space* k^X . It need not be a subring of K . In particular, pointwise multiplication on k^X is not Hahn multiplication. Boolean masks on X are idempotent for the former operation, not the latter. The product-functional theorem uses only the vector-space identification. [merge] This is the Hadamard product $\odot$ of (10.12) and the product warning after Proposition 10.8.

23.5 Countability detects admissibility, not the sum

Lemma 23.3 (Countable obstruction to summability [10]). *A set-indexed family of Hahn series is summable if and only if each of its countable subfamilies is summable.*

Proof. Only the reverse implication needs proof. If infinitely many members have a nonzero coefficient at one exponent, choose a countably infinite subfamily of them. This contradicts coefficientwise finiteness. If the union of supports is not reverse well ordered, dependent choice gives a strictly increasing sequence of exponents in that union. Choose for each exponent one family member containing it. These members form a countable subfamily whose union of supports contains the increasing sequence. The contradiction proves the claim. The witnesses use sets even when the ambient exponent group is **No**. $\square$

[merge] The omnific-preserving report proves a sharper form: a nonsummable family has a countable subfamily *and* a countably supported binary test y with $\text{ct}(yx_i) \neq 0$ for infinitely many i (opa:as:cor:countable, from opa:as:thm:detect). The lemma is the elementary half of that statement and is printed as 10’s because the arguments below use no pairing.

Corollary 23.4 (Countably strong maps preserve admissibility [10]). *Every countably strong map sends every set-indexed Hahn-summable family to a Hahn-summable family.*

Proof. Every countable subfamily of an image family is the image of a countable subfamily of the original one. Apply the hypothesis and Lemma 23.3. $\square$

[merge] For J this is the first assertion of Theorem 8.1, proved there by transfer.

Consequently, for a countably strong Ψ , a failure at an uncountable family is always a failure of the equality

$$\Psi\left(\sum_i x_i\right) = \sum_i \Psi(x_i). \quad (23.1)$$

The right side exists. It is not an undefined expression that has been mistaken for a counterexample.

Lemma 23.5 (Support localization [10]). *Let $\Psi : K \rightarrow K'$ be $<\nu$ -strong, where $K' = k((\omega^{\Gamma'}))$, and let $X \subseteq \Gamma$ be reverse well ordered. For each $b \in \Gamma'$, the functional*

$$\psi_{X,b} : k^X \rightarrow k, \quad \psi_{X,b}(f) = [\omega^b] \Psi\left(\sum_{s \in X} f(s)\omega^s\right) \quad (23.2)$$

commutes with every point-finite family of vectors indexed by a set of size less than ν ; its nonzero scalar values on that family are finite in number.

Proof. A point-finite family in k^X corresponds to a Hahn-summable family in $K[X]$: the total support is contained in the reverse well-ordered set X . Apply Ψ and then coefficient extraction. At one output exponent only finitely many image members contribute. $\square$

The adjective point-finite means that each coordinate belongs to the nonzero support of only finitely many vectors. It places no uniform bound on that finite number.

23.6 The product-functional theorem

This subsection is the classical engine, stated in the exact form needed for Hahn localization. The finite-ultrafilter factorization of the support filter is the mechanism of Bergman's Lemmas 3–4 [25]. 10 gives a null-ideal proof and then uses the special nature of real and complex coefficients to turn ultrapower factors into ordinary scalar evaluations. No novelty is claimed for this mechanism. [merge] The paper [7] cited in Section 10 is a different paper by the same author.

An ultrafilter $\mathcal{U}$ on a set I is ν -complete if it is closed under intersections of fewer than ν members. Countable completeness means $\aleph_1$ -completeness. A *measurable cardinal* is an *uncountable* cardinal κ carrying a nonprincipal κ -complete ultrafilter. [merge] This is the meaning used throughout this report; for the critical point it is Lemma 2.3.

Lemma 23.6 (Real and complex functions are almost constant [10]). *If $\mathcal{U}$ is a countably complete ultrafilter on I and $f : I \rightarrow k$, where $k = \mathbb{R}$ or $\mathbb{C}$, there is a unique $r \in k$ such that $\{i : f(i) = r\} \in \mathcal{U}$. Write $\text{ev}_{\mathcal{U}}(f) = r$. This is a unital k -algebra homomorphism for pointwise operations.*

Proof. First take real values. The sets $\{|f| \leq n\}$, $n \geq 1$, cover I . One belongs to $\mathcal{U}$, since otherwise the countable intersection of their complements would belong to $\mathcal{U}$ and be empty. On that set, repeatedly bisect a closed bounded interval containing the values. At each step at least one half has a preimage in $\mathcal{U}$. The resulting nested intervals have lengths tending to zero and a single common point r . Countable completeness puts the intersection of their preimages in $\mathcal{U}$, and on this intersection $f = r$. Uniqueness follows because disjoint sets cannot both belong to an ultrafilter. For complex values apply this to real and imaginary parts. Intersecting the finitely many relevant full-measure fibers proves the addition, multiplication, and scalar identities. $\square$

[merge] For the κ -complete derived measure U_j the same value is $j(f)(\kappa)$ (Theorem 10.7); 09's fiber argument in Proposition 10.8 finds the large fiber using $|\mathbb{R}| < \kappa$ and κ -completeness, whereas the bisection above needs only countable completeness. This is a genuinely different route to the same fiber.

Theorem 23.7 (Complete-ultrafilter representation [10]). *Let ν be an uncountable cardinal, I a set, and $k = \mathbb{R}$ or $\mathbb{C}$. A k -linear functional $\psi : k^I \rightarrow k$ preserves all point-finite sums indexed by sets of size less than ν , with only finitely many nonzero scalar images, if and only if*

$$\psi(f) = \sum_{p=1}^m c_p \text{ev}_{\mathcal{U}_p}(f), \quad (23.3)$$

where $m < \omega$, the $\mathcal{U}_p$ are distinct ν -complete ultrafilters on I , and $c_p \in k \setminus \{0\}$. For the zero functional take $m = 0$. The unordered list of pairs $(\mathcal{U}_p, c_p)$ is unique.

Proof. Define the support-null ideal

$$\mathcal{N}_\psi = \{B \subseteq I : \psi(f) = 0 \text{ whenever } \text{supp}(f) \subseteq B\}.$$

It is an ideal of subsets. It is closed under unions of fewer than ν sets: disjointify such a family (B_ξ) and decompose an arbitrary vector supported on its union into the corresponding disjoint pieces. These pieces are point-finite, and each has zero image.

There cannot be countably infinitely many pairwise disjoint sets outside $\mathcal{N}_\psi$. Otherwise choose on each such set a vector with nonzero image. Their family is point-finite but has infinitely many nonzero scalar images. It follows that the Boolean algebra $\mathcal{P}(I)/\mathcal{N}_\psi$ is finite. For completeness, every infinite Boolean algebra has a countably infinite disjoint nonzero family: either it has infinitely many atoms, or after removing finitely many atoms a nonzero atomless element can be repeatedly split. Representatives of a countable disjoint family modulo the ideal can be made actually disjoint by removing the finitely many earlier representatives.

If the quotient is zero, then $\psi = 0$. Otherwise represent its finitely many atoms by a partition $I = I_1 \sqcup \dots \sqcup I_m$. For each atom put

$$\mathcal{U}_p = \{B \subseteq I : I_p \setminus B \in \mathcal{N}_\psi\}.$$

The atom property makes $\mathcal{U}_p$ an ultrafilter; closure of the ideal under $<\nu$ unions makes it ν -complete. The ultrafilters are distinct because $I_p \in \mathcal{U}_p$ and $I_p \notin \mathcal{U}_{p'}$ for $p' \neq p$.

Given $f \in k^I$, use Lemma 23.6 to set $r_p = \text{ev}_{\mathcal{U}_p}(f)$. The subset of I_p where $f \neq r_p$ belongs to $\mathcal{N}_\psi$. Hence $f - \sum_p r_p \mathbf{1}_{I_p}$ is supported on an ideal member, and

$$\psi(f) = \sum_p \psi(\mathbf{1}_{I_p}) r_p.$$

Each coefficient $c_p = \psi(\mathbf{1}_{I_p})$ is nonzero: if it were zero, the same argument for all vectors supported on I_p would put I_p in the null ideal, contradicting the choice of an atom.

Conversely, suppose (23.3) holds. For a point-finite family $(f_\xi)_{\xi < \mu}$ with $\mu < \nu$, intersect the $\mathcal{U}_p$ -large fibers $f_\xi = \text{ev}_{\mathcal{U}_p}(f_\xi)$ over all ξ , separately for each p . On the nonempty intersection, point-finiteness says that only finitely many of these scalar values are nonzero. There the pointwise sum equals their finite sum. Its evaluation therefore has the required value. A finite sum over p proves the assertion for ψ .

Finally the null ideal determines its finite Boolean quotient and hence the ultrafilters. Equivalently, finitely many distinct ultrafilters can be separated by a finite partition, one cell in each ultrafilter. Applying ψ to the cell indicators determines the coefficients. This proves uniqueness. $\square$

Remark 23.8 (No regularity is used [10]). No regularity of ν is used in Theorem 23.7. Disjointifying a family of length $\mu < \nu$ still gives a point-finite family of that length. This is a statement about linear sums, not a claim that every support-restricted Hahn subspace is a field at singular cardinals.

Lemma 23.9 (The first incomplete partition [10]). *Let $\mathcal{U}$ be a nonprincipal countably complete ultrafilter on a set I . There is a least cardinal $\mu_{\mathcal{U}}$ for which I has a partition into $\mu_{\mathcal{U}}$ sets none belonging to $\mathcal{U}$. This $\mu_{\mathcal{U}}$ is measurable, and $\mathcal{U}$ is $\mu_{\mathcal{U}}$ -complete. If $\mathcal{U}$ is ν -complete, then $\mu_{\mathcal{U}} \geq \nu$. If $\mathcal{U}$ is κ -complete but not κ^+ -complete, then $\mu_{\mathcal{U}} = \kappa$.*

Proof. Singletons give some such partition, while countable completeness excludes countable partitions. For a family of $\mu < \mu_{\mathcal{U}}$ members of $\mathcal{U}$, suppose its intersection were not in $\mathcal{U}$. Disjointifying their complements and adding the remaining intersection would partition I into at most $\mu + 1 < \mu_{\mathcal{U}}$ small sets, a contradiction. Thus $\mathcal{U}$ is $\mu_{\mathcal{U}}$ -complete. Push $\mathcal{U}$ forward along a least partition $I \rightarrow \mu_{\mathcal{U}}$. The resulting ultrafilter on $\mu_{\mathcal{U}}$ is $\mu_{\mathcal{U}}$ -complete and contains no singleton, so $\mu_{\mathcal{U}}$ is measurable. The other statements follow from the same partition criterion for completeness. In the last statement, failure of κ^+ -completeness yields a small-set partition of size at most κ , whereas κ -completeness excludes a smaller one. $\square$

This elementary argument is the familiar complete-ultrafilter route to measurable cardinals; compare [25, 28]. It does not assume that every complete filter extends to a complete ultrafilter. Such an extension principle would be much stronger and is unnecessary here.

Remark 23.10 (The character theorem as an instance [merge]). Take j as in (1.1), $\Psi = J$, the support $X = \{-\alpha : \alpha < \kappa\}$ and the output exponent $b = -\kappa$. Then $\psi_{X,b}(f) = [\omega^{-\kappa}] J(h_f) = \chi_j(h_f)$ by (10.11) and (10.13), since $H_j(h_f) = h_f$ has no term at $\omega^{-\kappa}$. By Theorem 10.7 this is $\text{ev}_{U_j}(f)$: in (23.3), $m = 1$, $c_1 = 1$ and $\mathcal{U}_1 = U_j$, with no principal term because $J(\omega^{-\alpha}) = \omega^{-\alpha}$ has no coefficient at $\omega^{-\kappa}$. Thus 07's and 09's character theorem is the case of Theorem 23.7 in which the functional is also a Hadamard character. The converse half of Theorem 10.9 reads the ultrafilter off the values on idempotent masks; Theorem 23.7 reads it off the null ideal of an arbitrary linear functional, with no product at all (Corollary 26.6).

24 The least failure is measurable, and the canonical strong part [10]

Theorem 24.1 (Least failure is measurable [10]). *Let $K = k((\omega^\Gamma))$ and $K' = k((\omega^{\Gamma'}))$ be full Hahn fields over $k = \mathbb{R}$ or $\mathbb{C}$, with set-sized exponent groups or the class-sized interpretation of Section 23.1. Let $\Psi : K \rightarrow K'$ be countably strong and k -linear. If Ψ is not strong, the least cardinality κ of a family on which it fails to commute with the sum is a measurable cardinal.*

Proof. By Corollary 23.4, all the image families under consideration are summable. Choose a least failing cardinal κ and a witness $(x_i)_{i \in I}$ with $|I| = \kappa$. This least cardinal exists: one witness bounds a nonempty set of candidate cardinals, even when Ψ is a class map. It is uncountable, and Ψ is $<\kappa$ -strong.

Put $X = \bigcup_i \text{supp}(x_i)$, a reverse well-ordered set. Choose an exponent $b \in \Gamma'$ at which the nonzero defect has a nonzero coefficient. Under $K[X] \cong k^X$, let f_i be the coefficient vector of x_i . The functional $\psi = \psi_{X,b}$ of (23.2) satisfies

$$\psi\left(\sum_{i \in I} f_i\right) \neq \sum_{i \in I} \psi(f_i). \quad (24.1)$$

It is $<\kappa$ -strong for point-finite sums by Lemma 23.5. Apply Theorem 23.7 with $\nu = \kappa$:

$$\psi = \sum_{p=1}^m c_p \text{ev}_{\mathcal{U}_p},$$

where all $\mathcal{U}_p$ are κ -complete. If every $\mathcal{U}_p$ were κ^+ -complete, the converse direction of Theorem 23.7 would make ψ preserve the κ -indexed sum in (24.1). Therefore one $\mathcal{U}_p$ is κ -complete but not κ^+ -complete. It is nonprincipal. By Lemma 23.9, its least incomplete partition has size κ and yields a nonprincipal κ -complete ultrafilter on κ . Thus κ is measurable. $\square$

The theorem is a converse to a kind of behavior exhibited by large-cardinal embeddings, but it does not presuppose such an embedding. Its hypotheses concern one linear map and summation only. In particular, it does not merely show that some measurable cardinal is at most the support size of a witness: it identifies the first failing size itself. [merge] For $\Psi = J$ the first failure is $\kappa = \text{crit}(j)$ (Theorem 8.1 and Corollary 8.3), and the theorem re-derives that κ is measurable, which Lemma 2.3 obtains directly from U_j .

Corollary 24.2 (No-measurable-cardinal principle [10]). *If there are no measurable cardinals, every countably strong coefficient-linear map between full real or complex Hahn fields is strong. More generally, if there is no measurable cardinal at least ν , then every $<\nu$ -strong such map is strong.*

Proof. A first failure would be a measurable cardinal; in the second assertion it would also be at least ν . $\square$

Corollary 24.3 (Individual failure spectrum [10]). *For a countably strong coefficient-linear map, failure at a successor cardinal or at a singular cardinal cannot be the first failure. In particular, failure on some large family does not license a claim of a new obstruction at that family's cardinality without checking smaller families.*

Proof. Measurable cardinals are regular limit cardinals, a standard consequence of the completeness of their measures [28]. Apply Theorem 24.1. $\square$

24.1 A coefficient-by-coefficient description

The proof gives more than a cardinal bound. On every fixed reverse well-ordered support X and at every output exponent b , a countably strong map has a uniquely determined finite representation

$$[\omega^b] \Psi \left(\sum_{s \in X} f(s) \omega^s \right) = \sum_{p=1}^{m(X,b)} c_{X,b,p} \text{ev}_{\mathcal{U}_{X,b,p}}(f). \quad (24.2)$$

The ultrafilters are countably complete, and they are ν -complete when Ψ is $<\nu$ -strong. Principal ultrafilters give ordinary matrix coefficients. Nonprincipal ones are exactly the additional local terms invisible to individual coordinates.

This is a local representation, not a prescription for assembling an arbitrary family of ultrafilters into a global Hahn map. Compatibility when supports enlarge, reverse well-ordering of output supports, and coefficientwise finiteness across output rows still have to be checked. No converse global assembly theorem is asserted.

24.2 A canonical strong part and a monomial-invisible remainder

The preceding localization suggests separating the part determined by individual monomials from everything else. This can be done without assuming the absence of measurable cardinals.

Theorem 24.4 (Canonical strong part [10]). *Let $\Psi : K \rightarrow K'$ be countably strong and k -linear. The formula*

$$\Psi_{\text{at}}\left(\sum_{a \in X} c_a \omega^a\right) = \sum_{a \in X} c_a \Psi(\omega^a) \quad (24.3)$$

defines a strong k -linear map. It is the unique strong map agreeing with Ψ on all monomials. The remainder

$$\Psi_{\text{gh}} = \Psi - \Psi_{\text{at}}$$

is countably strong and vanishes on every monomial. If Ψ is $<\nu$ -strong, then so is Ψ_{gh} , and it vanishes on every series whose support has cardinality less than ν .

Proof. The family $(\omega^a)_{a \in X}$ is summable. By Corollary 23.4, its image family is summable. Multiplying its members by coefficient scalars preserves summability, so the right side of (24.3) exists. Uniqueness of normal forms makes the prescription unambiguous; finite linearity follows coefficientwise.

Let $(x_i)_{i \in I}$ be any summable family, with common reverse well-ordered support X and coefficients $c_{i,a}$. The doubly indexed family

$$(c_{i,a} \Psi(\omega^a))_{(i,a)}$$

is summable. Indeed, its total support is contained in the reverse well-ordered union of the supports of $(\Psi(\omega^a))_{a \in X}$. At a fixed output exponent only finitely many a contribute, and for each such a only finitely many i have $c_{i,a} \neq 0$. Finite coefficient sums can therefore be regrouped in either order. This gives

$$\Psi_{\text{at}}\left(\sum_i x_i\right) = \sum_{a \in X} \left(\sum_i c_{i,a}\right) \Psi(\omega^a) = \sum_i \Psi_{\text{at}}(x_i).$$

The same argument proves summability of the final family. Thus Ψ_{at} is strong. Any strong map with the prescribed monomial values must satisfy (24.3), proving uniqueness.

Differences of two $<\nu$ -strong linear maps are $<\nu$ -strong: the union of two summable output supports is reverse well ordered, and local finiteness and the sum identity survive subtraction. Finally, for a series with support of size less than ν , the defining normal-form sum itself is covered by the hypothesis on Ψ , so $\Psi = \Psi_{\text{at}}$ there. $\square$

We call Ψ_{gh} a *monomial-invisible remainder*, or briefly a ghost map. This is just terminology for a linear map that vanishes on all monomials; it asserts no extra geometric object. In the local formula (24.2), Ψ_{at} is exactly the sum of the principal ultrafilter terms, and Ψ_{gh} is the sum of the nonprincipal terms.

[merge] For $\Psi = J$: the uniqueness clause is that of Theorem 4.3, so $J_{\text{at}} = H_J$ and $J_{\text{gh}} = D_J$ (Definition 4.2); and the last clause, with $\nu = \kappa$, is the inclusion $\mathcal{S}_{<\kappa} \subseteq \text{Eq}(J, H_J)$ of Theorem 5.1. The reverse inclusion, and the exact missing-index formula, are special to J and are not consequences of Theorem 24.4. The theorem also extends Lemma 4.1 in the direction of uniqueness: a reindexing map is the canonical strong part of every countably strong map with its monomial values.

Proposition 24.5 (Algebra of the decomposition [10]). *The assignment $\Psi \mapsto \Psi_{\text{at}}$ is a linear idempotent projection from countably strong linear maps onto strong linear maps. Ghost maps form a left ideal under composition with countably strong endomorphisms. If a nonzero ghost map exists, this projection is not, in general, a ring homomorphism for composition.*

Proof. Linearity and idempotence follow from uniqueness in Theorem 24.4. If Υ vanishes on monomials, then $\Psi \circ \Upsilon$ does as well, and composition preserves countable strongness. For the last assertion choose y with $\Upsilon(y) \neq 0$ and let mul_y be multiplication by y , a strong map. Then

$$(\Upsilon \circ \text{mul}_y)_{\text{at}}(1) = \Upsilon(y) \neq 0, \quad \Upsilon_{\text{at}} \circ (\text{mul}_y)_{\text{at}} = 0.$$

Thus the projection cannot preserve this product. $\square$

24.3 Multiplicative maps and derivations

Theorem 24.6 (The canonical strong part respects finite algebra [10]). *If $E : K \rightarrow K'$ is a countably strong coefficient-fixing field embedding, then E_{at} is a strong coefficient-fixing field embedding. If $\partial : K \rightarrow K$ is a countably strong k -linear derivation, then ∂_{at} is a strong derivation, and $\partial - \partial_{\text{at}}$ is a derivation vanishing on all monomials.*

Proof. For monomials, multiplicativity gives $E(\omega^{a+b}) = E(\omega^a)E(\omega^b)$. For arbitrary x, y , expand both normal forms. The family of pairwise products of their image monomials is summable by Hahn multiplication and the summability already proved in Theorem 24.4. Regrouping finite contributions at each exponent therefore yields $E_{\text{at}}(xy) = E_{\text{at}}(x)E_{\text{at}}(y)$. Also $E_{\text{at}}(1) = 1$ and coefficients are fixed. A unital homomorphism out of a field is injective.

For a derivation, use on monomials the identity

$$\partial(\omega^{a+b}) = \omega^a \partial(\omega^b) + \partial(\omega^a) \omega^b.$$

The same summable double expansion extends this to arbitrary x, y for ∂_{at} . Subtracting two derivations gives the last assertion. $\square$

For a field embedding put $\Upsilon = E - E_{\text{at}}$. Its failure to be a derivation has the exact, finite identity

$$\Upsilon(xy) = E_{\text{at}}(x)\Upsilon(y) + \Upsilon(x)E_{\text{at}}(y) + \Upsilon(x)\Upsilon(y). \quad (24.4)$$

The quadratic final term must not be dropped. In particular, the additive examples below do not automatically integrate into field automorphisms. [merge] For $E = J$ one has $E_{\text{at}} = H_j$ and $\Upsilon = D_j$, and (24.4) is 06's and 09's symmetric product law (14.3), equivalent to 07's twisted law (14.2); manuscript 10 proves it for every countably strong coefficient-fixing field embedding.

Proposition 24.7 (Omnific preservation passes to the strong part [10]). *Suppose $E : \mathbf{No} \rightarrow \mathbf{No}$ is a countably strong $\mathbb{R}$ -linear field embedding and $E(\mathbf{Oz}) \subseteq \mathbf{Oz}$. Then E_{at} preserves and reflects $\mathbf{Oz}$. Both E and E_{at} preserve constant terms.*

Proof. Inside $\mathbf{Oz}$ one has

$$\mathfrak{P}_{\mathbf{No}} = \bigcap_{n \geq 1} n\mathbf{Oz}.$$

Indeed, a purely infinite series is divisible by every positive integer, whereas its ordinary integer constant term can be divisible by all such integers only if it is zero. Hence $E(\mathfrak{P}_{\mathbf{No}}) \subseteq \mathfrak{P}_{\mathbf{No}}$. Termwise summation shows the same for E_{at} .

Both embeddings fix $\mathbb{R}$ and preserve order, since the positive elements of a real-closed field are its nonzero squares. Thus they send ordinary infinitesimals to ordinary infinitesimals, which here means $\mathfrak{m}$ to itself. Applying either embedding to $x = x_{\mathfrak{p}} + r + x_{\mathfrak{m}}$ respects these three additive summands, so its constant term remains r . For $E_{\text{at}}(x)$ to be omnific requires $r \in \mathbb{Z}$ and $E_{\text{at}}(x_{\mathfrak{m}}) = 0$. Injectivity makes the latter equivalent to $x_{\mathfrak{m}} = 0$. This proves reflection as well as preservation. $\square$

[merge] The divisibility description of the purely infinite part is also the omnific-preserving report's `opa:lem:divisible`. The last paragraph of the proof applies verbatim to E , so E reflects $\mathbf{Oz}$ as well; 10 states reflection only for E_{at} . For $E = J$ both statements are Theorem 11.1.

25 Measurable transvections, set-sized workspaces and the surcomplex transfer [10]

We now realize every allowable first-failure cardinal explicitly. Unlike a universe ultrapower, the construction needs only a measure as a set parameter and coefficient extraction from ordinary set-supported normal forms.

25.1 A bounded support interval

Fix a measurable cardinal κ and a nonprincipal κ -complete ultrafilter $\mathcal{U}$ on κ . Choose $\delta \in \mathbf{No}$ with $\delta \geq 1$, and put, with $q_\alpha = \omega^{-\alpha}$ as in (8.4),

$$X_\delta = \{\delta + q_\alpha : \alpha < \kappa\}, \quad \delta < \delta + q_\alpha \leq \delta + 1. \quad (25.1)$$

The expressions $\omega^{-\alpha}$ inside the *exponents* use the Conway omega-map. The exponents $\delta + q_\alpha$ strictly decrease in α . Thus X_δ is a reverse well-ordered set of cardinality κ , contained in the bounded positive interval $(\delta, \delta + 1]$. There is no improper increasing-support Conway sum here: the Conway exponents $\delta + q_\alpha$ decrease along the index. (In 10's convention the t -exponents $d - \omega^{-\alpha}$, $d = -\delta$, increase; see Section 22.4.)

Define the scalar functional

$$\vartheta_{\mathcal{U}}(x) = \text{ev}_{\mathcal{U}}([\omega^{\delta+q_\alpha}]x)_{\alpha < \kappa}. \quad (25.2)$$

Existence and uniqueness of this real scalar follow from Lemma 23.6. No analytic limit is being taken.

Lemma 25.1 (The detector functional [10]). *The map $\vartheta_{\mathcal{U}} : \mathbf{No} \rightarrow \mathbb{R}$ is $\mathbb{R}$ -linear and $<\kappa$ -strong. It vanishes on every series having fewer than κ terms, and in particular on every monomial. For $A \subseteq \kappa$ the shifted omnific mask*

$$\omega^\delta Y_A = \sum_{\alpha \in A} \omega^{\delta+q_\alpha} \quad (25.3)$$

(09's mask Y_A of (10.9), multiplied by ω^δ) lies in $\mathfrak{P}_{\mathbf{No}} \subseteq \mathbf{Oz}$, and

$$\vartheta_{\mathcal{U}}(\omega^\delta Y_A) = \begin{cases} 1, & A \in \mathcal{U}, \\ 0, & A \notin \mathcal{U}. \end{cases}$$

Its first strong-sum failure has cardinality κ .

Proof. Coefficient restriction to X_δ carries every summable family to a point-finite family in $\mathbb{R}^\kappa$. The converse direction of Theorem 23.7, applied to the single κ -complete ultrafilter $\mathcal{U}$, proves $<\kappa$ -strongness. A nonprincipal κ -complete ultrafilter contains no set of cardinality less than κ ; intersect the complements of its singletons to see this. Hence a coefficient vector supported on fewer than κ indices has evaluation zero. The mask formula is evaluation of its indicator function. Each individual $\omega^{\delta+q_\alpha}$ has image zero, but their κ -term sum has image one, which proves the exact failure cardinal. $\square$

25.2 The symmetry and all its preservation properties

Set $N_{\mathcal{U}}(x) = \vartheta_{\mathcal{U}}(x) \omega^\delta$. Since $\delta \notin X_\delta$, one has $\vartheta_{\mathcal{U}}(\omega^\delta) = 0$ and therefore $N_{\mathcal{U}}^2 = 0$. For $c \in \mathbb{R}$ define

$$\Theta_{c, \mathcal{U}}(x) = x + c N_{\mathcal{U}}(x) = x + c \vartheta_{\mathcal{U}}(x) \omega^\delta. \quad (25.4)$$

Theorem 25.2 (Ordered omnific-preserving transvections [10]). *For every $c \in \mathbb{R}$, the map $\Theta_{c,\mathcal{U}}$ is an $\mathbb{R}$ -linear additive automorphism of $\mathbf{No}$, and*

$$\Theta_{c,\mathcal{U}}\Theta_{c',\mathcal{U}} = \Theta_{c+c',\mathcal{U}}, \quad \Theta_{c,\mathcal{U}}^{-1} = \Theta_{-c,\mathcal{U}}.$$

It preserves leading exponents, leading coefficients, order, constant terms, and each of $\mathfrak{P}_{\mathbf{No}}$, $\mathbb{R}$, $\mathfrak{m}$ and $\mathbf{Oz}$ setwise. It fixes every monomial and every series with fewer than κ terms. For $c \neq 0$ it has first strong-sum failure exactly κ and is not multiplicative. Its canonical strong part is the identity.

Proof. Nilpotence gives the group and inverse formulas by finite algebra. All these maps are $<\kappa$ -strong by Lemma 25.1. Suppose $\vartheta_{\mathcal{U}}(x) \neq 0$. Then x has a nonzero coefficient at some exponent $\delta + q_\alpha > \delta$, so its leading exponent exceeds δ . The correction in (25.4) has exponent δ when nonzero. It cannot change the leading term. If $\vartheta_{\mathcal{U}}(x) = 0$, the map does not change x at all. Thus all leading terms are preserved. Additivity then gives order preservation by applying the leading-sign assertion to differences.

The correction has positive exponent $\delta \geq 1$, so it sends $\mathfrak{P}_{\mathbf{No}}$ into itself and does not change the constant term. The functional is zero on real constants and on infinitesimals, so both of those summands are fixed pointwise. The same facts hold for the inverse. They prove setwise preservation of the three summands and of $\mathbf{Oz} = \mathbb{Z} \oplus \mathfrak{P}_{\mathbf{No}}$. Fixing of short-support series and monomials follows from Lemma 25.1.

Put $y = \omega^\delta Y_\kappa$. Then

$$\Theta_{c,\mathcal{U}}(y) = y + c \omega^\delta, \quad \sum_{\alpha < \kappa} \Theta_{c,\mathcal{U}}(\omega^{\delta+q_\alpha}) = y.$$

For $c \neq 0$ this is a failure of size κ , while smaller families are preserved. Fixing every monomial implies $(\Theta_{c,\mathcal{U}})_{\text{at}} = \text{id}$ by Theorem 24.4.

Finally $\omega^\delta y = \omega^{2\delta} Y_\kappa$ has support $\{2\delta + q_\alpha : \alpha < \kappa\} \subseteq (2\delta, 2\delta + 1]$, disjoint from $X_\delta \subseteq (\delta, \delta + 1]$ because $\delta \geq 1$ gives $2\delta \geq \delta + 1$. Thus $\vartheta_{\mathcal{U}}(\omega^\delta y) = 0$, and $\Theta_{c,\mathcal{U}}(\omega^\delta y) = \omega^\delta y$. On the other hand,

$$\Theta_{c,\mathcal{U}}(\omega^\delta) \Theta_{c,\mathcal{U}}(y) = \omega^\delta (y + c \omega^\delta) = \omega^\delta y + c \omega^{2\delta} \neq \omega^\delta y.$$

This proves nonmultiplicativity, rather than merely leaving it unspecified. $\square$

The image of every set-indexed summable family under $\Theta_{c,\mathcal{U}}$ is still summable. Its exact discrepancy is

$$\Theta_{c,\mathcal{U}}\left(\sum_i x_i\right) - \sum_i \Theta_{c,\mathcal{U}}(x_i) = c \omega^\delta \left(\vartheta_{\mathcal{U}}\left(\sum_i x_i\right) - \sum_i \vartheta_{\mathcal{U}}(x_i)\right). \quad (25.5)$$

The scalar sum on the right is finite. Indeed, if infinitely many $\vartheta_{\mathcal{U}}(x_i)$ were nonzero, a countably infinite subfamily would contradict countable strongness of $\vartheta_{\mathcal{U}}$.

Corollary 25.3 (Recovering the measure from the symmetry [10]). *For $c \neq 0$, the action of $\Theta_{c,\mathcal{U}}$ on the masks $\omega^\delta Y_A$ determines $\mathcal{U}$:*

$$A \in \mathcal{U} \iff \Theta_{c,\mathcal{U}}(\omega^\delta Y_A) - \omega^\delta Y_A = c \omega^\delta.$$

For fixed δ and nonzero c , different measures give different additive automorphisms.

Proof. Apply the mask formula of Lemma 25.1. $\square$

This report already recovers derived measures from normal-form defects of an elementary embedding (Theorem 10.1, with 06's and 09's codes in Theorem 10.2 and Theorem 10.3). Here the same coefficient philosophy is realized by a rank-one *additive automorphism* with fixed monomials and preserved order; it is not presented as a new construction of the measure itself. [merge] The detector reads 09's omnific masks shifted by ω^δ ; no embedding is involved, and the measure is a set parameter rather than a derived one.

25.3 Arbitrary set parameters can be fixed

Theorem 25.4 (Parameter avoidance [10]). *For any set $Z \subseteq \mathbf{No}$, the choice of δ in (25.1) can be made so that every $\Theta_{c,\mathcal{U}}$ fixes Z pointwise. All conclusions of Theorem 25.2 remain valid.*

Proof. The union E_Z of the supports of the elements of Z is a set of surreals. Every set of surreals has a strict upper bound; for example, an ordinal larger than the birthdays of its members, increased by one. Choose $\delta \geq 1$ strictly above every member of E_Z . Then $E_Z < \delta < X_\delta$, so the coefficient restriction of every $z \in Z$ to X_δ is zero. Consequently $\vartheta_{\mathcal{U}}(z) = 0$. $\square$

Corollary 25.5 (An additive definability obstruction [10]). *Assume a measurable cardinal. In the additive ordered structure $(\mathbf{No}, +, <, \mathbf{Oz})$, multiplication and the coefficient relation are not definable from any prescribed set of surreal parameters, even after naming all real scalars and all Conway monomials individually. The same is true after adding the real scalar-action operations.*

Proof. Choose a nontrivial parameter-fixing $\Theta_{c,\mathcal{U}}$ as in Theorem 25.4. It is an automorphism of all the indicated additive structure and fixes each named monomial and scalar. It is not multiplicative, so a definable multiplication would not be invariant under this automorphism. For the coefficient relation, $y = \omega^\delta Y_\kappa$ has coefficient zero at ω^δ , whereas $\Theta_{c,\mathcal{U}}(y) = y + c \omega^\delta$ has coefficient c there; the monomial ω^δ and both scalars are fixed. Thus that relation is not invariant either. Naming a proper class of constants is understood schematically: every individual formula still uses only finitely many constant symbols. $\square$

This is a result about the *additive* reduct. The omnific-preserving report reconstructs much more from the full ring or field pair [30]; no contradiction is involved. [merge] Without a measurable cardinal a witness of the same kind would have to fail a countable sum: an $\mathbb{R}$ -linear countably strong map fixing every monomial has canonical strong part the identity (Theorem 24.4), so by Corollary 24.2 it is the identity. This bears on Question 26.18 without answering it.

25.4 Set-sized workspaces and a sharp cardinal bound

The measurable obstruction is not an artifact of proper classes. It is already visible in ordinary Hahn fields, provided their allowed supports are sufficiently long.

Theorem 25.6 (Exact full-Hahn workspace criterion [10]). *Let Γ be a set-sized ordered abelian group and $K = k((\omega^\Gamma))$, where $k = \mathbb{R}$ or $\mathbb{C}$. The following are equivalent:*

- (i) *every countably strong k -linear functional $K \rightarrow k$ is strong;*
- (ii) *every countably strong k -linear map from K into any full Hahn field over k is strong;*
- (iii) *no reverse well-ordered subset $X \subseteq \Gamma$ has cardinality at least an uncountable measurable cardinal.*

More precisely, a measurable cardinal κ occurs as the first failure of a countably strong k -linear map on K if and only if Γ contains a reverse well-ordered subset of cardinality κ .

Proof. Assume (iii). For any fixed support X , a nonprincipal countably complete ultrafilter on X would, by Lemma 23.9, yield a measurable cardinal at most $|X|$. Thus every ultrafilter in each local representation (24.2) is principal. Each output coefficient is then a finite coordinate functional on $K[X]$, so it commutes with every point-finite family supported on X . Together with Corollary 23.4, this proves (ii). Clearly (ii) implies (i).

If (iii) fails, take a measurable κ and a reverse well-ordered support with at least κ elements. Its first κ elements in the decreasing enumeration form a support $X_0 = \{s_\alpha : \alpha < \kappa\}$. Choose a nonprincipal κ -complete $\mathcal{U}$ on κ and define

$$\vartheta(x) = \text{ev}_{\mathcal{U}}([\omega^{s_\alpha}]x)_{\alpha < \kappa}.$$

The proof of Lemma 25.1 works without the special shape of X_δ : ϑ is $<\kappa$ -strong, vanishes on monomials, and takes value one on $\sum_{\alpha < \kappa} \omega^{s_\alpha}$. Hence its first failure is κ , disproving (i).

For the final necessity assertion, localize a least κ -failure as in Theorem 24.1. A κ -complete nonprincipal ultrafilter appears on the witness support X , so $|X| \geq \kappa$. Taking an initial segment gives a reverse well-ordered subset of cardinality κ . $\square$

For example, an ordered subgroup of $\mathbb{R}$ has only countable reverse well-ordered subsets. To see this, choose a rational number strictly between each nonfinal element of such a subset and its successor in the decreasing enumeration; the chosen rational interval witnesses are distinct. Hence all the implications above hold for those exponent groups in ZFC, even if the universe contains measurable cardinals elsewhere. The relevant issue is the size of an allowed support, not simply the cardinality of the ambient universe.

Theorem 25.7 (Sharp minimal size [10]). *Fix a measurable cardinal κ . Among full real Hahn fields admitting a countably strong linear map whose first failure is κ , the least possible field cardinality is 2^κ . This size is attained by a real-closed Hahn field that admits an order- and leading-term-preserving additive automorphism stabilizing its canonical omnific-type ring and fixing every monomial, with first failure κ .*

Proof. Necessity in Theorem 25.6 gives a reverse well-ordered support of cardinality κ . All assignments of zero or one to that support are Hahn series, so the field has at least 2^κ elements.

For attainment, take $\delta = 1$ in (25.1) and let

$$G_\kappa = \text{span}_{\mathbb{Q}}(\{\delta\} \cup X_\delta) = \text{span}_{\mathbb{Q}}\{1 + q_\alpha, 1 : \alpha < \kappa\} = \bigoplus_{\alpha < \kappa} \mathbb{Q}q_\alpha \subseteq (\mathbf{No}, +),$$

where the last equality uses $q_0 = \omega^0 = 1$. This is a divisible ordered abelian *set* group of cardinality κ : the generators already contain κ distinct elements and finite rational combinations give at most κ elements. The field $K_\kappa = \mathbb{R}((\omega^{G_\kappa}))$ has cardinality at most $|\mathbb{R}|^\kappa = 2^\kappa$, and the support X_1 gives the reverse inequality. The real closedness is the classical Hahn-field theorem for a real-closed coefficient field and divisible value group; equivalently, the full complex Hahn field on that group is algebraically closed, and its real part is real closed [2, 29].

The proof of Theorem 25.2 takes place entirely inside K_κ : all detector exponents $1 + q_\alpha$, the correction exponent 1 and the exponents $2 + q_\alpha$ of the product test belong to G_κ , and the map adds only one allowed monomial. It therefore provides the asserted additive automorphism of the field and of its omnific-type ring $\mathcal{O}_{G_\kappa} = \mathbb{Z} \oplus \mathfrak{P}_{K_\kappa}$. $\square$

[merge] 10 defines its group as the rational span of d and the $d - \omega^{-\alpha}$ with $d = -1$ in the t -convention; after the sign change of Section 22.4 it is exactly 09's G_κ , and 10's field is 09's K_κ of Theorem 13.5. That theorem proves $|K_\kappa| = 2^\kappa$ under $2^{<\kappa} = \kappa$, which holds at a measurable cardinal; the upper bound above, $(2^{\aleph_0})^\kappa = 2^\kappa$, needs no hypothesis. The same field therefore carries both 09's 2^κ algebraically independent omnific masks and 10's additive transvections.

No assertion that this set-sized ring has the entire Hahn field as its fraction field is needed. That property can fail for set-sized exponent groups (`opa:as:ex:smallfrac`) and must not be imported from the full surreal case.

25.5 Surcomplex and Gaussian-omnific consequences

All arguments through Theorem 25.6 apply directly to $\mathbf{SC} = \mathbb{C}((\omega^{\mathbf{No}}))$. Countable completeness makes every complex coefficient function almost constant, by Lemma 23.6.

Theorem 25.8 (Surcomplex transfer [10]). *Every countably strong $\mathbb{C}$ -linear map $\Psi : \mathbf{SC} \rightarrow \mathbf{SC}$ is either strong or has measurable first failure. Every measurable κ is realized by a $\mathbb{C}$ -linear additive automorphism Θ that preserves the Hahn valuation and leading coefficients, stabilizes $\mathbf{Oz}[i]$, fixes all ordinary complex numbers and every monomial, and first fails at κ . The automorphism can commute with conjugation and fix any prescribed set of surcomplex parameters.*

Proof. The first assertion is Theorem 24.1. For the construction replace real coefficient evaluation in (25.2) by complex evaluation and keep $\delta \geq 1$. The maps

$$\Theta_{c,\mathcal{U}}(z) = z + c \vartheta_{\mathcal{U}}(z) \omega^\delta$$

are $\mathbb{C}$ -linear additive automorphisms for $c \in \mathbb{C}$. The valuation argument still proves leading-term preservation. Their corrections are purely infinite complex series, so they preserve $\mathbb{Z}[i] \oplus \mathfrak{P}_{\mathbf{SC}}$ in both directions. Take $c \in \mathbb{R}$ to make the maps commute with conjugation; this follows either from almost-constant fibers or from coefficientwise real and imaginary parts. Parameter avoidance uses the union of the supports of the prescribed parameters, exactly as in Theorem 25.4. The same mask and product calculations prove exact failure and nonmultiplicativity. $\square$

These are not field homomorphisms, and no unitarity assertion is made. In fact they are locally translations on infinitesimal neighborhoods: for $v(h) > 0$, one has $\vartheta_{\mathcal{U}}(h) = 0$ and $\Theta_{c,\mathcal{U}}(z + h) = \Theta_{c,\mathcal{U}}(z) + h$. Thus even this local affine regularity is compatible with failure of a large Hahn sum. No global entire-series representation is asserted. Preservation of leading coefficients is much weaker than preservation of a complex modulus or an inner product. In particular, this theorem does *not* settle the omnific-preserving report's question whether every Gaussian-omnific *ring* automorphism preserves the natural valuation (`opa:as:q:gaussian`). [merge] The transfer of J and H_j to $\mathbf{SC}$ is Theorem 12.1; the maps here are not of that kind.

26 Field embeddings, target tests, and what manuscript 10 answers [10]

This section combines the lower bound with the elementary-embedding construction of the preceding sections of this report. The ultrapower construction, its normal-form interpretation, and its strong companion are not new here. The new role is that the general first-failure theorem makes their measurable hypothesis optimal among embeddings that are countably strong.

26.1 The elementary-embedding input, recalled

Proposition 26.1 (Existing construction [06, 07, 09, 10]). *A measurable cardinal κ gives a coefficient-fixing ordered-field embedding $J : \mathbf{No} \rightarrow \mathbf{No}$ that preserves and reflects $\mathbf{Oz}$, is $<\kappa$ -strong, and first fails strong summation at κ . On monomials it satisfies $J(\omega^a) = \omega^{J(a)}$. Its canonical strong part in Theorem 24.4 is the strong companion H_j .*

Proof pointers. Take the normal-measure ultrapower $j : V \rightarrow M$, an instance of (1.1). The embedding, fixing of $\mathbb{R}$ and the monomial formula are Proposition 2.10; preservation and reflection of $\mathbf{Oz}$ are Theorem 11.1; $<\kappa$ -strongness is Theorem 8.1. For the first failure, $J(h_\kappa) = h_{j(\kappa)}$ while $\sum_{\alpha < \kappa} J(\omega^{-\alpha}) = h_\kappa$ (Theorem 7.1);

the coefficient at $\omega^{-\kappa}$ is one on the left and zero on the right (Corollary 8.3). The termwise formula (1.3) for H_j is exactly (24.3), and uniqueness of the strong extension proves the last assertion (Theorem 4.3). $\square$

10 recalls the same construction with its witness (its Proposition 10.1: apply j to surreal sign codes; the sign-code arithmetic and normal-form absoluteness package of Lemma 2.6 identifies $\mathbf{No}^M$ with a subfield of $\mathbf{No}$; a family of fewer than κ members has $j(I) = j \text{ “ } I$) and says it is not new; it is printed once, here with pointers.

The class notation in this construction does not posit an elementary self-embedding $V \rightarrow V$. The codomain is the transitive ultrapower class M . Nor does it turn the proper embedding J into an automorphism. The first-failure and additive-transvection theorems of the earlier sections do not depend on the imported absoluteness package.

Theorem 26.2 (Exact boundary for countably strong embeddings [10]). *Over the stated class-theoretic conventions, the following are equivalent:*

- (i) *there are no measurable cardinals;*
- (ii) *every countably strong $\mathbb{R}$ -linear map $\mathbf{No} \rightarrow \mathbf{No}$ is strong;*
- (iii) *every countably strong $\mathbb{R}$ -linear ordered additive automorphism of $\mathbf{No}$ stabilizing $\mathbf{Oz}$ and fixing every monomial is strong;*
- (iv) *every countably strong coefficient-fixing field embedding $\mathbf{No} \rightarrow \mathbf{No}$ preserving $\mathbf{Oz}$ is strong.*

Moreover, for each uncountable ν , absence of measurable cardinals at least ν is equivalent to the corresponding universal strongness statement for $<\nu$ -strong linear maps. The same $<\nu$ equivalence holds for the field embeddings in (iv).

Proof. The first-failure theorem gives (i) $\Rightarrow$ (ii), and (ii) implies (iii) and (iv). A measurable cardinal contradicts (iii) by Theorem 25.2, and contradicts (iv) by Proposition 26.1. For the $<\nu$ statements, any first failure must be a measurable $\kappa \geq \nu$. Conversely a measure on such a κ supplies a $<\kappa$ -strong, hence $<\nu$ -strong, counterexample of either type. $\square$

[merge] Under (i) every map in (iii) is in fact the identity: it is strong, $\mathbb{R}$ -linear and fixes every monomial, so it equals its canonical strong part, which is the identity (Theorem 24.4).

Remark 26.3 (The hypothesis not removed [10]). The theorem does not say that every nonstrong omnific-preserving field embedding requires a measurable cardinal. It says this for embeddings that already preserve every countable Hahn sum. A counterexample failing at a countable family would fall outside the theorem. Deciding whether such a coefficient-fixing omnific-preserving field embedding exists is a separate question, left open here (Question 26.8).

[merge] *The countable hypothesis is not removable, and there is no conflict with Kaplan–Krapp–Serra.* Their Example 4.4 [5] builds, in NBG with global choice and without large cardinals, a field automorphism σ of $\mathbf{No}$ that fixes every real number and every monomial ω^y but is not strongly additive. Its failing family is countable: σ moves a sum a of an ω -indexed family of monomials, the pseudolimit of the ω -sequence of its partial sums over the Puiseux field, to $a + \omega^{-\omega}$ while fixing every summand. So σ is $\mathbb{R}$ -linear and not countably strong, $\text{fail}(\sigma) = \aleph_0$ in the sense of (22.1), and Theorem 24.1 does not apply to it. It shows that the countable hypothesis cannot be dropped even for field automorphisms fixing every monomial. By `opa:as:thm:main` (and `opa:as:cor:monodet`) σ does not preserve $\mathbf{Oz}$, so it does not bear on Question 26.8. (As printed in [5, Example 4.4] the family is $\sum_n \omega^{-1/n}$, whose exponents increase and so do not form a Conway normal form; a sign slip such as $\sum_n \omega^{1/n}$ appears to be meant. In any reading the pseudo-Cauchy sequence is indexed by ω ,

so the failing family is countable.) The statements “one should not infer a large cardinal from an arbitrary non-strong field map” in Sections 1.4 and 19 are consistent with this and are re-scoped there.

26.2 An explicit unrepresentable target test

For a coefficient-linear embedding E , a target element y has a *source multiplier representative* if there exists $a \in \mathbf{No}$ such that

$$\text{ct}(yE(x)) = \text{ct}(ax) \quad \text{for every } x \in \mathbf{No}. \quad (26.1)$$

Question `opa:as:q:targettests` of the omnific-preserving report asks whether all target tests can be represented for the $\mathbb{R}$ -linear embeddings preserving the omnific predicate.

Theorem 26.4 (Conditional negative answer for target tests [10]). *Assume a measurable cardinal and take J from Proposition 26.1. The target element $y = \omega^\kappa$ has no source multiplier representative. The pulled-back functional*

$$\Lambda_j(x) = \text{ct}(\omega^\kappa J(x)) = [\omega^{-\kappa}] J(x)$$

is $<\kappa$ -strong, vanishes on every monomial, and takes value one on h_κ .

Proof. No surreal g satisfies $J(g) = \kappa$. If it did, elementarity and absoluteness of the property of a sign code being an ordinal would imply that g is an ordinal. But an ordinal below κ is fixed, whereas an ordinal at least κ has image at least $j(\kappa) > \kappa$. Thus κ , and hence $-\kappa$, is not in the range of J on exponents.

Since $J(\omega^g) = \omega^{J(g)}$, it follows that $\Lambda_j(\omega^g) = 0$ for every g . On the other hand $J(h_\kappa) = h_{j(\kappa)}$ gives $\Lambda_j(h_\kappa) = 1$. If (26.1) held for some a , testing all ω^g would give $[\omega^{-g}] a = 0$ for every g , and hence $a = 0$. Testing h_κ would then say $1 = 0$. The asserted $<\kappa$ -strongness follows by composing the $<\kappa$ -strong J with coefficient extraction. $\square$

Second route to $-\kappa \notin J(\mathbf{No})$ [merge]. By Theorem 5.1 applied to h_κ , the exponent $-\kappa$ of the leading term of $D_j(h_\kappa)$ ((7.4)) is a missing-image exponent and lies outside $\Gamma_j = J(\mathbf{No})$. $\square$

This resolves the universal target-test assertion negatively *under a measurable-cardinal hypothesis*. It is not a ZFC construction without that hypothesis, and it does not settle the narrower question with *strong* field embeddings in place of merely coefficient-fixing embeddings. The distinction also separates this example from the omnific-preserving report’s strongly linear proper-class functional that has no representing surreal (`opa:as:thm:classdefect`): Λ_j here is not strong and is invisible on all monomials, whereas that earlier functional is strong and has proper-class monomial data.

[merge] Three remarks place the theorem in this report. First, $\text{supp}(H_j(x)) \subseteq \Gamma_j \not\ni -\kappa$, so $\Lambda_j(x) = [\omega^{-\kappa}] D_j(x)$ for every x : Λ_j is the $\omega^{-\kappa}$ -row of the ghost map $J_{\text{gh}} = D_j$, and on the masks it is 07’s functional, $\Lambda_j(h_f) = \chi_j(h_f)$ (10.13). So Λ_j extends χ_j from masks to all of $\mathbf{No}$, and on $K[\{-\alpha : \alpha < \kappa\}]$ it is ev_{U_j} (Remark 23.10); recovery of the measure from masks is Theorem 10.1 and Theorem 10.3. Second, the negative answer to `opa:as:q:targettests` already followed from results recorded before 10’s pin: if every target test of J had a source multiplier representative, the representatives would form a constant-term adjoint, and “adjoint implies strong” holds for $\mathbf{No}$ (the note after `opa:as:thm:adjoint`), whereas J is not strongly additive (Remark 11.6, recorded in the omnific-preserving report’s status note on `opa:as:q:embeddingstrong`). What 10 adds is the explicit test ω^κ , its invisibility on every monomial, and its identification with the measure functional. Third, the sharper question for strong embeddings is Question 26.10.

26.3 Consequences for the questions of Section 21 [merge]

Manuscript 10 did not see 06's and 09's questions (Section 22.4); the consequences below are deduced here from its theorems and are recorded as status notes at the questions.

Corollary 26.5 (A numerical criterion without coefficient algebra [merge]). *Let κ be an uncountable cardinal. The following are equivalent.*

- (i) *The cardinal κ is measurable.*
- (ii) *There is an $\mathbb{R}$ -linear map $E : \mathbf{No} \rightarrow \mathbf{No}$ such that*
 - (a) $[\omega^{q\kappa}] E(\tau_\alpha) = 0$ *for every $\alpha < \kappa$ (for instance $E(\tau_\alpha) = \tau_\alpha$);*
 - (b) *E preserves every strongly summable family of cardinality less than κ ;*
 - (d) $[\omega^{q\kappa}] E(Y_\kappa) = 1$.

The letters are those of Theorem 10.10; its condition (c) is omitted.

No order, multiplicativity, omega-map compatibility or Hadamard condition is required. Under (i) the map J of a normal-measure ultrapower satisfies (ii), and so do all four conditions of Theorem 10.10.

Proof. (i) $\Rightarrow$ (ii) is the forward direction of Theorem 10.10. Assume (ii). Since κ is uncountable, (b) makes E countably strong. The family $(\tau_\alpha)_{\alpha < \kappa}$ is strongly summable with sum Y_κ ((8.5)). By Corollary 23.4 its image family is strongly summable, and by (a) and finiteness of coefficient sums its sum has coefficient zero at τ_κ , whereas $E(Y_\kappa)$ has coefficient one there by (d). So E fails on a family of size κ , while by (b) it preserves all smaller families. Thus $\text{fail}(E) = \kappa$, which is measurable by Theorem 24.1. $\square$

Corollary 26.6 (The mask criterion without the Hadamard product [merge]). *An uncountable cardinal κ is measurable if and only if there is an $\mathbb{R}$ -linear functional $\chi : \mathcal{B}_\kappa \rightarrow \mathbb{R}$ that vanishes on every $h_{\{\alpha\}}$, satisfies $\chi(h_\kappa) \neq 0$, and carries every point-finite family $(h_{f_\xi})_{\xi < \mu}$ with $\mu < \kappa$ to a scalar family with finitely many nonzero members and the sum $\chi(h_{\sum_\xi f_\xi})$.*

Proof. If κ is measurable, χ_j of a normal-measure ultrapower is ev_{U_j} under $h_f \leftrightarrow f$ (Theorem 10.7); the converse half of Theorem 23.7 gives the summation property, and $\chi_j(h_{\{\alpha\}}) = 0$, $\chi_j(h_\kappa) = 1$. Conversely, $\mathcal{B}_\kappa \cong \mathbb{R}^\kappa$ by (10.11), so Theorem 23.7 with $\nu = \kappa$ writes $\chi = \sum_p c_p \text{ev}_{\mathcal{U}_p}$ with κ -complete $\mathcal{U}_p$ and $m \geq 1$ because $\chi(h_\kappa) \neq 0$. A principal $\mathcal{U}_p$ at α would give $\chi(h_{\{\alpha\}}) = c_p \neq 0$. So $\mathcal{U}_1$ is a nonprincipal κ -complete ultrafilter on κ . $\square$

This trades the Hadamard homomorphism of Theorem 10.9 for preservation of all small point-finite sums of masks, not only unions of disjoint indicator masks; neither criterion is first-order in the field language.

Proposition 26.7 (Pairs of embeddings with a countably strong member [merge]). *Let κ be uncountable.*

- (a) *Let $P, Q : \mathbf{No} \rightarrow \mathbf{No}$ be field embeddings fixing $\mathbb{R}$ that agree on every monomial, with Q strongly $\mathbb{R}$ -linear and $\text{Eq}(P, Q) = \mathcal{S}_{<\kappa}$. If P is countably strong, then $Q = P_{\text{at}}$, P is not strong, and $\text{fail}(P)$ is a measurable cardinal at most κ , on whose short-support field $\mathcal{S}_{<\text{fail}(P)}$ the two maps agree. If P is moreover $<\kappa$ -strong, then $\text{fail}(P) = \kappa$ is measurable.*
- (b) *Let $P : \mathbf{No} \rightarrow \mathbf{No}$ be an ordered field embedding fixing $\mathbb{R}$ and commuting with the omega-map, and let T_P be its Hahn reindexing map (Lemma 4.1). If P is countably strong, then $T_P = P_{\text{at}}$; either $P = T_P$, or $\text{fail}(P)$ is measurable and $\mathcal{S}_{<\text{fail}(P)} \subseteq \text{Eq}(P, T_P)$. Without measurable cardinals every countably strong such P equals T_P .*

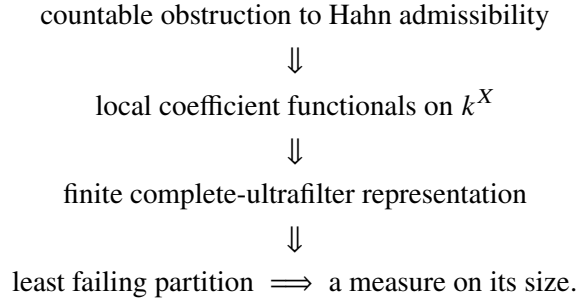
- (c) For every countably strong $\mathbb{R}$ -linear $\Psi : \mathbf{No} \rightarrow \mathbf{No}$ that is not strong, the null ideal of one output coefficient of Ψ on the support of a least failing family carries a nonprincipal $\text{fail}(\Psi)$ -complete ultrafilter on $\text{fail}(\Psi)$; the only structure used is summation.

Proof. (a) Q is strong and agrees with P on monomials, so $Q = P_{\text{at}}$ by uniqueness in Theorem 24.4. The mask h_κ has κ terms, so $h_\kappa \notin \mathcal{S}_{<\kappa} = \text{Eq}(P, Q)$ and $P(h_\kappa) \neq Q(h_\kappa) = \sum_{\alpha < \kappa} P(\omega^{-\alpha})$: P fails on a family of size κ . Hence P is not strong, $\text{fail}(P) \leq \kappa$, and $\text{fail}(P)$ is measurable by Theorem 24.1. Agreement on $\mathcal{S}_{<\text{fail}(P)}$ is the last clause of Theorem 24.4. If P is $<\kappa$ -strong, then $\text{fail}(P) \geq \kappa$. (b) Commutation with the omega-map gives $T_P(\omega^a) = \omega^{P(a)} = P(\omega^a)$, and T_P is strong, so $T_P = P_{\text{at}}$; apply Theorem 24.4, Theorem 24.1 and Corollary 24.2. (c) is the proof of Theorem 24.1 with Lemma 23.9. $\square$

Part (a) does not decide whether κ itself must be measurable when P is only countably strong: the equalizer condition bounds the first failure by κ but does not obviously force P to be $<\kappa$ -strong. Nor does it say anything about embeddings that fail a countable sum. Part (b) does not classify the equalizers asked for in Question 21.2; it only bounds them below.

26.4 Proof audit and limits of the conclusions [10]

The core route is



The additive converse uses the independent route

$$\mathcal{U} \implies \vartheta_{\mathcal{U}} \implies N_{\mathcal{U}}^2 = 0 \implies \Theta_{c, \mathcal{U}} = \text{id} + cN_{\mathcal{U}}.$$

Only the field-embedding equivalence and its particular target-test counterexample use the j -construction and its normal-form absoluteness. No field-automorphism classification is needed in the proof of the lower bound or in the additive construction.

Result	Status and input	Boundary
Product functional theorem	Specialized proof of the classical finite-ultrafilter mechanism; Bergman is credited.	Not a new general product-factorization theorem.
Least-failure spectrum	Proposed Hahn/surreal contribution; proved from support localization and the product theorem.	Requires coefficient linearity and countable strongness.
Canonical strong part	Proposed generalization of the particular companion construction H_j .	Projection is linear, generally not multiplicative for composition.
Ordered additive transvections	Explicit construction from a measurable cardinal; all preservation and failure claims proved.	They are deliberately not field homomorphisms.

Result	Status and input	Boundary
Workspace criterion and 2^κ bound	Exact linear criterion and matching example.	Full Hahn fields are essential; not arbitrary subfields.
Field-embedding equivalence	New lower bound combined with the existing J, H_j construction.	The unrestricted embedding problem is not settled.
Target-test obstruction	Explicit consequence for J , with a monomial test proof of nonrepresentation.	Conditional on measurability; not a counterexample among strong field embeddings.
Gaussian transfer	Direct complex-linear transfer of the core results.	No unrestricted Gaussian ring-automorphism theorem.

(10’s table; 10 wrote “the repository’s particular companion construction” for H_j .)

Common pitfalls explicitly excluded. The scalar target in the product-functional theorem has only finite strong sums. Replacing this by ordinary convergence of scalar series would change the theorem and could introduce real-valued measures rather than the complete ultrafilters used here. All source supports are sets. No formal sum indexed by the proper class of all ordinals is an element of **No**. The exponents $\delta + q_\alpha$ decrease; reversing a convention without changing signs would destroy well-ordering. The product space k^X is a coefficient vector space, not a Cauchy-product subring. A Boolean mask is not an idempotent of the surreal field. The ultrafilter evaluations in the proof are multiplicative only for the pointwise coefficient product. Nothing therefore turns the rank-one transvections into field homomorphisms. The word “measurable” denotes an uncountable cardinal carrying a κ -complete nonprincipal ultrafilter. 10 proves equivalences and conditional constructions, not the existence or consistency of such cardinals. The least-failure theorem gives a genuine measurable cardinal in the ambient universe, not merely one in an unspecified inner model. Strong summation is not ordinary valuation-topological convergence. The order-preserving counterexamples preserve valuation and leading terms; that does not force them to commute with all Hahn sums. Countable strongness is a substantial extra assumption even for a linear map that is order-preserving.

Verification supplied and not supplied. The companion program code/10-measurable-spectrum m-finite_regression.py uses exact rational coefficients and finite Laurent supports, written in 10’s t -convention. It tests the nilpotent transvection identities, inverses, leading-term preservation, constant-term preservation, preservation of the finite omnific-type subring, and an explicit failure of multiplicativity. It also checks finite weighted-coordinate formulas. These are regression tests for finite algebraic steps, not tests of the main large-cardinal statements. A finite set has no nonprincipal ultrafilter. Consequently no finite simulation can verify the monomial-invisible phenomenon, a measurable first failure, the normality of an infinite measure, a class elementary embedding, or the arbitrary-support assertions; the finite functional is a principal coefficient projection and does not vanish on every monomial. The central results require independent mathematical review; none has been formalized in Lean. [merge] The program was rerun on a copy for this merge: 11,955 assertions in 18 categories, output identical to the recorded files (Appendix D).

10’s checklist for independent proof review. A reviewer can check the core in the following order. First verify that countable subfamilies detect both clauses of Hahn summability. Next verify that the support-null ideal in Theorem 23.7 is closed under $<\nu$ unions and has a finite Boolean quotient; this must use the requirement that only finitely many scalar images are nonzero. Check that complete-ultrafilter evaluation of real and complex functions is actual almost-constant evaluation, not a limit requiring boundedness. In the

least-failure argument, verify that a κ -complete but not κ^+ -complete ultrafilter really appears, rather than just some nonprincipal countably complete ultrafilter. For the construction, check the convention $X_\delta \subseteq (\delta, \delta + 1]$, the fact that $\delta \notin X_\delta$, and $N_{\mathcal{U}}^2 = 0$. Leading-term preservation depends on the strict inequality between the leading exponent of x and δ whenever $\vartheta_{\mathcal{U}}(x) \neq 0$. The product counterexample depends on $\delta \geq 1$. Parameter avoidance chooses a new detector interval and must not try to choose a measure on a proper class. For the field applications, verify the imported normal-form absoluteness interface (Lemma 2.6) independently; those applications are the only part that needs it.

26.5 Questions of manuscript 10 [10]

The first four questions locate the remaining conceptual boundaries. The others suggest concrete extensions of the proofs. They are proposed research questions, not a list of named historical conjectures.

Question 26.8 (Remove the countable hypothesis for field embeddings). Must a coefficient-fixing field embedding $E : \mathbf{No} \rightarrow \mathbf{No}$ preserving $\mathbf{Oz}$ be countably strong? If not, construct one with a countable failure and determine the assumptions needed.

A positive answer would extend Theorem 26.2 to all such embeddings; the present proof does not supply that answer. [merge] The Kaplan–Kraep–Serra automorphism (Remark 26.3) fails a countable sum but does not preserve $\mathbf{Oz}$. A positive answer would also answer the first two clauses of `opa:as:q:embeddingstrong` for coefficient-fixing embeddings in universes without measurable cardinals.

Question 26.9 (The unrestricted Gaussian stabilizer). Does every ring automorphism of $\mathbf{Oz}[i]$ preserve the positive-support component of its fraction field?

This is the omnific-preserving report’s still-open question `opa:as:q:gaussian`. The complex additive examples here do not address it, since they fail multiplication.

Question 26.10 (Target tests for strong proper embeddings). For a strongly $\mathbb{R}$ -linear omnific-preserving field embedding, does every pulled-back constant-term test have a source multiplier representative?

Theorem 26.4 only answers the version allowing nonstrong embeddings. The proper-class scalar dual obstruction (`opa:as:thm:classdefect`) shows why a blanket linear duality argument is insufficient.

Question 26.11 (Global assembly of local ultrafilter data). Give necessary and sufficient compatibility conditions on the finite representations (24.2) for them to arise from one countably strong Hahn map. Then impose the finite multiplicative identities needed for a field embedding.

The local theorem alone does not control output supports or consistency under enlargement of X .

Question 26.12 (Classification of measurable transvections). Classify the maps $\Theta_{c,\mathcal{U}}$ up to conjugacy by strong omnific-preserving maps. Determine which equivalence relations on measures, such as pushforward equivalence of ultrafilters, are detected by this conjugacy.

The mask action already recovers $\mathcal{U}$ when the detector support is named (Corollary 25.3). [merge] Compare Questions 21.4 and 21.26, which ask for relations between measures through the defect functionals of J .

Question 26.13 (Nonstrong derivations). Can a measurable cardinal produce a nonzero countably strong derivation of $\mathbf{No}$ that vanishes on all monomials and on $\mathbb{R}$?

The canonical strong part of a derivation exists by Theorem 24.6, and a nonzero remainder would have measurable first failure. The rank-one additive construction does not satisfy the Leibniz rule. [merge] The defect D_j satisfies the twisted law (14.2), not the Leibniz rule; compare Questions 21.8 and 21.18.

Question 26.14 (Compatibility with exponential and differential structure). Which countably strong proper embeddings preserving $\mathbf{Oz}$ also commute with the Gonshor exponential, the Conway omega-map, or a chosen surreal derivation?

The critical-point construction supplies particular examples and constraints; a general classification should use the ultrafilter data rather than infer compatibility from additive behavior. [merge] For J and H_j the exponential and omega-map loci are Theorem 15.3 and Corollary 5.6; the derivation half is Questions 21.8, 21.18 and 21.30.

Question 26.15 (Support-restricted subfields). Extend the workspace criterion to truncation-closed or support-restricted subfields that do not contain every coefficient function on an allowed support. Identify precisely which product fragments still force a complete-ultrafilter factorization, especially at singular support bounds.

The full product k^X is crucial in the present proof. [merge] The short-support field $\mathcal{S}_{<\kappa}$ is such a subfield (Section 3); compare Question 21.28.

Question 26.16 (Other coefficient fields). For a general coefficient field k , replace scalar ultrafilter evaluation by linear maps from the ultrapowers $k^X/\mathcal{U}$ to k . Determine useful hypotheses under which a comparably explicit local representation and first-failure spectrum remain available.

The real and complex cases are special because countably complete ultrafilters make every coefficient function almost constant. [merge] Compare Question 21.29.

Question 26.17 (Cancellation and composition of defects). Describe how first-failure cardinals change under sums and compositions of countably strong maps.

They cannot become a nonmeasurable first failure, but different local nonprincipal terms may cancel. The canonical projection is not a composition homomorphism, as Proposition 24.5 shows.

Question 26.18 (Additive definability without measurability). Determine whether the additive nondefinability statement of Corollary 25.5 has a ZFC proof without measurable cardinals.

The current witnesses satisfy the much stronger countable-sum condition; a different additive automorphism may require less set theory. [merge] By the note after Corollary 25.5, in a universe without measurable cardinals such a witness fixing every monomial and commuting with the real scalar action must fail a countable sum.

Question 26.19 (Formalization and foundational weakening). Formalize the null-ideal/product theorem and the set-sized Hahn first-failure theorem first, and then the sharp workspace construction. Separate the use of dependent choice, ordinary choice, and class replacement.

The class-surreal applications should follow from an explicit normal-form interface rather than from an informal collection of all class maps. [merge] Compare Questions 21.12 and 21.34.

26.6 What manuscript 10 does not claim [10]

Every limitation stated by manuscript 10 is kept; the delivered proof-status and audit files are shipped unchanged. Manuscript 10 does not claim:

- 10.1 that every non-strong omnific-preserving field embedding requires a measurable cardinal: an embedding failing a countable sum is outside Theorem 24.1, and the countable hypothesis is essential (Remark 26.3, Question 26.8);
- 10.2 anything about unrestricted Gaussian-omnific ring automorphisms or their valuation preservation (`opa:as:q:gaussian`); its complex additive examples fail multiplication;
- 10.3 a target-test theorem for strong proper field embeddings: Theorem 26.4 is conditional on a measurable cardinal and concerns the non-strong J ;
- 10.4 a classification of support-restricted or truncation-closed Hahn subfields: full Hahn fields are essential to the workspace criterion;
- 10.5 a global assembly of local ultrafilter data: the local representation (24.2) has no converse;
- 10.6 that the additive maps $\Theta_{c,\mathcal{U}}$ preserve multiplication, the omega-map as a function, the Gonshor exponential or the Berarducci–Mantova derivation; no unitarity is asserted in the complex case;
- 10.7 that fixing every monomial as an element is compatibility with the omega-map as a function on moved exponents;
- 10.8 a global entire-series representation: the additive examples are only locally translations on infinitesimal neighborhoods;
- 10.9 that a set-sized canonical omnific-type ring $\mathcal{O}_\Gamma$ has the full Hahn field as its fraction field;
- 10.10 novelty for Bergman’s finite complete-ultrafilter mechanism or for the first-incomplete-partition argument: Theorem 23.7 is a specialized proof, not a new product-factorization theorem;
- 10.11 the existence or consistency of measurable cardinals: 10 proves equivalences and conditional constructions;
- 10.12 independence of Section 26 from this report’s normal-form absoluteness and elementary-embedding construction, which need independent review; Sections 23 to 25 do not use them;
- 10.13 any computational verification of an ultrafilter, a measurable cardinal, a transfinite sum, a proof or novelty: the finite checks use a principal functional that does not vanish on every monomial;
- 10.14 refereeing, a Lean formalization, or certified priority: it is an AI-assisted research draft, and its repository and literature comparisons were targeted, not exhaustive; the repository was not modified;
- 10.15 any contradiction with automatic strongness of **Oz**-preserving field automorphisms (`opa:as:thm:main`): the $\Theta_{c,\mathcal{U}}$ are additive, not multiplicative;
- 10.16 any infinite scalar summation: the scalar sums in the product theorem are finite, and replacing them by convergent series would change the theorem;
- 10.17 more than an additive-reduct statement in Corollary 25.5; the full ring or field pair reconstructs much more;
- 10.18 that its questions are published open problems;

10.19 that strong summation is valuation-topological convergence; countable strongness is a substantial extra assumption even for order-preserving maps.

Added by the merge:

- M10.1 Corollaries 26.5 and 26.6 and Proposition 26.7 are deductions made for this merge from 10's theorems; 10 does not state them. Corollary 26.5 does not decide whether condition (c) of Theorem 10.10 follows from the other conditions; it shows that the criterion does not need it.
- M10.2 The partial answers of Proposition 26.7 assume countable strongness; for embeddings that fail a countable sum, and for whether κ itself is measurable in (a), nothing is claimed.
- M10.3 The comparison with Kaplan–Krapp–Serra's Example 4.4 is a reading of that example, not a new theorem; the note on its displayed formula concerns notation only.

27 Conclusion

The difference between $j(I)$ and $j \text{ “ } I$ becomes a literal normal-form difference in surreal arithmetic. Comparing the elementary-induced map with the strongly linear map having the same monomial values exposes that difference without sacrificing finite algebra.

The outcome is stronger than a single counterexample to summation preservation. There is an exact equalizer, an exact intersection, a canonical isomorphism of two incomparable immediate image fields, and an intrinsic positive-family escape threshold. Their common field has two different canonical enlargements: logarithmic closure produces the elementary image, while strong-sum closure produces its Hahn companion. One coefficient of the difference recovers the derived normal measure; shifting the code makes it purely infinite and omnific. The same mechanism works over complex coefficients and at supercompact seeds.

These statements answer the structural questions posed in this article under a measurable-cardinal hypothesis. They are offered for independent verification and comparison with the literature, not as a claim that the large-cardinal hierarchy or a named longstanding conjecture has been settled. The strongest next step is to test the converse: how much of the set-theoretic embedding can be forced by the arithmetic signature alone?

[06] The more global issue is not whether the termwise series exists—it always exists in the ambient surreal field—but whether it exists in the target model. For every individual normal form that question is exactly seed membership. In a set-ultrapower it is exactly sequence closure. Normal measures give the strict/non-strict two-threshold picture, while fine and normal fine measures give support covers and exact descent at larger cardinals. The constructions remain meaningful in the surcomplex field and in the omnific-integer rings, where explicit coefficient codes recover normal measures. Thus large-cardinal hypotheses have a precise manifestation in arithmetic of canonical infinite supports, without invoking a new finite algebraic law, a blanket automorphism theorem, or an unproved consistency claim.

[09] The short-support field is not merely an incidental agreement set. It is real closed, has a canonical omnific integer part, and admits an explicit large algebraically independent family of long-support omnific extensions. At a measurable cardinal these are simultaneously independent witnesses to the failure of full strong additivity, and the measure itself appears as a coefficient of the same sort of number. The remaining challenge is to reduce the extra language needed to observe these effects. Normal-form support, coefficientwise multiplication, truncation trees and seed data make the set-theoretic information visible. The pure field language does not. The exact boundary between those two levels is a concrete research program rather than a reason to conflate ordinary surreal arithmetic with the entire large-cardinal hierarchy.

[10] The large-cardinal examples suggest that uncountable normal forms can contain information invisible to their individual monomials. Manuscript 10 identifies an exact converse in the countably strong linear setting. Every first loss of a sum has measurable size, every measurable size occurs, and the phenomenon can coexist with order, valuation, leading terms, the omnific predicate, and complete fixing of the monomial system—provided multiplication is not demanded of the additive symmetry. The decisive step is not a new consistency principle. It is localization: a single coefficient on a fixed support is a product functional, and its nonprincipal complete-ultrafilter components precisely account for what monomial data miss. This yields a canonical strong part, sharp set-sized realization bounds, and a conditional resolution of the unrestricted target-test question. The remaining multiplicative and Gaussian questions are now separated from this linear obstruction rather than obscured by it.

A Proof dependencies and a formalization plan

A.1 Dependency order

The proof does not use the repository’s unreviewed set-theoretic bi-interpretation or quotient theorems as assumptions. Its main dependency chain is

canonical surreal/Hahn theory + elementary embedding with critical point
 $\implies$ absolute normal forms and the missing-index lemma
 $\implies J, H_j$, exact defect and exponent-range exclusion
 $\implies$ equalizer, intersection, coefficient projection, strong escape
 $\implies$ omnific, surcomplex, and mask-coefficient consequences.

The logarithmic-defect homomorphism adds only the standard near-one Hahn logarithm. The exact exponential locus and two-closure theorem add the canonical Gonshor exponential and its purely infinite normal-form formula. The supercompact seed theorem adds the separate λ -sequence-closure hypothesis on M .

A.2 A proposed Lean interface, not a formalization claim

A realistic implementation should begin with a universe-bounded Hahn workspace, then state separately how that workspace represents canonical surreal normal forms. The following components are logically distinct.

Component	Obligation	Verification status here
Exponent reindexing	An additive order embedding preserves reverse well-ordered support, finite convolution, and summable families.	Written proof; repository describes related infrastructure.
Canonical bridge	Sign sequences, Conway arithmetic, the omega-map, and normal-form reconstruction agree with the Hahn representation.	Classical input; not rebuilt or kernel-checked here.
Universe absoluteness	Old normal forms and their strong sums are unchanged on passage from M to V .	Written reduction to the constructive sign-expansion theorem.
Embedding interface	Images of index functions; fixed coefficients; $j(\theta) \setminus j \text{ “ } \theta$; critical-index and range-reflection lemmas.	Written proofs; no Lean set-theory implementation supplied.

Component	Obligation	Verification status here
Defect theorem	Split a transformed summable family into image and missing-image indices; prove first-term and support exclusion.	Written proof; finite deletion identity checked separately.
Image geometry	Exact intersection, projection isomorphism, real closedness, immediacy, and strong-sum escape.	Written proofs only.
Measure interface	Indicator masks, Hadamard characters, completeness, normality, and supercompact seeds.	Written proofs; finite Boolean characters checked only as a finite analogue.
Exponential interface	Absolute recursive exponential, monomial values on purely infinite inputs, and infinitesimal power-series evaluation.	Classical input and written consequences; not formalized here.

The measurable-cardinal assumption must not be inserted as an unnamed “surreal summation axiom.” A useful interface can abstract the exact properties of j for the algebraic proofs, but the application to a normal ultrapower still needs an explicit model of the set-theoretic construction. That boundary is essential for avoiding a formal proof whose interesting mathematics has merely been assumed.

A.3 06’s dependency map and formalization route [06]

06’s main proof chain is

canonical normal forms and absoluteness $\implies (J, H_j)$ and their array formulas
 $\implies$ equalizer and missing-index defect,
array formulas and internal support extraction $\implies$ universal landing criterion,
landing criterion and ultrapower seed lemma $\implies$ sequence-closure detector,
landing criterion and index-size obstruction $\implies$ normal-measure trichotomy.

The coefficient-recovery theorem uses the array formula directly. The strong-compactness criterion uses support extraction and the classical cover seed; the supercompactness criterion adds exact seed membership. The exponential formula uses the equalizer theorem, infinitesimal formal evaluation, and the classical decomposition of Gonshor exponentiation. The surcomplex and omnific statements are restrictions or coefficient-field extensions of these arguments.

No Lean proof or automated verification of the new large-cardinal claims is included. The repository’s existing Hahn layer is relevant infrastructure, but its presence is not formal verification. A useful formalization should isolate four interfaces. First, prove the algebraic reindexing theorem for a Hahn field over a set-sized coefficient field and an ordered-group embedding, together with the finite-fiber proof of strong additivity. Second, formalize the array statement: retaining an injected subset of a nonzero ordered coefficient array gives equality precisely when no indices were lost, and the least omitted index determines the leading term of the defect. These are independent of large-cardinal foundations. Third, give an explicit normal-form/sign-sequence bridge with the absoluteness theorem for transitive universes. This is the most foundationally sensitive interface and should not be replaced by an unproved coercion between internal and external Hahn fields. Fourth, formalize ultrapower seeds, the representative-function proof of sequence closure, and the index-size union obstruction. The final theorems then become short compositions of these interfaces. Finite truncation experiments can check coefficient masking and convolution identities, but they cannot validate the

critical-point or nonprincipal-measure assertions: a finite set does not carry a nonprincipal ultrafilter. For that reason 06 includes no finite simulation advertised as evidence for the large-cardinal steps.

A.4 09’s dependency ledger and formalization boundary [09]

09’s main argument is organized so that neither a conjectural exponential identity nor a repository bi-interpretation theorem is needed.

Result	Required ingredients	Status in 09
Absolute normal forms, Lemma 2.6 (second route)	Absolute sign-prefix recursion; Conway normal forms; truncations are simpler	Written proof relative to standard foundations
Short-support field, Theorem 3.1	Real closed divisible-group Hahn fields; finite coefficient support bounds	Standard construction, proof included
Strong lift, Theorem 4.3	Increasing additive exponent embedding; finite Hahn convolution	Standard lifting mechanism, proof included
Exact discrepancy, Theorem 5.1	Normal-form transport; subseries partition; first missing index	Main proposed contribution
Summation threshold and omega obstruction	Exact discrepancy; explicit ordinal families	Proposed consequences with proofs
Independence, Theorem 13.3	Small rational coefficient span; private coordinates; maximum total degree	Main proposed contribution; no large cardinal required
2^κ family, Theorem 13.5	Independence theorem; binary-branch almost-disjoint family	Proposed application with explicit cardinal hypothesis
Normal and seed coefficient formulas	Absolute mask construction; standard seed ultrafilters	Explicit coding results; seed theory credited
Numerical measurability criterion, Theorem 10.10	Strong additivity on disjoint masks; Hadamard compatibility; measurable-cardinal construction	Infinitary criterion, not a bare-field equivalence
Strong/supercompactness interpretation	Fine and normal-fine ultrafilter characterizations	Established set theory transported to the codes
Omnific prefix principle	Binary coding of a κ -tree; inaccessible tree characterization	Established weak compactness transported to the codes

09’s mathematical proofs are written proofs, not Lean proofs. No finite program can test the existence of a measurable cardinal by enumerating examples, and no such test is offered as evidence. The repository contains checked Hahn and algebraic infrastructure, but 09’s claims have not been imported into that development or built against its Lean version. A future formalization should state the elementary-embedding interface as assumptions and separately verify which assumptions arise from the chosen set-theoretic construction.

B Verification, limitations, and an audit checklist

B.1 Finite regression checks

The accompanying program code/`finite_regression.py` uses only the Python standard library and exact rational arithmetic. It tests finite Laurent-polynomial reindexing, the deletion-of-indices coefficient identity,

the twisted product identity, a finite exponent shift, and the exhaustive Boolean-algebra characters of a four-point set. Results are saved in `data/finite_regression_results.json`. [merge] These are the delivery names. In this directory the program is `code/07-critical-point-defects-finite_regression.py` and its recorded outputs are `data/07-critical-point-defects-finite_regression_results.json` and `data/07-critical-point-defects-finite_regression_console.txt`. Run without arguments, the program writes to the delivery name `data/finite_regression_results.json`; rerun it on a copy with an explicit `-output` path. 06 and 09 ship no programs.

These tests can detect an incorrect sign, a wrong coefficient position, or confusion between pointwise and convolution products in a finite identity. They do not model a nontrivial elementary embedding of the universe. On a finite Boolean algebra every ultrafilter is principal; the exhaustive check confirms that fact and cannot certify the existence of a nonprincipal complete ultrafilter. There is no numerical experiment for measurability in this package.

The PDF is compiled from the standalone source and visually inspected. No Lean kernel, Wolfram kernel, or other proof assistant was used to verify this article’s new theorems. The source and finite program are provided so that those limited forms of verification can be reproduced.

B.2 Points requiring particular attention in an independent review

The most important review target is the passage from the internal normal form of $j(x)$ to its external canonical normal form. That passage uses Lemma 2.6 and the constructive, not arbitrary, Hahn identification. Next, verify that the missing-image exponents cannot lie in $J(\mathbf{No})$; this range-reflection argument is the source of both the intersection theorem and the intrinsic escape theorem.

Further checks concern the separation of outer support from birthday; the restriction to positive nonzero terms in the exact family-cardinality criterion; the distinction between a zero coefficient probe and a zero whole defect; and the use of Hadamard multiplication in the character theorem. In the exponential section, check the uniform recursive absoluteness and the monomial value of a purely infinite exponential. The supercompact theorem requires its actual seed to belong to M , not just an external set of the right apparent cardinality.

Every assertion about classes in this article reduces to a specified class map and set-indexed operations, or to a universal property of such maps. No proof uses a proper-class normal-form support. No claim is made that $H_j(\mathbf{No}) \subseteq \mathbf{No}^M$, or that $J(\mathbf{No}) = \mathbf{No}^M$; neither identification is part of the construction. All image and intersection statements occur inside the fixed ambient $\mathbf{No}^V$. [merge] Both identifications in fact fail for a normal-measure ultrapower: $H_j(x) \notin \mathbf{No}^M$ once x has more than κ terms (Theorem 9.7), and the ordinal $\kappa \in \mathbf{No}^M$ is not in $J(\mathbf{No})$ (Lemma 2.5).

B.3 06’s audit checklist [06]

Potential mistake	Correction used in the proofs
Replacing $j(\lambda)$ by $j \restriction \lambda$	They coincide exactly below the critical point. The first missing index for larger domains is κ .
Assuming termwise images belong to M	This is the conclusion of the landing theorem, not part of the definition of H_j .
Zero coefficients in a purported normal form	All input coefficients are nonzero. This makes every omitted term detectable and is essential to the equalizer theorem.
Identifying support size with birthday	A monomial has support size one regardless of the complexity of its exponent.

Potential mistake	Correction used in the proofs
Using arbitrary coefficientwise convergence as strong summation	Strong summability requires finite incidence at every exponent and reverse well-ordered union of supports.
Inferring a normal measure from any non-strong field embedding	The measure recovery theorem assumes an elementary universe embedding and its canonical transport.
Using external size in the cover condition	The bound $ s < j(\kappa)$ is evaluated in M . The union obstruction is then proved with external representatives.
Claiming $j(\kappa)$ is inaccessible in V	Only its inaccessibility in M is used; the power-set agreement yields the external lower bound needed in the proof.
Applying the closure converse to arbitrary embeddings	A set-ultrapower representation is explicitly required in Lemma 9.3.
Reading coefficientwise inclusion as numerical order	Binary supports, not their leading values, define $\preccurlyeq_{\text{coeff}}$.
Calling either lift an automorphism	Explicit omitted range elements prove that both are proper embeddings.
Treating a classical reformulation as a new consistency theorem	Cover and exact-descent criteria are proved equivalent to the corresponding fine-ultrafilter principles.

06's delivered PDF was compiled, checked for unresolved references and overfull boxes, and visually inspected; that document verification is not a machine verification of the mathematics.

B.4 09's specific error checks [09]

The crucial checks are mathematical, not numerical sampling. All normal-form terms in (5.1) are nonzero; otherwise an additional index might disappear and the equalizer conclusion would fail. The transformed exponent sequence is strictly decreasing, so the complement in (5.3) cannot cancel. The index κ really is the first missing one, even when the length is much larger than κ . The support bound for the coefficient span in Theorem 13.3 uses only finitely many coefficient supports, so it remains valid at singular uncountable κ . The branch code uses coefficients one and two, not zero and one, to retain the node's length. The ultrafilter code instead uses zero and one, where zero deletion is intended. Coefficientwise multiplication is always explicitly separated from ordinary surreal multiplication. The equalizer is explicitly separated from a fixed field. Finally, all constructions use set-sized supports and set-sized families, although their ambient fields are proper classes. 09's delivered PDF was compiled from its source; 09 reports no undefined references, missing citations, overfull boxes or LaTeX warnings, and visual review, and states that these checks concern document production, not the proofs.

B.5 Claim summary

Under (1.1), the exact defect, support equalizer, image intersection, projection isomorphism, positive escape, measure coefficients, omnific and surcomplex transfers, exponential locus, and two closures are proved in the text, relative to the stated classical inputs. The supercompact construction is proved with its additional closure hypothesis. The mask criterion is a proved reformulation of the classical measure characterization. Publication priority, consistency-strength converses, joint algebraic independence, differential compatibility, and all twelve research questions remain outside the claims of this article.

[merge] For the merged report: 06's landing criterion, landing spectrum, one-number test and normal-measure trichotomy are proved under their stated hypotheses (a set-ultrapower presentation for sequence

closure, an index set of size κ for the trichotomy); its support-cover and exact-descent characterizations and 09's seed, covering-seed and prefix-tree statements are proved translations of classical principles; 09's short-support field, almost-disjoint independence and 2^κ family are proved in ZFC (the last under $2^{<\kappa} = \kappa$); 09's compatible-field-embedding criterion is a proved infinitary equivalence. The 34 research questions of the three sources remain outside the claims, except the exponential half of Question 21.30. Appendix C lists every non-claim.

[merge] Batch 32: manuscript 10's first-failure theorem, local representation, canonical strong part, transvections, workspace criterion and 2^κ bound, and surcomplex transfer are proved in ZFC or NBG with global choice, the constructions under a measurable cardinal; its field-embedding equivalence and target test are proved relative to this report's absoluteness lemma. The consistency-strength converse is now proved for countably strong maps only (Section 26); its twelve questions, and the unanswered parts of the questions it bears on, remain outside the claims.

C Non-claims, by source [merge]

Every limitation, disclaimer and priority caveat stated by any source is kept, with its place in this report; the delivered proof-status files are shipped unchanged. The lists are grouped by source; items are paraphrased only where the source's own words appear elsewhere in the report.

Manuscript 07 (base).

- 07.1 Not refereed, not Lean-verified; publication priority not certified; no solution of a named published conjecture and no new consistency proof for a large-cardinal axiom (title page, Section 1.4).
- 07.2 No claim that arbitrary non-strong surreal embeddings require a measurable cardinal; the two embeddings are not identified as automorphisms; equality on monomials does not imply equality on a completed field (Section 1.4).
- 07.3 The proofs are not theorem-prover output; the finite checks test finite identities only and cannot test an elementary embedding or a nonprincipal complete ultrafilter (Section 1.4, Appendix B).
- 07.4 The literature and repository comparison is targeted, not an exhaustive priority search; "candidate original" means not located in the material inspected, not that no specialist has observed it; a specialist comparison is appropriate before any priority claim (Section 20).
- 07.5 Strong sums are not order-topological limits (Remark 2.2).
- 07.6 The sign-expansion theorem is a classical input, not reproved; an arbitrary Hahn-field identification would not suffice (Remark 2.8).
- 07.7 Hahn reindexing is standard; the derived measure, its properties, ultrapowers and the fine-normal characterization of supercompactness are classical; the mask criterion's ultrafilter characterization is not new (Sections 4, 10, 16, 20).
- 07.8 The equalizer $\mathcal{S}_{<\kappa}$ and the common image C_j are not identified (Remark 6.2); exact intersection gives transcendence, not linear disjointness or joint algebraic independence of the image fields.
- 07.9 No unconditional cardinality criterion for signed families (remark after Theorem 8.1); a zero probe does not mean a zero defect; the character theorem concerns the Hadamard product only; the mask criterion is not a characterization in the pure ordered-field language.

- 07.10 No real-valued probability measure on all surreal sets is introduced (after Theorem 10.1); no new factorization theorem for set-sized quotient rings (Section 11).
- 07.11 The surcomplex maps fix a chosen orientation i and say nothing about arbitrary automorphisms of the pure surcomplex field (Section 12).
- 07.12 D_j satisfies a twisted identity; it is not claimed to be a Berarducci–Mantova or strongly additive derivation; only the near-one logarithm is used for L_j , and no hyperclass or class-sized quotient is formed (Section 14).
- 07.13 The supercompact construction depends on the enumeration; no new large-cardinal equivalence; an enumeration-free seed geometry is a question (Section 16).
- 07.14 No exception to Kunen-type obstructions; the inclusion is elementary only in the ordered-field reduct; the information hierarchy’s arrows mean forgetting, not extension (Section 19).
- 07.15 Choice is used only for sets: no basis of $\mathbf{No}$, no class Zorn lemma, no collection of all class endomorphisms (Section 19).
- 07.16 No claim to settle a numbered question of Kaplan–Krapp–Serra; results assuming bijectivity or pointwise preservation of the value group must not be applied to this pair unchecked (Section 20).
- 07.17 The repository’s Lean Hahn infrastructure does not mean any theorem here is checked; the proposed Lean interface is not a formalization claim; the measurable-cardinal assumption must not become an unnamed axiom (Section 20, Appendix A).
- 07.18 No claim that $H_j(\mathbf{No}) \subseteq \mathbf{No}^M$ or $J(\mathbf{No}) = \mathbf{No}^M$ (Appendix B).
- 07.19 The twelve questions are proposals, not famous open problems; consistency-strength converses, joint algebraic independence, differential compatibility, recovery of the ambient embedding, classification of image pairs and an enumeration-free supercompact invariant are not established (Section 21, delivered proof status).
- 07.20 The repository inspection covered selected root documentation, the catalogue, two report READMEs and a source preamble; it was not a full audit or exhaustive novelty search; no repository theorem is an unproved premise; the repository was not modified, no Lean files were written and the repository build was not rerun (Section 20, delivered README and proof status).
- 07.21 The finite models are deliberately not finite elementary embeddings; there is no finite analogue of the large-cardinal assumption; the program does not verify elementarity, measurability, class recursion, normal-form absoluteness, transfinite summation, supercompactness, or the article (delivered proof status and recorded data).

Manuscript 06.

- 06.1 An AI-assisted draft with written proofs, not refereed or machine-checked; the proposed contributions are the comparison, defect and descent statements, not the Hahn-lifting or ultrapower-seed machinery; not a priority certification; no named published conjecture resolved (Section 1.4).
- 06.2 All large-cardinal implications retain their hypotheses; no consistency strength beyond the classical ultrafilter principle; no large-cardinal axiom is claimed proved in ZFC (Section 1.4, Section 17).
- 06.3 The repository’s birthday, hereditary-set, embedding and forcing work is not repackaged as new; no new proof needs an unreviewed repository theorem; mere existence of a non-strong map is not a contribution (Section 20).

- 06.4 The absoluteness lemma is an interface, not a claim that the Hahn fields of M and V coincide (Remark 2.7).
- 06.5 The maps are proper embeddings, not automorphisms; the Kaplan–Krapp–Serra automorphism classification (their Question 5.6) is not answered; no conflict with Kunen (after Proposition 7.4).
- 06.6 The “one number” test is relative to j and M ; no pure-field formula defines supercompactness (after Corollary 9.4).
- 06.7 The seed–closure converse is proved only for set-ultrapowers; the no-descent result above κ uses an index set of size κ and is not a theorem about every embedding with critical point κ ; $j(\kappa)$ is inaccessible in M , not asserted in V (Sections 9, 19).
- 06.8 Coefficient extraction and the monomial scale are specified extra structure, not claimed definable in the pure field language (after Corollary 10.5).
- 06.9 The compactness criteria are equivalences with classical principles; a measurable cardinal is not shown to fail a stronger property because of its normal-measure ultrapower’s landing spectrum (Section 17).
- 06.10 No global surcomplex exponential or logarithm is assumed (after Theorem 12.3).
- 06.11 Preserving **Oz** is not preserving it pointwise; not a theorem that all endomorphisms of **Oz** come from large cardinals (Section 11).
- 06.12 No automorphism classification, Kunen inconsistency result, new consistency theorem or first-order characterization in the pure surreal field; no large cardinal may be inferred from an arbitrary non-strong field map (Section 19).
- 06.13 No Lean proof; the repository’s Hahn layer is not verification of 06; finite experiments cannot validate the critical-point or measure assertions, so none is supplied; the archive contains no purported formal proof (Appendix A).
- 06.14 The literature check was targeted; the Conway and Gonshor books were not retrieved in full; a Wikipedia page served as orientation only; not performed: peer review, Lean or other proof-assistant verification, computation of a nonprincipal ultrafilter (delivered proof status).
- 06.15 The questions are proposed directions, not published open problems (Section 21); the repository inspection was a scope and overlap check, not a full audit (Section 20).

Manuscript 09.

- 09.1 An unrefereed AI-assisted manuscript; its contributions are proposed, not certified priority claims; not formally verified; no established named open problem is claimed settled (Section 1.4, delivered README).
- 09.2 Merely constructing a Hahn lift, observing that embeddings fail to preserve long sequences, or coding an ultrafilter numerically would not by itself be the result (Section 1.3).
- 09.3 No existence or consistency of any large cardinal; no nontrivial $(V, \in) \rightarrow (V, \in)$; the structure questions for all surreal exponential automorphisms are not settled (Section 1.4).
- 09.4 The maps are embeddings, not automorphisms; the short-support field construction is not claimed new; no claim to have initiated the large-cardinal and generalized-analysis connection; repository bi-interpretation results are not used as assumptions (Section 20).

- 09.5 Definability alone does not give absoluteness; the reals remark does not put every set of reals or of old surreals into M (Section 2).
- 09.6 Support cardinality, birthday and order-bounded support are different bounds; individual transcendence is much weaker than independence; the equalizer is not a fixed field (Sections 3, 5).
- 09.7 D_j is not an ordinary derivation; continuity is an interval statement about a class order; not every family of size κ fails; the omega incompatibility concerns a specified monomial action only (Sections 5, 8, 14).
- 09.8 No conflict with repository claims about maps to set-sized rings: the embeddings have proper-class codomains (Section 11).
- 09.9 The independence theorem is not a large-cardinal theorem (Section 13).
- 09.10 Measure recovery is not a proof that an arbitrary field embedding yields a measure; a zero coefficient is not a zero defect; masks are not ordinary idempotents; the numerical criterion is infinitary and not first-order in the bare field or omnific-ring language (Section 10).
- 09.11 A derived ultrafilter need not determine j or M ; the seed and compactness statements are numerical presentations of classical mechanisms, not new consistency equivalences (Section 16).
- 09.12 The prefix principle is an exact coding of a classical tree principle, not a new relative consistency proof; the branch is not an order-topological limit; the Carl–Galeotti–Löwe hypotheses must not be replaced (Section 18).
- 09.13 Parameter-free field sentences cannot detect large cardinals; the exact expressive strength of the birthday predicate is not resolved (Section 19).
- 09.14 Every sum is set-indexed; there are no proper-class sums; the omega-map is not the Gonsior exponential; pure field elementarity does not give elementarity in an enriched birthday language or a self-embedding of V (delivered proof status, items 1–10).
- 09.15 The proofs are written, not Lean proofs; no finite program tests a measurable cardinal; the claims were not imported into the repository’s Lean development; the literature check does not certify that no equivalent result exists elsewhere; the repository comparison was targeted and no repository file was modified (Appendix A, delivered README).
- 09.16 The questions include genuine gaps and are not an exhaustive list of published problems (Section 21).

Added by the merge.

- M.1 No statement of this report has a Lean implementation mapping; independent proof review and formalization are pending.
- M.2 Remark 2.9 covers same-ordinal inner models with the same reals only; the classification asked in `dsn:q:support` stays open.
- M.3 Remark 11.6 is conditional on a measurable cardinal (or a class embedding with critical point); whether an **Oz**-reflecting non-strong embedding exists without large cardinals, and the clauses on dense images, are not addressed. (Batch 32: such an embedding fixing $\mathbb{R}$ would have to fail a countable sum, Theorem 26.2; existence stays open.)
- M.4 Of the 34 questions of 06, 07 and 09 only the exponential half of Question 21.30 was answered by the three-source merge (batch 32 adds the status notes listed in M.7); Theorem 13.3 does not decide Question 21.6.

- M.5 The merge found no contradiction between the three sources; its corrections (Appendix D) are bibliographic and concern the repository identifier, not mathematics.
- M.6 A clean build and the rerun finite regression certify no transfinite statement.
- M.7 Batch 32: the status notes at Questions 21.1, 21.2, 21.13, 21.22 and 21.23 record answers or partial progress for countably strong maps only; the parts they name stay open. The re-scoped sentences of Sections 1.4 and 19, and items 07.2 and 06.12 above, remain true of arbitrary maps.

Manuscript 10 (batch 32). Its non-claims are items 10.1–10.19 of Section 26.6, with three merge items M10.1–M10.3 there; in brief: the countable hypothesis is essential and embeddings failing a countable sum are not covered; nothing on Gaussian ring automorphisms or target tests for strong embeddings; full Hahn fields only; no global assembly of local data; the transvections are additive, not multiplicative, and fix monomials without respecting the omega-map as a function; Bergman’s mechanism is classical; no existence or consistency of measurable cardinals; Section 26 depends on this report’s absoluteness lemma; the finite checks verify no ultrafilter; unrefereed, no Lean, priority not certified.

D Provenance and merge decisions [merge]

Sources. The report is built from three manuscripts of batch 30, committed as archives in 190d301 and placed in 21375f8. Each archive was matched to its manuscript number by the hashes of its proof-status file.

No.	Archive	Pin	Shipped here
06	Surreal_Large_Cardinals_Critical_Supports.zip (27-page PDF)	0865f04	proof status, build helper (prefixed 06-critical-supports-)
07	Surreal_Large_Cardinals_Research.zip (32-page PDF)	0865f04	this article’s base text; proof status, build helper, finite regression and its three recorded outputs (prefixed 07-critical-point-defects-)
09	surreal_large_cardinals_article.zip (30-page PDF)	0865f04	proof status, build helper (prefixed 09-support-geometry-)
10	surreal_measurable_spectrum.zip (24-page PDF; batch 32, in aa268a4)	7af8a30	proof status, source audit, build helper, finite regression and its four recorded outputs (prefixed 10-measurable-spectrum-)

All three pin the same commit 0865f043aec113c14c69ef45006bbc7546a4e75a (“Merge reviewed documentation after angular multiplicity validation”), 55 commits before the placement. The manuscripts’ .tex, PDFs and READMEs of 06 and 09, and 07’s PDF, are not shipped; 07’s delivered README was replaced by the collection README.

Where the merge had to choose.

- *Base.* 07, as decided at placement: on the shared theorems 06 and 07 have the weakest hypotheses and 07 is broader. 09’s headline assumes a normal-measure ultrapower, though its arguments need no more than 07’s.

- *Printed once.* The first missing index, the absoluteness lemma, the induced embedding, the strong companion, the defect formula and equalizer, the leading-term corollary, the omega-map locus, the small-family threshold and first failure cardinal, the derived normal measure, the distinguishing corollary, the zero-probe warning, the character theorem, the omnific equalizer and constant terms, the surcomplex transfer, the twisted product law, exponential transfer and the exponential locus. Each carries its sources.
- *Second routes.* 09’s proof of the absoluteness lemma (limit cylinders and truncation simplicity); 09’s direct real-closedness of $\mathcal{S}_{<\kappa}$ beside 07’s proof through the embeddings; 06’s direct support counting for the short fields; 09’s fiber argument in the weighted evaluation; 06’s formal-evaluation proposition for the infinitesimal factor.
- *Kept separately, as different statements.* The three measure codes h_A , $\tilde{h}_A$, Y_A and their omnific shifts; the mask criterion (07) and the compatible-field-embedding criterion (09); 07’s supercompact seed coefficient, 06’s exact-descent and cover characterizations and 09’s covering and exact seeds; the three sources’ questions, with overlaps noted.
- *Printed from a source where 07 took the fact as background.* 06’s proof of the critical-point facts (Lemma 2.3).
- *Headline theorems.* 06’s and 09’s summary theorems are printed in Section 1.3 with proof pointers, not as separate proofs.

Discrepancies found and corrected.

- 06 records the snapshot 0865f043 . . . as a tree object, “not asserted to be a commit”. It is a commit; its message is quoted above. The delivered 06 proof status is unchanged.
- 07 gave Gonshor’s book as Lecture Note Series 124; 06 and 09 give 110, which is correct. The bibliography now says 110. (The definable-surreals report records the same correction for another manuscript.)
- 06’s bibliography links the universes and birthday-cutoff reports on the main branch; 07’s and 09’s link them at the snapshot. The merged bibliography keeps 07’s entries.
- 06 states the counting proof of the short fields with “regularity and uncountability”; only uncountability is used (note after Corollary 3.2). This is a remark, not a correction of a false statement.
- 07’s text names its program and outputs by their delivery paths; the shipped paths are given in Appendix B.

No false theorem and no contradiction between the sources was found.

Manuscript 10 (batch 32). Manuscript 05 of batch 32, local number 10, archive surreal_measurable_spectrum.zip delivered in aa268a4 and placed in 7d04483; pinned to 7af8a3026c8440944e94df89020d42ca08f4e3dd, which precedes the three-source merge de45cee. It contributes Sections 22–26, placed after Section 21 so that no existing number changes; its .tex, PDF and README are not shipped. Where the merge chose:

- *Exponent convention.* 10’s t -exponents are converted to this report’s ω -exponents with a sign change throughout (Section 22.4); 10’s detector $d \leq -1$ is $\delta = -d \geq 1$, and its masks are 09’s Y_A shifted by ω^δ .
- *Printed once.* 10’s Proposition 10.1 (the construction of J , its $<\kappa$ -strongness, first failure at κ , $\mathbf{Oz}$ reflection and $J_{\text{at}} = H_j$) is Proposition 26.1 with pointers to Proposition 2.10, Theorem 11.1, Theorem 8.1,

Corollary 8.3 and Theorem 4.3; 10 says it is not new. 10’s group and field of Theorem 8.2 are 09’s G_κ and K_κ (Theorem 13.5). 10’s Lemma 3.1 is the elementary half of `opa:as:cor:countable`; its product law for $F - F_{\text{at}}$ is 06’s and 09’s (14.3) for J .

- *Second routes.* The exclusion $-\kappa \notin J(\mathbf{No})$ in Theorem 26.4 (10’s elementarity argument and the defect theorem); the existence of the U_j -large fiber (10’s bisection, Lemma 23.6, and 09’s cardinality argument in Proposition 10.8).
- *Merge deductions*, marked [merge]: Remark 23.10, Corollaries 26.5 and 26.6, Proposition 26.7, the reflection clause after Proposition 24.7, the Kaplan–Krapp–Serra comparison in Remark 26.3, and the status notes in Section 21.

Stale against the tree: 10 cites this report as 07’s text and its “Sections 2–4”; it did not see 06’s and 09’s questions that it answers; its claim to give a conditional negative answer to `opa:as:q:targettests` was already implied at its pin (the note after Theorem 26.4). 10 states reflection of **Oz** only for the strong part; its proof gives it for the embedding too (not a false statement). No false theorem was found. The delivered finite regression was rerun on a copy under Python 3.14.4: 11,955 assertions in 18 categories, console and JSON identical to the recorded files.

References

- [1] John H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001; first edition, Academic Press, 1976. Classical source for surreal arithmetic, normal forms, and omnific integers.
- [2] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. Especially Chapters 5 (normal forms) and 10 (exponentiation). [merge] 07 gave the series number as 124; 06 and 09 give 110.
- [3] Lou van den Dries and Philip Ehrlich, “Fields of surreal numbers and exponentiation,” *Fundamenta Mathematicae* **167**(2) (2001), 173–188. doi:10.4064/fm167-2-3. Publisher-hosted text consulted: <https://www.impan.pl/shop/en/publication/transaction/download/product/89226>.
- [4] Philip Ehrlich, “Conway names, the simplicity hierarchy and the surreal number tree,” *Journal of Logic & Analysis* **3**:1 (2011). <https://logicandanalysis.org/index.php/jla/article/download/88/34>.
- [5] Elliot Kaplan, Lothar Sebastian Krapp, and Michele Serra, “Decomposing the automorphism group of the surreal numbers,” arXiv:2509.22374, version 3, consulted 23 September 2026. <https://arxiv.org/html/2509.22374v3>. Used for the contemporary Hahn and strong-map framework, not as a premise for a particular non-strong automorphism example. [merge] Manuscript 10 consulted the same version; its Example 4.4 is discussed in Remark 26.3.
- [6] Itay Neeman, “Ultrafilters and Large Cardinals,” author-hosted manuscript, Section 1, consulted 23 September 2026. <https://www.math.ucla.edu/~ineeman/uflc.pdf/>. Includes the standard elementary-embedding, derived-measure, and supercompact-seed constructions.
- [7] George M. Bergman, “Homomorphisms on infinite direct products of groups, rings and monoids,” arXiv:1406.1932, version 2, consulted 23 September 2026. <https://arxiv.org/html/1406.1932v2>. Context for ultrafilters and homomorphisms of infinite product algebras. [merge] Not the paper [25] cited by manuscript 10.
- [8] Vladimir Reshetnikov and contributors, *Surreal*, repository snapshot 0865f043aec113c14c69ef45006bbc7546a4e75a, root README.md and docs/README.md, consulted 23 September 2026. <https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a>. Repository statements and formalization coverage are distinguished from independently checked proofs in the present article.

- [9] *Surreal Fields Across Set-Theoretic Universes*, repository research report, README and source preamble inspected at the snapshot in [8].
<https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a/docs/foundations-and-computation/surreal-fields-across-universes>. Research context only; its new theorems are not premises here.
- [10] *Birthday Cutoffs and Hereditary Sets*, repository research report, README inspected at the snapshot in [8].
<https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets>. The README credits the full-class bi-interpretation announcement to Chen–Hamkins–Yang. That announcement is not claimed as new here.
- [11] *Set-Sized Quotients of Omnific Integers*, catalogue entry in docs/README.md at the snapshot in [8]. Relevant report directory: <https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a/docs/surreal/set-sized-quotients-of-omnific-integers>. Cited only to identify the repository’s existing constant-term theme.
 [merge] Entries added from manuscripts 06 and 09 (as delivered; 06’s *conway*, *gonshor*, *kks*, *neeman*, *repo*, *birthday*, *universes* and 09’s *Conway*, *Gonshor*, *DE*, *KKS*, *Repo*, *RepoUniverses*, *RepoBirthdays* are the entries above).
- [12] K. J. Williams, *Math655 Lecture Notes: Part 1.2. Supercompact cardinals and beyond*, 25 February 2019. In particular Definition 1, Fact 7, and the seed characterization.
<https://juliakw.net/teaching/2019/math655/part1.2.pdf>. [06]
- [13] A. Berarducci, S. Kuhlmann, V. Mantova, and M. Matusinski, *Exponential fields and Conway’s omega-map*, Proceedings of the American Mathematical Society **151** (2023), 2655–2669. Author version arXiv:1810.03029v2, especially Definition 2.10 and the surreal specialization. <https://arxiv.org/html/1810.03029v2>. [06]
- [14] A. Berarducci, *Surreal numbers, exponentiation and derivations*, arXiv:2008.06878, 2020. In particular, Theorems 10.2 and 19.2. <https://arxiv.org/abs/2008.06878>. [09]
- [15] S. Kuhlmann and S. Shelah, κ -bounded exponential-logarithmic power series fields, Annals of Pure and Applied Logic 136 (2005), 284–296. <https://arxiv.org/abs/math/0512220>. [09]
- [16] S. Kuhlmann and M. Serra, *The automorphism group of a valued field of generalised formal power series*, Journal of Algebra, 2022, <https://doi.org/10.1016/j.jalgebra.2022.04.023>; preprint version 3, <https://arxiv.org/abs/2107.03362v3>. [09]
- [17] J. Holland, *An Introduction to Measures and Ultrapowers*, GOST Seminar Notes, 2020, updated 11 October 2022. <https://sites.math.rutgers.edu/~jch258/assets/Measures%20and%20Ultrapowers.pdf>. [09]
- [18] A. Kanamori, *The Higher Infinite: Large Cardinals in Set Theory from Their Beginnings*, second edition, Springer, 2003; corrected softcover printing, 2009. <https://doi.org/10.1007/978-3-540-88867-3>. [09]
- [19] J. Bagaria and G. Goldberg, *Reflecting measures*, author preprint, 6 September 2024. Standard seed and normal-fine ultrafilter facts are used here, not its new reflection results.
<https://math.berkeley.edu/~goldberg/Papers/ReflectingMeasures.pdf>. [09]
- [20] D. Marker, *Model Theory: An Introduction*, Graduate Texts in Mathematics 217, Springer, 2002.
<https://doi.org/10.1007/b98860>. [09]
- [21] M. Carl, L. Galeotti and B. Löwe, *The Bolzano–Weierstrass theorem in generalised analysis*, Houston Journal of Mathematics 44 (2018), no. 4, 1081–1109. Preprint: <https://eprints.illc.uva.nl/id/document/10115>. Journal contents: <https://www.math.uh.edu/~hjm/Vol44-4.html>. [09]
- [22] J. D. Hamkins, *The elementary theory of surreal arithmetic is bi-interpretable with set theory*, announcement of joint work with J. Chen and R. Yang, Kobe talk, September 2025. Cited as an announcement, not as a fully audited published proof. <https://jdh.hamkins.org/theory-of-surreal-arithmetic-bi-interpretable-with-set-theory-japan-september-2025/>. [09]
- [23] *Bounded-Support Hahn Arithmetic: Cofinal Descent, Optimal Support Thresholds, and Maximal Transcendence*, repository research report, introduction and definitions at the snapshot in [8].
<https://github.com/VladimirReshetnikov/Surreal/blob/0865f043aec113c14c69ef45006bbc75>

- 46a4e75a/docs/surreal/transcendence-over-bounded-support/article.tex. [09]
- [24] Wikipedia contributors, *Surreal number*, user-supplied overview, consulted 23 September 2026. Technical arguments use the primary sources rather than this overview. [09]; 06 used the same page as orientation only. https://en.wikipedia.org/wiki/Surreal_number.
- [merge] Entries added from manuscript 10 (batch 32; its *Conway*, *Gonshor* and *KKS* are the entries above, and its *RepoCrit* is this report).
- [25] George M. Bergman, “Families of ultrafilters, and homomorphisms on infinite direct product algebras,” *Journal of Symbolic Logic* **79** (2014), 223–239. In particular Lemmas 3–4 of the preprint. <https://doi.org/10.1017/jsl.2013.5>; <https://arxiv.org/abs/1301.6383>. [10]
- [26] George M. Bergman and Nazih Nahlus, “Linear maps on k^I , and homomorphic images of infinite direct product algebras,” *Journal of Algebra* **356** (2012), 257–274. <https://doi.org/10.1016/j.jalgebra.2012.01.004>; <https://arxiv.org/abs/0910.5183>. [10]
- [27] Richard Blute, Robin Cockett, Pierre-Alain Jacqmin, and Philip Scott, “Finiteness spaces and generalized power series,” *Electronic Notes in Theoretical Computer Science* **341** (2018). <https://doi.org/10.1016/j.entcs.2018.11.002>; <https://arxiv.org/abs/1805.09836>. [10]
- [28] Thomas Jech, *Set Theory: The Third Millennium Edition, Revised and Expanded*, Springer Monographs in Mathematics, Springer, 2003. <https://doi.org/10.1007/3-540-44761-X>. [10]
- [29] Irving Kaplansky, “Maximal fields with valuations,” *Duke Mathematical Journal* **9** (1942), 303–321. <https://doi.org/10.1215/S0012-7094-42-00922-0>. [10]
- [30] *Omnific-Preserving Automorphisms*, repository research report, especially its Part III, “Automatic strongness and proper embeddings of the omnific pair,” and Questions `opa:as:q:targettests` and `opa:as:q:gaussian`; inspected by manuscript 10 at snapshot `7af8a3026c8440944e94df89020d42ca08f4e3dd`. <https://github.com/VladimirReshetnikov/Surreal/blob/7af8a3026c8440944e94df89020d42ca08f4e3dd/docs/surreal/omnific-preserving-automorphisms/article.tex>. Now [docs/surreal/omnific-preserving-automorphisms](https://github.com/VladimirReshetnikov/Surreal/blob/7af8a3026c8440944e94df89020d42ca08f4e3dd/docs/surreal/omnific-preserving-automorphisms). [10]